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Emre Coşkun

Projective Geometry

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ISSN 2296-4568 ISSN 2296-455X (electronic)
Compact Textbooks in Mathematics
ISBN 978-3-032-02195-3 ISBN 978-3-032-02196-0 (eBook)
https://doi.org/10.1007/978-3-032-02196-0

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Preface

This book arose from my lecture notes for a course on projective geometry given at the Middle East Technical University in spring 2023 and spring 2024 semesters. It provides an introduction to the basic concepts of projective geometry using first an analytical and later an axiomatic (or synthetic) treatment, before moving on to the more advanced topics of conics.

I chose the topics in this book out of a certain dissatisfaction with what other books had to offer. There are certainly many excellent textbooks on projective geometry, ones from which I have learned a lot—the books by Coxeter [2] and Hartshorne [6] deserve special mention here; some of the ideas in the latter have been adopted here simply because I found it impossible to improve upon them. Klein's classic book [9], too, is always useful and a pleasure to read. However, most introductions to projective geometry use either the axiomatic approach or the analytic approach exclusively. In contrast to this situation, in the first two chapters of the book, I have tried to provide both points of view. Thus, Chap. 2 defines projective planes over fields and investigates their properties. Particular attention is given to symmetry groups of objects of projective geometry such as planes or lines, as well as to using such symmetries to simplify calculations and prove theorems such as those of Desargues and Pappus. Chapter 2 starts with axioms for affine and projective planes and investigates many of the same properties of projective planes as in Chap. 1, but this time using an axiomatic approach. It ends with the coordinatization of projective planes on which Desargues' and Pappus' theorems hold. The main results are that a projective plane where Desargues' theorem holds is a projective plane over a unique division ring, and if Pappus' theorem holds as well, this division ring is in fact a field. Thus, at the end of Chap. 2, we come full circle. Chapters 1 and 2 are independent of each other, making the book useful for courses where the instructor wishes to emphasize one approach or the other.

Chapter 3 treats conic sections in projective planes, a topic that I feel is often neglected in previous textbooks on projective geometry. I have followed an analytic approach here, as most authors who treat the topic do, but the reader should be aware that it is possible to treat conics from a synthetic point of view, as Coxeter does [2]. However, Coxeter works on projective planes where Pappus' Theorem holds; in other words, his projective planes are really projective planes over fields. Hence, any results he proves are also true in the analytic setting.

Treating conics from an analytic point of view gives an analytic slant to the book overall. This is due to my own biases as an algebraic geometer and to my hope that after reading this book the reader will be well equipped to move on to more advanced works on the algebraic geometry of plane curves, such as the books by Fulton [5], Kirwan [8], and others.[1]

The book is intended for undergraduate students in mathematics who have completed at least two years of study. More precisely, in addition to analytic geometry, the reader is expected to be familiar with not only linear algebra but also abstract algebra including the basic properties of groups and fields. Experience with field theory is not necessary; what little is used in Chap. 3 (quadratic field extensions) can certainly be learned quickly by the reader if needed. In any case, the reader who is not comfortable working over arbitrary fields can safely assume that the base field is **R** or **C**.

In writing this book, I have received assistance and encouragement from various people. My thanks go to Dorothy Mazlum, editor of the Compact Textbooks in Mathematics series, for her patience in answering my endless questions about the publishing process. I also thank production editor Fairle Thattil for her help in the production of the book, and the anonymous reviewers for their comments. My students Filiz Badici, Atakan Erdem, Javad Aliyev, Sema Yıldırım, and Ayşe Akımsar pointed out errors when I was using the draft of the book to teach projective geometry. My dear friend Phillip Kurzeja has been a source of support and friendship for me over many years. Finally and above all, I am grateful to my dear wife Elif, who has never ceased to encourage me in the conception and writing of this book.

Ankara, Turkey Emre Coşkun
June 2025

[1] In the review process of this book, it was pointed out that the draft contained no historical information about the development of projective geometry. This is a deliberate decision on my part as I have no detailed knowledge of that subject. In any case, the reader who is interested in the history of projective geometry can now consult the recent monograph by Del Centina and Gimigliano [3].

Declarations

Competing Interests The author has no competing interests to declare that are relevant to the content of this manuscript.

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List of Notations

$\mathbb{A}^2(F)$	the affine plane over the field F
$P, Q, R, \ldots$	points
$P = (x_0, y_0)$	the point in $\mathbb{A}^2(F)$ with x-coordinate x_0 and y-coordinate y_0
$\ell, m, n, \ldots$	lines
$\ell : ax + by + c = 0$	the affine line with equation $ax+by+c = 0$, where $a, b \in F$ are not both zero
ℓ_x, ℓ_y	the x- and y-axes
PQ	the join of P and Q, defined as the unique line passing through P and Q, used in both the affine and the projective setting
$\ell \parallel m$	ℓ and m are parallel
$\mathbb{P}^2(F)$	the projective plane over the field F
$P = [x_0 : y_0 : z_0]$	the point in $\mathbb{P}^2(F)$ with projective coordinates x_0, y_0, z_0
$\ell : ax + by + cz = 0$	the projective line with equation $ax + by + cz = 0$, where $a, b, c \in F$ are not all zero
ℓ_∞	the line at infinity, with equation $z = 0$
$\ell_1 \cdot \ell_2 = P$	two (affine or projective) lines intersecting in a unique point P
$(\mathbb{P}^2(F))^*$	the dual projective plane to $\mathbb{P}^2(F)$
$\mathrm{Aut}(\mathbb{P}^2(F))$	the group of automorphisms of $\mathbb{P}^2(F)$
$\mathrm{Aut}(\mathbb{A}^2(F))$	the group of automorphisms of $\mathbb{A}^2(F)$
$\mathrm{GL}_n(F)$	the group of $n \times n$ invertible matrices
$\mathrm{PGL}_n(F)$	the group defined as $\mathrm{GL}_n(F)/F^*$
$\mathrm{Aut}(F)$	the group of automorphisms of the field F
ABC	the triangle with vertices $A, B, C \in \mathbb{P}^2(F)$
$X \barwedge x$	elementary correspondence between point X and line x
$X \overset{=}{\barwedge} X'$ or $X \overset{P}{\overset{=}{\barwedge}} X'$	perspectivity (with center P) mapping X to X'
$\mathrm{PJ}(\ell)$	the group of projectivities of the line ℓ to itself
$(a, b; c, d)$	the cross-ratio of four distinct points $a, b, c, d \in F \cup \{\infty\}$

$(A, B; C, D)$	the cross-ratio of four distinct points A, B, C, D on a line
$\mathbb{A}^2$	an affine plane
$\mathbb{P}^2$	a projective plane
$(\mathbb{P}^2)^*$	the dual projective plane to $\mathbb{P}^2$
$\mathbb{P}^3$	a projective 3-space
$\mathcal{P}, \mathcal{Q}, \mathcal{R}, \ldots$	planes in a projective 3-space
$\ell \cdot \mathcal{P}$	the unique intersection point of a line and a plane in a projective 3-space
$\mathcal{P} \cdot \mathcal{Q}$	the unique line of intersection of two planes in a projective 3-space
$\mathbb{P}^3(F)$	the projective 3-space over the field F
$P = [x_0 : y_0 : z_0 : w_0]$	the point in $\mathbb{P}^3(F)$ with projective coordinates x_0, y_0, z_0, w_0
$\mathcal{P} : ax + by + cz + dw = 0$	the projective plane with equation $ax + by + cz + dw = 0$, where $a, b, c, d \in F$ are not all zero
$\mathbb{P}^2(D), \mathbb{P}^3(D)$	the projective plane and projective 3-space over the division ring D
$\mathrm{Dil}(\mathbb{A}^2)$	the group of dilations of $\mathbb{A}^2$
$\mathrm{Tran}(\mathbb{A}^2)$	the group of translations of $\mathbb{A}^2$
$\mathrm{Tran}_\ell(\mathbb{A}^2)$	the group of translations of $\mathbb{A}^2$ that map the line ℓ to itself
τ_{PQ}	the unique translation that maps P to Q
$\mathrm{Scal}_P(\mathbb{A}^2)$	the group of scalings of $\mathbb{A}^2$ with center P
σ^P_{QR}	the unique scaling with center P that maps Q to R
ℓ_P	the polar of point P
P_ℓ	the pole of line ℓ
$C(F)$	the set of points of the conic C over the field F
t^C_P	the tangent line to the conic C at the point P



The Analytic Approach: Projective Planes over a Field

1

Abstract

Projective geometry is developed analytically. Affine and projective planes are defined over fields, followed by an exposition of duality. We then investigate automorphisms of the projective plane over a field and determine the structure of the automorphism group. Some of these properties are then used to prove Desargues' and Pappus' Theorems. Projectivities and harmonic points are also covered.

In this chapter, we develop projective geometry from an analytical viewpoint. This means defining affine and projective planes over fields using affine and projective coordinates (Sects. 1.1 and 1.2). Affine versions of Desargues' and Pappus' theorems are also included (Theorems 1.1.16 and 1.1.17) to introduce, as early as possible, two of the most important results in projective geometry.

Duality (Sect. 1.3) is treated immediately after projective spaces as its explanation is very simple and intuitive once one has defined projective spaces over fields and projective lines in them.

Section 1.4, while technically not very demanding, forms the heart of the chapter, since the automorphisms of $\mathbb{P}^2(F)$ are often used later to prove a number of results. Among these are Desargues' and Pappus' theorems in Sect. 1.6. We also cover projectivities and harmonic points in Sects. 1.7 and 1.8.

Throughout this chapter, F denotes an arbitrary but fixed field, whose elements are called *scalars*. Vector spaces are over F, unless stated otherwise. The affine plane $\mathbb{A}^2(F)$ and the projective plane $\mathbb{P}^2(F)$ are over F. As stated in the preface, the reader can safely assume that the base field is $\mathbf{R}$ or $\mathbf{C}$ if (s)he is not comfortable working over an arbitrary field.

To avoid repetitiveness, we will not mention that scalars lie in the base field F, or points and lines lie on the affine plane $\mathbb{A}^2(F)$ or the projective plane $\mathbb{P}^2(F)$, whenever such information is clear from the context.

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E. Coşkun, *Projective Geometry*, Compact Textbooks in Mathematics,
https://doi.org/10.1007/978-3-032-02196-0_1

1.1 Affine Planes over a Field

Affine planes over fields are defined in analogy with the Euclidean plane over $\mathbb{R}$. We begin this chapter with a discussion of affine planes for two reasons: first, they are already likely to be familiar to the student and second, projective planes, which are the topic of the next section, can be considered to be an extension of affine planes in a natural manner. After defining affine planes over fields and studying some of their elementary properties, we introduce changes of coordinates and use them to prove affine versions of Desargues' and Pappus' theorems (Theorems 1.1.16 and 1.1.17). The reader is therefore introduced at the outset to the idea of using changes of coordinates to simplify analytic proofs, an idea that is used frequently in this chapter.

Definition 1.1.1 The *affine plane* $\mathbb{A}^2(F)$ *over the field* F is $F \times F$, the set of ordered pairs of elements of F. Elements of $\mathbb{A}^2(F)$ are called *points*. A subset of $\mathbb{A}^2(F)$ defined by an equation of the form $ax + by + c = 0$, where a and b are not both zero, is called a *line*. The elements $x_0, y_0 \in F$ are called the *coordinates* of the point $(x_0, y_0) \in \mathbb{A}^2(F)$ (Fig. 1.1).

Throughout the book, we adopt the convention of denoting points by uppercase letters, such as P, Q, R, and lines by lowercase letters, such as ℓ, m, n. A line ℓ with the equation $ax + by + c = 0$ will be denoted by $\ell : ax + by + c = 0$.

Let P and ℓ be a point and a line, respectively. We say that P *lies on* ℓ (or simply P is *on* ℓ) and ℓ *passes through* P, if $P \in \ell$.

Remark 1.1.2 The points on a given line ℓ are in one-to-one correspondence with the elements of F; hence, there exist at least two points on ℓ. We leave the proof to the reader.

Proposition 1.1.3 *Let* $\ell : ax + by + c = 0$ *be a line. For any nonzero scalar* λ, ℓ *can also be described with the equation* $\lambda ax + \lambda by + \lambda c = 0$. *Conversely, if* ℓ *can be described with the equation* $a'x + b'y + c' = 0$, *there exists a nonzero scalar* λ *such that* $a' = \lambda a$, $b' = \lambda b$, $c' = \lambda c$.

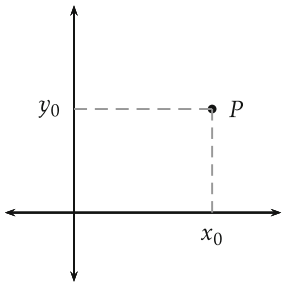


Fig. 1.1 A point $P = (x_0, y_0) \in \mathbb{A}^2(F)$ is defined by its coordinates $x_0, y_0 \in F$

Proof The first statement is clear since $(x, y) \in \ell$ if and only if $ax + by + c = 0$, if and only if $\lambda ax + \lambda by + \lambda c = 0$.

For the second statement, suppose that ℓ can also be described with the equation $a'x + b'y + c' = 0$. Then the system of linear equations

$$ax + by = -c,$$
$$a'x + b'y = -c',$$

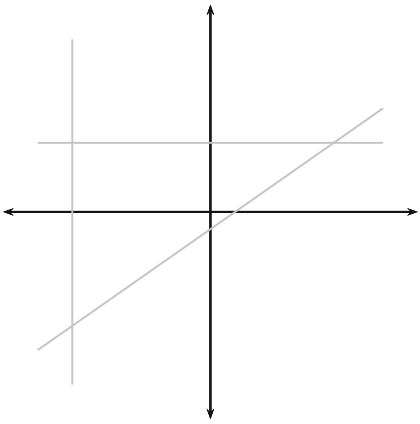
has more than one solution since there exist at least two points on ℓ. Hence, $ab' - a'b = 0$, which implies that there exists a nonzero scalar λ such that $a' = \lambda a$ and $b' = \lambda b$. It is now easily checked that $c' = \lambda c$ as well. □

Suppose that we are given the line $\ell : ax + by + c = 0$. If $b = 0$, then $a \neq 0$, and the equation of ℓ can be written in the form $x = k$, where $k = -\frac{c}{a}$. By Proposition 1.1.3, k is well-defined. If $b \neq 0$, then the equation of ℓ can be written in the form $y = mx + n$, where $m = -\frac{a}{b}, n = -\frac{c}{b}$. We call m the *slope* of the line. By Proposition 1.1.3, m and n are well-defined.[1]

We call lines whose equations can be written in the form $y = k$ *horizontal* and lines whose equations can be written in the form $x = k$ *vertical*. We also call the line $y = 0$ the *x-axis* and the line $x = 0$ the *y-axis*; they are denoted by ℓ_x and ℓ_y, respectively. The slope of horizontal lines is 0, and the slope of vertical lines is defined (symbolically) as ∞ (Fig. 1.2).

Proposition 1.1.4 *Given two distinct points P and Q, there exists a unique line that passes through both.*

Fig. 1.2 Lines in $\mathbb{A}^2(F)$



[1] The fact that the letter m is also used sometimes to denote lines causes no confusion in practice.

Proof We divide the proof into two cases. First, if $P = (k, y_1)$ and $Q = (k, y_2)$, with $y_1 \neq y_2$, it can be seen that the equation of ℓ cannot be of the form $y = mx + n$ since a given value of x would correspond to only one value of y. Hence, ℓ must be the line with equation $x = k$.

Otherwise, let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$, with $x_1 \neq x_2$. It is clear that the equation of a line ℓ passing through P and Q cannot be of the form $x = k$, so it must be of the form $y = mx + n$. Since $P, Q \in \ell$, we have

$$y_1 = mx_1 + n,$$
$$y_2 = mx_2 + n.$$

Solving this system for m and n, we see that these equations have the unique solution

$$m = \frac{y_2 - y_1}{x_2 - x_1},$$
$$n = \frac{y_1 x_2 - y_2 x_1}{x_2 - x_1}.$$

□

Definition 1.1.5 Let P and Q be two distinct points. The unique line passing through P and Q, whose existence is guaranteed by Proposition 1.1.4, is called the *join* of P and Q, and is denoted by PQ.

We note that there is usually more than one way to write a given line as the join of two points.

Definition 1.1.6 Two lines ℓ and m are called *parallel*, written $\ell \parallel m$, if $\ell = m$ or $\ell \cap m = \emptyset$.

Proposition 1.1.7 *The two lines given by equations* $ax + by + c = 0$ *and* $dx + ey + f = 0$ *are parallel if and only if* $ae = bd$.

Proof The proof is similar to that of Proposition 1.1.3. The system of linear equations

$$ax + by = -c,$$
$$dx + ey = -f,$$

describing the intersection of the two given lines has no solution, a unique solution, or more than one solution. A necessary and sufficient condition for either the first or the third case is that $ae - bd = 0$. If the system has more than one solution, in other words, if the lines intersect at more than one point, Proposition 1.1.4 implies that the two lines coincide. □

Corollary 1.1.8 *Two lines are parallel if and only if their equations can be written as* $ax + by + c_1 = 0$ *and* $ax + by + c_2 = 0$. *Two lines are identical if both of their equations can be written identically. Two lines are parallel if and only if they have the same slope.*

Proposition 1.1.9 *Two lines that are not parallel intersect in a unique point.*

Proof By Proposition 1.1.7, two lines that are not parallel have equations $ax+by+c = 0$ and $dx + ey + f = 0$, where $ae \neq bd$. From linear algebra it follows that the system consisting of these two equations has a unique solution. □

If the lines ℓ_1 and ℓ_2 intersect in a unique point P, we write $\ell_1 \cdot \ell_2 = P$.

We will see in Sect. 1.2 that a projective plane over a field can be written as the disjoint union of an affine plane over the same field and a "line at infinity." The line at infinity is composed of the equivalence classes of the relation of parallelism, which is an equivalence relation.

Proposition 1.1.10 *Parallelism is an equivalence relation on the set of lines in* $\mathbb{A}^2(F)$.

Proof Reflexivity and symmetry follow from Definition 1.1.6.

For transitivity, take three lines $\ell_i : a_i x + b_i y + c_i = 0$, $i = 1, 2, 3$, such that $\ell_1 \parallel \ell_2$ and $\ell_2 \parallel \ell_3$. By Proposition 1.1.7, we have $a_1 b_2 = a_2 b_1$ and $a_2 b_3 = a_3 b_2$. First, it can be easily seen that if any $b_i = 0$, then all other b_i's are also 0. In this case, we can write the equations of ℓ_i as $x = k_i$ for some $k_i \in F$, and the transitivity follows.

Otherwise, assume that $b_i \neq 0$ for $i = 1, 2, 3$. Multiplying both sides of the equation $a_2 b_3 = a_3 b_2$ by b_1 and using $a_1 b_2 = a_2 b_1$, we obtain $b_2 a_1 b_3 = a_3 b_2 b_1$. Canceling b_2 yields $a_1 b_3 = a_3 b_1$ and $\ell_1 \parallel \ell_3$ by Proposition 1.1.7. □

Definition 1.1.11 Equivalence classes of lines under parallelism are called *parallelism classes*.

Points that lie on the same line are called *collinear*. Lines that pass through the same point are called *concurrent*.

The next theorem summarizes some properties of $\mathbb{A}^2(F)$, which will be taken as axioms for the definition of an affine space in Chap. 2.

Theorem 1.1.12 $\mathbb{A}^2(F)$ *satisfies the following properties:*

A1. *Given two distinct points, there is a unique line that passes through both.*
A2. *Given a line* ℓ *and a point* P, *there is a unique line that passes through* P *and is parallel to* ℓ.
A3. *There exist three noncollinear points.*

Proof **A1** was proved in Proposition 1.1.4. For **A3**, it suffices to note that the points $(0, 0)$, $(1, 0)$, $(0, 1)$ are noncollinear.

For **A2**, let $\ell : ax + by + c = 0$ and $P = (x_1, y_1)$. Let $c_1 = -ax_1 - by_1$. If we define the line $m : ax + by + c_1 = 0$, then $P \in m$, and m is parallel to ℓ by Proposition 1.1.7. It is clear that m is the unique line with these properties: if there is another line m' that passes through P and is parallel to ℓ, then $m \parallel m'$ by Proposition 1.1.10; since $m \cap m' \neq \emptyset$, from Definition 1.1.6 it follows that $m = m'$. □

Remark 1.1.13 **A3** implies that $\mathbb{A}^2(F)$ is not equal to any line in it.

We have defined $\mathbb{A}^2(F)$ as the set of ordered pairs of elements of F, where the first element of the pair is the x-coordinate and the second element is the y-coordinate. Together, these coordinates form a *coordinate system*. This raises the following question: do there exist other coordinate systems on $\mathbb{A}^2(F)$? The answer is "yes," and we will now describe a method to change the coordinate system on $\mathbb{A}^2(F)$.

Consider an invertible matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

and expressions

$$x' = ax + by + m,$$
$$y' = cx + dy + n.$$

Note that

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = A \cdot \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} m \\ n \end{bmatrix}.$$

Since A is invertible, to every pair (x, y) of elements of F, there corresponds a unique pair (x', y') of elements of F, and vice versa. Indeed, one can obtain x and y from x' and y' by the formulas

$$x = px' + qy' + u,$$
$$y = rx' + sy' + v,$$

where

$$\begin{bmatrix} p & q \\ r & s \end{bmatrix} = A^{-1}$$

and

$$\begin{bmatrix} u \\ v \end{bmatrix} = -A^{-1} \begin{bmatrix} m \\ n \end{bmatrix}.$$

It is an easy exercise to prove that a change of coordinate systems maps lines to lines. Hence, changes of coordinate systems preserve *incidence*: if P lies on ℓ, then P' lies on ℓ', where the primes indicate the images of objects under the change of coordinates. In particular, intersections are preserved: if $\ell_1 \cdot \ell_2 = P$, then $\ell_1' \cdot \ell_2' = P'$.

The main advantage of being able to use different coordinate systems is the ability to simplify calculations in analytic proofs. Typically, a geometric configuration is at hand during a proof, and with a suitable change in coordinates, three of the points in the configuration can be mapped to $(0, 0)$, $(1, 0)$, $(0, 1)$, provided they are noncollinear. The following theorem is the main tool for this.

Theorem 1.1.14 *Given three noncollinear points P, Q, R, there exists a unique change of coordinates that maps these points to $(0, 0)$, $(1, 0)$, $(0, 1)$, respectively.*

Proof Let $P = (p_1, p_2)$. Since P, Q, R are noncollinear, the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ are linearly independent. Then there exists a unique invertible 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

that maps $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ to the standard unit vectors $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$, respectively. It can now be easily checked that the required change in coordinates is given by

$$\begin{aligned} x' &= ax + by + m, \\ y' &= cx + dy + n, \end{aligned}$$

where

$$\begin{bmatrix} m \\ n \end{bmatrix} = -A \begin{bmatrix} p_1 \\ p_2 \end{bmatrix}.$$

□

We now give two examples to illustrate the point that a suitably chosen coordinate change can simplify proofs of theorems considerably. The following theorem is a special case of the affine version of Desargues' Theorem. We shall prove Desargues' Theorem for projective planes later (Theorem 1.6.5). First, we need a definition.

Definition 1.1.15 A *triangle* in $\mathbb{A}^2(F)$ is a collection of three noncollinear points, called the *vertices* of the triangle, together with the joins of pairs of these points, called the *sides* of the triangle, which are then three nonconcurrent lines. The triangle with vertices A, B, C is denoted by ABC, or by any permutation of these letters. The sides of the triangle ABC are the lines AB, AC, BC.

The reader may have noticed the difference of Definition 1.1.15 from the usual concept of a triangle, where the edges are line segments. However, in order to be able to define the concept of line segments in $\mathbb{A}^2(F)$, we would have to assume that F is an *ordered field*. We will not discuss this here; the interested reader may consult [7].

Theorem 1.1.16 (Big Desargues' Theorem in the Affine Plane) *Suppose two triangles ABC and $A'B'C'$ are* perspective from the point *O, that is, AA', BB', CC' are concurrent at the point O. Suppose further that OA, OB, OC are distinct lines; A and A' are different from O; B and B' are different from O; and C and C' are different from O. Suppose finally that $AB \parallel A'B'$ and $AC \parallel A'C'$. Then $BC \parallel B'C'$.*

Proof The hypotheses imply that O, A, B are not collinear. Hence, using Theorem 1.1.14 we may assume, without loss of generality, that $O = (0, 0)$, $A = (0, 1)$, $B = (1, 0)$. Now let $A' = (0, a)$, $B' = (b, 0)$, $C = (c_1, c_2)$, $C' = (kc_1, kc_2)$ for some nonzero a, b, c_1, c_2, k. We now calculate slopes:

$$
\begin{aligned}
m_{AB} &= -1, \\
m_{A'B'} &= -\frac{a}{b}, \\
m_{AC} &= \frac{c_2 - 1}{c_1}, \\
m_{A'C'} &= \frac{kc_2 - a}{kc_1}.
\end{aligned}
$$

It follows that $a = b = k$. Now we find

$$
\begin{aligned}
m_{BC} &= \frac{c_2}{c_1 - 1}, \\
m_{B'C'} &= \frac{kc_2}{kc_1 - b},
\end{aligned}
$$

and it is clear that $m_{BC} = m_{B'C'}$. Hence, $BC \parallel B'C'$ by Corollary 1.1.8 (Fig. 1.3). □

Similarly, one can prove the affine version of Pappus' Theorem (Theorem 1.6.6). We leave the proof to the reader.

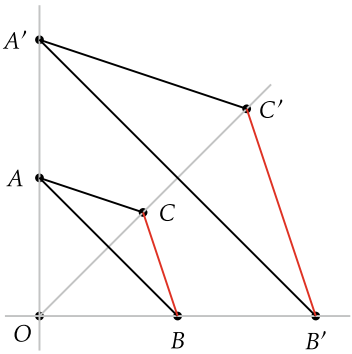


Fig. 1.3 Big Desargues' Theorem in the affine plane (Theorem 1.1.16)

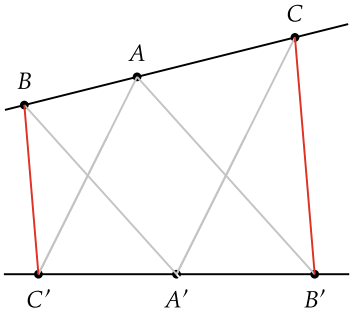


Fig. 1.4 Pappus' Theorem, affine version (Theorem 1.1.17)

Theorem 1.1.17 (Pappus' Theorem, Affine Version) *Let A, B, C be three distinct points on one line, and let A', B', C' be three distinct points on another line. Suppose that none of these six points lies on both lines, $AB' \parallel A'B$, and $AC' \parallel A'C$. Then $BC' \parallel B'C$ (Fig. 1.4).*

Exercises

1. Prove the statements in Remark 1.1.2. Prove also that there exist four distinct points in $\mathbb{A}^2(F)$, no three of which are collinear.
2. Prove Corollary 1.1.8.
3. Let F be a finite field. Calculate the number of points in $\mathbb{A}^2(F)$. Prove that no affine plane $\mathbb{A}^2(F)$ can have 36 points. Calculate the number of lines in $\mathbb{A}^2(F)$. Determine the parallelism classes in $\mathbb{A}^2(F)$, and their number.
4. Prove that the lines in two parallelism classes in $\mathbb{A}^2(F)$ are in one-to-one correspondence with each other.
5. Let $R = \mathbb{Z}/6\mathbb{Z}$, and let

$$\mathbb{A}^2(R) = R \times R = \{(x, y) \mid x, y \in R\}$$

be the set of ordered pairs of elements of R. Define a *point* to be an element of $\mathbb{A}^2(R)$ and a *line* to be a subset of $\mathbb{A}^2(R)$ defined by an equation of the form $ax + by + c = 0$ with $a, b, c \in R$ and not both of a and b zero. Prove that, given the points $(1, 2), (3, 4) \in \mathbb{A}^2(R)$, there exists more than one line passing through both.

6. Prove that a change of coordinate systems in $\mathbb{A}^2(F)$ maps lines to lines.
7. Prove that Definition 1.1.15 is equivalent to the definition where a triangle is defined as a collection of three nonconcurrent lines and the three points of intersection of pairs of these lines.
8. Prove Small Desargues' Theorem in the affine plane: Let there be three distinct parallel lines ℓ, m, n in $\mathbb{A}^2(F)$, and let there be given six distinct points $A, A' \in \ell$, $B, B' \in m$, $C, C' \in n$. Suppose further that $AB \parallel A'B'$ and $AC \parallel A'C'$. Then $BC \parallel B'C'$.
9. Prove Theorem 1.1.17 (Pappus' Theorem, affine version).
10. Let $P = (x_1, y_1)$, $Q = (x_2, y_2)$, and $R = (x_3, y_3)$ be three points in $\mathbb{A}^2(F)$. Prove that P, Q, and R are collinear if and only if

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$$

Use this to prove that the equation of the line passing through P and Q can be written in the form

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0.$$

11. Find out whether the given triples of points in $\mathbb{A}^2(F)$ are collinear. (Assume that $\mathrm{char}(F) = 0$.)

- $(3, 7), (-1, -1), (0, 1)$.
- $(2, -3), (-2, 4), (-2, 7)$.
- $(\frac{7}{4}, \frac{9}{16}), (-2, -\frac{9}{4}), (1, 0)$.

1.2 Projective Planes over a Field

We start with two equivalent definitions of the projective plane. We will see later on in the section that the affine plane can be considered a subset of the projective plane in a natural (but not canonical) manner.

Definition 1.2.1 The *projective plane* $\mathbb{P}^2(F)$ is the set of 1-dimensional subspaces of the vector space F^3.

Another, equivalent, definition can be given using the following equivalence relation:

Definition 1.2.2 Given two points $P = (x_1, y_1, z_1)$, $Q = (x_2, y_2, z_2) \in F^3 \setminus \{0\}$, we say that P *is equivalent to* Q, denoted by $P \sim Q$, if there exists $\lambda \neq 0$ such that $x_2 = \lambda x_1$, $y_2 = \lambda y_1$, $z_2 = \lambda z_1$.

Proposition 1.2.3 *$P \sim Q$ as defined in Definition 1.2.2 is an equivalence relation.*

Definition 1.2.4 The *projective plane* $\mathbb{P}^2(F)$ *over* F is defined as the set of equivalence classes of points in $F^3 \setminus \{0\}$ under the equivalence relation given in Definition 1.2.2. Elements of $\mathbb{P}^2(F)$ are called *points*. A point P in $\mathbb{P}^2(F)$ that is the equivalence class of $(x_0, y_0, z_0) \in F^3 \setminus \{0\}$ will be denoted by $[x_0 : y_0 : z_0]$. The elements $x_0, y_0, z_0 \in F$ are called the *projective coordinates* of P.

Remark 1.2.5 Definitions 1.2.1 and 1.2.4 can be stated more generally for any 3-dimensional vector space V. In that case, we define the projective plane $\mathbb{P}(V)$ as the set of 1-dimensional subspaces of V, or equivalence classes of nonzero vectors in V under the equivalence relation given by $\mathbf{v} \sim \mathbf{w}$ if $\mathbf{w} = \lambda\mathbf{v}$ for some $\lambda \neq 0$.[2] To define projective coordinates on $\mathbb{P}(V)$, we choose a basis for V and express the projective coordinates with respect to that basis. More concretely, let a basis of V be given by the vectors $\mathbf{e_1}, \mathbf{e_2}, \mathbf{e_3} \in V$. Then any nonzero vector $\mathbf{v} \in V$ can be written uniquely as $\mathbf{v} = x_0\mathbf{e_1} + y_0\mathbf{e_2} + z_0\mathbf{e_3}$ for $x_0, y_0, z_0 \in F$, not all zero, and we can use these coefficients as projective coordinates of the point $[\mathbf{v}] = [x_0 : y_0 : z_0] \in \mathbb{P}(V)$.

Of course, $[x_0 : y_0 : z_0] = [\lambda x_0 : \lambda y_0 : \lambda z_0]$ for any nonzero λ. The fact that the same point can be represented in different ways using projective coordinates has an important consequence: when defining subsets of $\mathbb{P}^2(F)$ by the vanishing of polynomials, we must use *homogeneous* polynomials.

Definition 1.2.6 A polynomial $f(x, y, z) \in F[x, y, z]$ is called *homogeneous of degree* d if every nonzero term in f has degree d.

Let $P = [x_0 : y_0 : z_0] \in \mathbb{P}^2(F)$. Given a homogeneous polynomial $f(x, y, z) \in F[x, y, z]$ of degree d, it is easy to check that $f(\lambda x, \lambda y, \lambda z) = \lambda^d f(x, y, z)$. Hence, the statement that "f vanishes at P," written $f(P) = 0$, is well-defined. For this reason, we use homogeneous polynomials in defining subsets of the projective plane.

[2] There is no need to indicate dimension in this notation since the dimension of V, and hence the dimension of $\mathbb{P}(V)$, is known. The base field F is implicit.

Lines are defined by homogeneous polynomials of degree 1.

Definition 1.2.7 A *line* in $\mathbb{P}^2(F)$ is a subset defined by an equation of the form $ax + by + cz = 0$, where a, b, c are not all zero.

Remark 1.2.8 A line ℓ with equation $ax + by + cz = 0$ will be simply denoted by $\ell : ax + by + cz = 0$.

Let $\ell : ax + by + cz = 0$ be a line in $\mathbb{P}^2(F)$. We note that the union of all the 1-dimensional subspaces of F^3 corresponding to the points in ℓ under Definition 1.2.1 forms a 2-dimensional subspace in F^3 and that this subspace is also defined by the equation $ax + by + cz = 0$.

The ability to use the description of points and lines in the projective plane as vector subspaces of F^3 of dimensions 1 and 2, respectively, can be very useful. The proof of the next proposition, which is left to the reader, is a case in point.

Proposition 1.2.9 *Given two distinct points in $\mathbb{P}^2(F)$, there exists a unique line that passes through both.*

Definition 1.2.10 Let $P, Q \in \mathbb{P}^2(F)$ be two distinct points. The unique line passing through P and Q, whose existence is guaranteed by Proposition 1.2.9, is called the *join* of P and Q and it is denoted by PQ.

Again, there is more than one way of writing a given line as the join of two points.

Proposition 1.2.11 *Two distinct lines in $\mathbb{P}^2(F)$ intersect in a unique point.*

Proof Since the two given lines correspond to two distinct 2-dimensional subspaces of F^3 by Remark 1.2.8, the result follows from the fact that these subspaces have a 1-dimensional intersection. □

Hence, there exists no concept of "parallelism" in $\mathbb{P}^2(F)$, which is the most important property that differentiates it from $\mathbb{A}^2(F)$.

The following criterion for determining when three points in $\mathbb{P}^2(F)$ are collinear is extremely useful:

Proposition 1.2.12 *Three points $P_i = [x_i : y_i : z_i] \in \mathbb{P}^2(F)$, $i = 1, 2, 3$, are collinear if and only if*

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = 0.$$

Proof The projective coordinates of the points P_1, P_2, P_3 give three vectors $\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3} \in F^3$. The statement that P_1, P_2, P_3 are collinear is equivalent to the statement that $\mathbf{v_1}$, $\mathbf{v_2}$, $\mathbf{v_3}$ are contained in a 2-dimensional subspace of F^3; in other words, that they are linearly dependent. The result now follows by linear algebra. □

The affine plane $\mathbb{A}^2(F)$ can be *embedded* in the projective plane $\mathbb{P}^2(F)$ as a subset.

Proposition 1.2.13 *The function that maps a point* $(x_0, y_0) \in \mathbb{A}^2(F)$ *to the point* $[x_0 : y_0 : 1] \in \mathbb{P}^2(F)$ *establishes a one-to-one correspondence between the affine points and the projective points* $[x_0 : y_0 : z_0]$ *with* $z_0 \neq 0$.

Henceforth, we shall consider $\mathbb{A}^2(F)$ as a subset of $\mathbb{P}^2(F)$ through this correspondence. We note that there are other ways to consider $\mathbb{A}^2(F)$ as a subset of $\mathbb{P}^2(F)$, using the coordinates x or y, or indeed using coordinate transformations to define $\mathbb{A}^2(F)$ as the set of points where a homogeneous linear polynomial does not vanish. We will not use these in this book.

Let now $\ell : ax + by + c = 0$ be an affine line. Since the equation defining it is not homogeneous, we cannot use it directly to define a projective line, but we can *homogenize* it with respect to a third variable z. *Homogenization* is the process of multiplying the constant c by z to obtain the homogeneous polynomial $ax+by+cz$, which can then be used to define the projective line $m : ax + by + cz = 0$. We call m the *projectivization* of ℓ.

Intersecting m with $\mathbb{A}^2(F)$ produces an affine line whose equation is $ax + by + c = 0$; in other words, $m \cap \mathbb{A}^2(F) = \ell$, the line we started with. Regarding the intersection of m with the complement of $\mathbb{A}^2(F)$ in $\mathbb{P}^2(F)$, let us note that this complement is simply the line $\ell_\infty : z = 0$, called the *line at infinity*. Then $m \cap \ell_\infty = \{[b : -a : 0]\}$ (recall that a and b cannot both be 0). And it is clear that every point on ℓ_∞ can be written in this form; therefore, every point on ℓ_∞ can be obtained by the procedure above starting from an affine line.

Definition 1.2.14 An *ideal point* is an equivalence class of parallel lines in $\mathbb{A}^2(F)$.

We see that there is a one-to-one correspondence between ideal points and points on ℓ_∞, since an ideal point is determined by a and b only by Corollary 1.1.8.

Let us now consider two examples.

Example 1.2.15 Consider the three lines $\ell_1 : x + 2y - 5 = 0$, $\ell_2 : x + 2y + 1 = 0$, $\ell_3 : 2x - y - 3 = 0$ in $\mathbb{A}^2(\mathbb{R})$. The projectivizations of ℓ_1 and ℓ_2 are $m_1 : x + 2y - 5z = 0$ and $m_2 : x + 2y + z = 0$, respectively. Note that, as $\ell_1 \parallel \ell_2$ in $\mathbb{A}^2(\mathbb{R})$, the ideal points of ℓ_1 and ℓ_2 are the same: $m_1 \cdot \ell_\infty = m_2 \cdot \ell_\infty = [2 : -1 : 0]$. The projectivization of ℓ_3 is $m_3 : 2x - y - 3z = 0$. Since ℓ_3 is not parallel to ℓ_1 or ℓ_2, its ideal point is different: $m_3 \cdot \ell_\infty = [1 : 2 : 0]$. This example shows that distinct parallel affine lines should be thought of as intersecting in their common

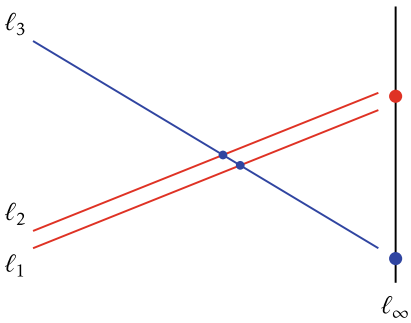


Fig. 1.5 Three lines in $\mathbb{A}^2(\mathbb{R})$ and their ideal points. The region to the left of ℓ_∞ should be thought of as $\mathbb{A}^2(\mathbb{R})$. The red dot is the ideal point $[2 : -1 : 0]$ of ℓ_1 and ℓ_2; the blue dot is the ideal point $[1 : 2 : 0]$ of ℓ_3

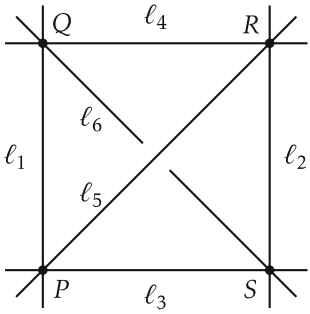


Fig. 1.6 $\mathbb{A}^2(\mathbb{F}_2)$, an affine plane with four points. ℓ_1 and ℓ_2 should be thought to intersect "at infinity" at the ideal point A; ℓ_3 and ℓ_4 at the ideal point B; ℓ_5 and ℓ_6 at the ideal point C

ideal point after projectivization. If two distinct affine lines intersect already in the affine plane, then they are not parallel, and their ideal points are different. In both cases, we see that two affine lines, when projectivized, intersect at exactly one point, already implied by Proposition 1.2.11. See Fig. 1.5.

Example 1.2.16 There are four points in $\mathbb{A}^2(\mathbb{F}_2)$: $P = (0, 0)$, $Q = (0, 1)$, $R = (1, 1)$, $S = (1, 0)$. The six pairs of points chosen from these four determine six lines: $\ell_1 : x = 0$, $\ell_2 : x = 1$, $\ell_3 : y = 0$, $\ell_4 : y = 1$, $\ell_5 : y = x$, $\ell_6 : y = 1 - x = x + 1$ (recall that $\mathrm{char}(\mathbb{F}_2) = 2$); these are the only lines in $\mathbb{A}^2(\mathbb{F}_2)$. The nontrivial relations of parallelism between the lines are $\ell_1 \parallel \ell_2$, $\ell_3 \parallel \ell_4$, $\ell_5 \parallel \ell_6$, giving three ideal points. See Fig. 1.6.

In projective coordinates, we have $P = [0 : 0 : 1]$, $Q = [0 : 1 : 1]$, $R = [1 : 1 : 1]$, $S = [1 : 0 : 1]$ and the ideal points determined by ℓ_1, ℓ_3, ℓ_5 are $A = [0 : 1 : 0]$, $B = [1 : 0 : 0]$, $C = [1 : 1 : 0]$, respectively. Therefore, there are a total of seven points in $\mathbb{P}^2(\mathbb{F}_2)$. $\mathbb{P}^2(\mathbb{F}_2)$, the so-called *Fano plane*, is conventionally represented as in Fig. 1.7.

In the affine plane, the geometric meaning of the statement "given two distinct points, there exists a unique line that passes through both" is clear. What is the corresponding meaning in the case of the projective plane? First, if the two distinct points $P, Q \in \mathbb{P}^2(F)$ both lie in $\mathbb{A}^2(F)$, then the projective line through P

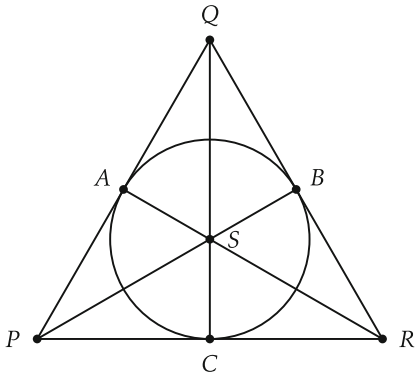


Fig. 1.7 The Fano plane $\mathbb{P}^2(\mathbb{F}_2)$

and Q consists of the affine line through P and Q together with the ideal point corresponding to that line. If one of them, say Q, is an ideal point, then the projective line through P and Q consists of the affine line through P that has Q as its ideal point, plus Q itself; note that since Q determines the parallelism class of this line, it must be unique. Finally, if both P and Q are ideal points, then the projective line through P and Q is simply ℓ_∞, the line at infinity.

Again, we summarize some properties of $\mathbb{P}^2(F)$ in the next theorem. They will be taken as axioms for the definition of a projective space in Chap. 2.

Theorem 1.2.17 $\mathbb{P}^2(F)$ *satisfies the following properties:*

P1. *Given two distinct points, there exists a unique line that passes through both.*
P2. *Two distinct lines intersect in a unique point.*
P3. *There exist three noncollinear points.*
P4. *There exist at least three points on any line.*

Proof **P1** and **P2** were proved in Propositions 1.2.9 and 1.2.11, respectively. For **P3**, it suffices to note that the points $[1:0:0]$, $[0:1:0]$, $[0:0:1] \in \mathbb{P}^2(F)$ are noncollinear by Proposition 1.2.12.

For **P4**, we recall that a given line $\ell \subset \mathbb{P}^2(F)$ corresponds to a 2-dimensional subspace V of F^3 by Remark 1.2.8. Finding three distinct points on ℓ is then equivalent to finding three nonzero vectors in V such that any two of them are linearly independent. Now, if $\mathbf{v_1}$ and $\mathbf{v_2}$ are two basis vectors for V, it can be seen that any two of $\mathbf{v_1}$, $\mathbf{v_2}$, $\mathbf{v_1} + \mathbf{v_2}$ are linearly independent. □

Remark 1.2.18 Similarly as in the affine case, **P3** implies that $\mathbb{P}^2(F)$ is not equal to any line in it.

Remark 1.2.19 When two affine or projective lines ℓ_1 and ℓ_2 intersect in a unique point P, we will write $\ell_1 \cdot \ell_2 = P$.

We note that **P1** and **P3** have their affine counterparts in **A1** and **A3**, respectively. **P4** has its affine counterpart in the fact that there exist at least two points on any affine line by Remark 1.1.2. However, **P2** has no affine counterpart since it excludes the existence of parallel lines.

Exercises

1. Prove Propositions 1.2.3, 1.2.9, 1.2.13.
2. Prove that Definitions 1.2.1 and 1.2.4 are equivalent.
3. Do there exist fields F such that a nonhomogeneous polynomial vanishes at all points of $\mathbb{P}^2(F)$, for all possible representations of these points in projective coordinates?
4. Prove that $\mathbb{A}^2(F)$ can be embedded in $\mathbb{P}^2(F)$ by the maps $(x, y) \mapsto [x : 1 : y]$ and $(x, y) \mapsto [1 : x : y]$. Work out the details of the discussion after Proposition 1.2.13 for these two cases.
5. Given a line $\ell \subset \mathbb{P}^2(F)$, prove that if $\ell \cap \mathbb{A}^2(F) \neq \emptyset$, then $\ell \cap \mathbb{A}^2(F)$ is an affine line, and if $\ell \cap \mathbb{A}^2(F) = \emptyset$, then $\ell = \ell_\infty$.
6. Calculate the number of points and lines in $\mathbb{P}^2(F)$ for a finite field F.
7. Prove that the points on two given lines in $\mathbb{P}^2(F)$ are in one-to-one correspondence with each other. Calculate the number of points on a line in $\mathbb{P}^2(F)$ for a finite field F.
8. Work out the lines in $\mathbb{A}^2(\mathbb{F}_3)$ and their parallelism classes in the manner of Example 1.2.16.
9. Let $P = [x_1 : y_1 : z_1]$ and $Q = [x_2 : y_2 : z_2]$ be two distinct points in $\mathbb{P}^2(F)$. Prove that the equation of the line passing through P and Q can be written in the form

$$\begin{vmatrix} x & y & z \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix} = 0.$$

10. Find the equation of the projective line that passes through $[1 : -2 : 6]$ and $[-8 : 1 : 1]$. (In this exercise and the next, assume that $\mathrm{char}(F) = 0$.)
11. Find out whether the given triples of points in $\mathbb{P}^2(F)$ are collinear.

- $[1 : 0 : 0]$, $[0 : 1 : 0]$, $[0 : 0 : 1]$.
- $[1 : 2 : 1]$, $[-2 : -3 : -1]$, $[9 : 12 : 3]$.
- $[2 : 1 : 0]$, $[-2 : -2 : -2]$, $[-3 : -1 : 1]$.

1.3 Duality Between Points and Lines

Let us recall the first two properties of $\mathbb{P}^2(F)$ that we stated in Theorem 1.2.17. We slightly change the phrasing of **P2** to emphasize the resemblance between the two statements.

P1. Given two distinct points, there exists a unique line that passes through both.
P2. Given two distinct lines, there exists a unique point that lies on both.

Stated in this form, **P1** and **P2** illustrate the principle of *duality between points and lines* (henceforth simply "duality") on $\mathbb{P}^2(F)$. According to this principle, if one takes a definition or proposition that involves nothing but configurations of points, lines, and incidence relations between these points and lines, and if certain words of this definition or proposition are changed according to the dictionary

$$\begin{aligned} \text{point} &\longleftrightarrow \quad \text{line}, \\ \text{lies on} &\longleftrightarrow \text{passes through}, \end{aligned}$$

we obtain another definition or proposition. For example, it is clear that the definitions of *join of two distinct points* and *intersection of two distinct lines*, namely "the line that passes through two distinct points" and "the point that lies on two distinct lines" are dual to each other. Another pair of definitions dual to each other is given by *collinear* and *concurrent*, namely "points that lie on the same line" and "lines that pass through the same point." To come back to the examples given at the beginning of this section, **P1** and **P2** are dual to each other. If the dual of a definition or a proposition is itself, we will call that definition or proposition *self-dual*. We will see examples of self-dual definitions and propositions later. Since much of projective geometry is concerned with definitions or propositions that involve nothing but configurations of points and lines, and incidence relations between these points and lines, the principle of duality has a very wide range of application.

Why is the principle of duality valid? Recall that, by Definition 1.2.7, a line is a subset of $\mathbb{P}^2(F)$ defined by an equation of the form $ax + by + cz = 0$, where a, b, c are not all zero. Two lines

$$\begin{aligned} \ell_1 &: a_1x + b_1y + c_1z = 0, \\ \ell_2 &: a_2x + b_2y + c_2z = 0, \end{aligned}$$

are identical if and only if there is a nonzero λ such that $\lambda a_1 = a_2$, $\lambda b_1 = b_2$, $\lambda c_1 = c_2$. This implies that a line $a_0x+b_0y+c_0z = 0$ in $\mathbb{P}^2(F)$ is uniquely identified by the point $[a_0 : b_0 : c_0] \in \mathbb{P}^2(F)$; in other words, there is a one-to-one correspondence between the lines in $\mathbb{P}^2(F)$ and the points in $\mathbb{P}^2(F)$. Let us now consider another copy of $\mathbb{P}^2(F)$, denoted by $(\mathbb{P}^2(F))^*$ and whose projective coordinates are a, b, c. With this notation, the lines in $\mathbb{P}^2(F)$ are in one-to-one correspondence with the points in $(\mathbb{P}^2(F))^*$. Similarly, given a point $[x_0 : y_0 : z_0] \in \mathbb{P}^2(F)$, we can

construct the line $x_0a + y_0b + z_0c = 0$ in $(\mathbb{P}^2(F))^*$. Once again, this is a one-to-one correspondence, this time between the points in $\mathbb{P}^2(F)$ and the lines in $(\mathbb{P}^2(F))^*$. Given a line $\ell \subset \mathbb{P}^2(F)$, the corresponding point in $(\mathbb{P}^2(F))^*$ is denoted by P_ℓ; given a point $P \in \mathbb{P}^2(F)$, the corresponding line in $(\mathbb{P}^2(F))^*$ is denoted by ℓ_P.

Suppose now that $P = [x_0 : y_0 : z_0] \in \mathbb{P}^2(F)$ lies on $\ell : a_0x + b_0y + c_0z = 0$. Then we have $a_0x_0 + b_0y_0 + c_0z_0 = 0$. But this is equivalent to the statement that $P_\ell \in (\mathbb{P}^2(F))^*$ lies on $\ell_P \subset (\mathbb{P}^2(F))^*$, which is given by the equation $x_0a_0 + y_0b_0 + z_0c_0 = 0$. It is therefore clear that duality preserves incidence relations between points and lines in $\mathbb{P}^2(F)$.

The dictionary given above can be extended using pairs of dual definitions. Some examples are as follows:

point	$\longleftrightarrow$	line
lies on	$\longleftrightarrow$	passes through
collinear	$\longleftrightarrow$	concurrent
join	$\longleftrightarrow$	intersection

From now on, the reader should get into the habit of dualizing the definitions and propositions encountered throughout the book.

Exercise

1. Prove that two lines $\ell_i : a_ix + b_iy + c_iz = 0$, $i = 1, 2$, are identical if and only if there is a nonzero λ such that $\lambda a_1 = a_2$, $\lambda b_1 = b_2$, and $\lambda c_1 = c_2$.

1.4 Automorphisms

Many mathematical theories include the concept of *isomorphism* between the objects studied in that theory. Usually, one considers a bijective function from one object to another that preserves some additional structure on the objects. For example, a group is defined as a nonempty set with a binary operation that satisfies some properties. An isomorphism from one group G to another group H is a bijective function $f : G \to H$ that preserves the binary operation. An *automorphism* is an isomorphism from an object to itself. Since we work in the projective plane $\mathbb{P}^2(F)$ and the only structure on $\mathbb{P}^2(F)$ is the set of lines, it is obvious that we must require an automorphism of $\mathbb{P}^2(F)$ to map lines to lines. Parts of the exposition in this section follow [6, Chapter 3], although only the case of the base field $\mathbb{R}$ is treated there.

Definition 1.4.1 An *automorphism* of $\mathbb{P}^2(F)$ is a bijective map $\phi : \mathbb{P}^2(F) \to \mathbb{P}^2(F)$ that maps lines to lines. The set of automorphisms of $\mathbb{P}^2(F)$ is denoted by $\mathrm{Aut}(\mathbb{P}^2(F))$.

As happens with many mathematical theories, the set of automorphisms of $\mathbb{P}^2(F)$ forms a group.

Proposition 1.4.2 *$Aut(\mathbb{P}^2(F))$ is a group under composition.*

Proof It is clear that $\text{Aut}(\mathbb{P}^2(F)) \neq \emptyset$ since the identity on $\mathbb{P}^2(F)$ is an automorphism of $\mathbb{P}^2(F)$. We must show, therefore, that the composition of two automorphisms of $\mathbb{P}^2(F)$ and the inverse of an automorphism of $\mathbb{P}^2(F)$ are automorphisms of $\mathbb{P}^2(F)$.

Let $\phi, \psi \in \text{Aut}(\mathbb{P}^2(F))$. Since ϕ and ψ are bijective, $\phi\psi$ is also bijective. And since ϕ and ψ map lines to lines, $\phi\psi$ also maps lines to lines. Hence, $\phi\psi \in \text{Aut}(\mathbb{P}^2(F))$.

Finally, we must show that $\phi^{-1} \in \text{Aut}(\mathbb{P}^2(F))$. It is clear that ϕ^{-1} is bijective. Let $\ell \subset \mathbb{P}^2(F)$ be a line; we wish to show that $\phi^{-1}(\ell)$ is a line. Since ϕ establishes a bijection between ℓ and $\phi^{-1}(\ell)$, we can choose two distinct points $P, Q \in \phi^{-1}(\ell)$. Let $m \subset \mathbb{P}^2(F)$ be the unique line through P and Q. Then $\phi(m)$ is a line, since $\phi \in \text{Aut}(\mathbb{P}^2(F))$, and it contains two distinct points, namely $\phi(P)$ and $\phi(Q)$, that lie on ℓ. Therefore, $\phi(m) = \ell$. Hence, $\phi^{-1}(\ell) = \phi^{-1}(\phi(m)) = m$, a line. □

We do not yet know whether $\mathbb{P}^2(F)$ has nontrivial automorphisms. Since the points of $\mathbb{P}^2(F)$ can be thought of as 1-dimensional subspaces of F^3, natural candidates for automorphisms of $\mathbb{P}^2(F)$ are given by vector space automorphisms of F^3. Equivalently, once we have fixed a basis for F^3, we can consider 3×3 invertible matrices over F. The set of such matrices is denoted by $\text{GL}_3(F)$. It is in fact a group under matrix multiplication, called the *general linear group of degree 3*. This construction can also be generalized to invertible $n \times n$ matrices over F.

Proposition 1.4.3 *$GL_n(F)$, defined as the set of $n \times n$ matrices over F with nonzero determinant, is a group under matrix multiplication.*

A given matrix $A \in \text{GL}_3(F)$ can be used to define a map $\phi_A : \mathbb{P}^2(F) \to \mathbb{P}^2(F)$. Let $P = [x : y : z] \in \mathbb{P}^2(F)$. Using the projective coordinates of P, we define a vector

$$\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in F^3.$$

The matrix product $A\mathbf{v}$ is then another vector in F^3, which is nonzero since $\mathbf{v} \neq \mathbf{0}$ and A is invertible. Hence, we get a point $[A\mathbf{v}] \in \mathbb{P}^2(F)$. We define $\phi_A(P) = [A\mathbf{v}]$. It is clear that ϕ_A is well-defined.

Proposition 1.4.4 *Given $A \in GL_3(F)$, ϕ_A is an automorphism of $\mathbb{P}^2(F)$. The map $\Phi : GL_3(F) \to Aut(\mathbb{P}^2(F))$, defined as $A \mapsto \phi_A$ for all $A \in GL_3(F)$, is a group homomorphism.*

Proof From the definition of ϕ_A above, it follows that $\phi_I \in \mathrm{Aut}(\mathbb{P}^2(F))$ is the identity, and given $A, B \in \mathrm{GL}_3(F)$, $\phi_{AB} = \phi_A\phi_B$. Using these facts, it is easy to prove that $\phi_{A^{-1}}$ is the inverse of ϕ_A. Hence, ϕ_A is bijective for any $A \in \mathrm{GL}_3(F)$. It is also clear that Φ is a group homomorphism.

The last requirement is to prove that ϕ_A maps lines to lines. Let $\ell : ax+by+cz = 0$ be a line in $\mathbb{P}^2(F)$. We can rewrite this equation in the form $\mathbf{n}^t\mathbf{v} = 0$, where

$$\mathbf{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in F^3.$$

If now we let

$$\mathbf{n}' = (A^{-1})^t\mathbf{n} = \begin{bmatrix} a' \\ b' \\ c' \end{bmatrix} \in F^3,$$

it is clear that $\mathbf{n}' \neq \mathbf{0}$ and the image $\phi_A(\ell)$ is the line $\ell' : a'x + b'y + c'z = 0$. □

We now potentially have a wealth of automorphisms of $\mathbb{P}^2(F)$. However, we note that different matrices in $\mathrm{GL}_3(F)$ can give rise to the same automorphism of $\mathbb{P}^2(F)$. In other words, Φ is not injective. A very simple example is given by matrices that are scalar multiples of each other: given $A \in \mathrm{GL}_3(F)$ and a nonzero $\lambda \in F$, it is clear that $\phi_A = \phi_{\lambda A}$. Fortunately, it turns out that this is the only way different matrices in $\mathrm{GL}_3(F)$ can give rise to the same automorphism of $\mathbb{P}^2(F)$.

Until the end of the section, we define the four points

$$P_1 = [1 : 0 : 0], P_2 = [0 : 1 : 0], P_3 = [0 : 0 : 1], Q = [1 : 1 : 1] \in \mathbb{P}^2(F).$$

By Proposition 1.2.12, no three of these four points are collinear.

Proposition 1.4.5 *Let $A \in GL_3(F)$. Then ϕ_A is the identity if and only if $A = \lambda I$ for some nonzero λ. In other words, the kernel of the group homomorphism Φ is $ker(\Phi) = F^*$, where F^* is considered as a subgroup of $GL_3(F)$ via the scalar matrices. Hence, for $A, B \in GL_3(F)$, $\phi_A = \phi_B$ if and only if $A = \lambda B$ for some nonzero $\lambda \in F$.*

Proof It suffices to prove the first statement only. The "if" direction is clear.

For the "only if" direction, suppose that ϕ_A is the identity for some $A \in \mathrm{GL}_3(F)$. The four points $P_1, P_2, P_3, Q \in \mathbb{P}^2(F)$ correspond to the vectors

$$\mathbf{v_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{v_2} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v_3} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \mathbf{v_4} = \mathbf{v_1} + \mathbf{v_2} + \mathbf{v_3} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \in F^3.$$

Since each of the points P_1, P_2, P_3, Q is mapped to itself by ϕ_A, we have

$$\begin{aligned} A\mathbf{v_1} &= \lambda_1 \mathbf{v_1}, \\ A\mathbf{v_2} &= \lambda_2 \mathbf{v_2}, \\ A\mathbf{v_3} &= \lambda_3 \mathbf{v_3}, \\ A\mathbf{v_4} &= \lambda_4 \mathbf{v_4} = \lambda_1 \mathbf{v_1} + \lambda_2 \mathbf{v_2} + \lambda_3 \mathbf{v_3} = \lambda_4(\mathbf{v_1} + \mathbf{v_2} + \mathbf{v_3}), \end{aligned}$$

for nonzero $\lambda_1, \lambda_2, \lambda_3, \lambda_4$. The last equation implies $\lambda_1 = \lambda_2 = \lambda_3$ and hence $A = \lambda_1 I$. □

We define the *projective general linear group of degree 3* as $\mathrm{PGL}_3(F) = \mathrm{GL}_3(F)/F^*$ and, by a slight abuse of notation, we will represent elements of $\mathrm{PGL}_3(F)$ as matrices. By Proposition 1.4.5, Φ induces an injective group homomorphism,

$$\Phi : \mathrm{PGL}_3(F) \to \mathrm{Aut}(\mathbb{P}^2(F)),$$

which is also denoted by Φ since there is usually no danger of confusion. We will consider $\mathrm{PGL}_3(F)$ to be a subgroup of $\mathrm{Aut}(\mathbb{P}^2(F))$ via Φ. Similarly, for $A \in \mathrm{PGL}_3(F)$, $\Phi(A)$ is also denoted by ϕ_A. The following proposition shows that an automorphism of $\mathbb{P}^2(F)$ of the form ϕ_T for $T \in \mathrm{PGL}_3(F)$ is completely determined by its action on four points, no three of which are collinear.

Proposition 1.4.6 *Given $A, B, C, D \in \mathbb{P}^2(F)$ and $A', B', C', D' \in \mathbb{P}^2(F)$, such that no three points in either collection of four points are collinear, there exists a unique $T \in PGL_3(F)$ such that $\phi_T(A) = A'$, $\phi_T(B) = B'$, $\phi_T(C) = C'$, $\phi_T(D) = D'$. In particular, there exists a unique $T \in PGL_3(F)$ such that $\phi_T(P_1) = A$, $\phi_T(P_2) = B$, $\phi_T(P_3) = C$, $\phi_T(Q) = D$. Any automorphism of $\mathbb{P}^2(F)$ of the form ϕ_T for $T \in PGL_3(F)$ that fixes A, B, C, D is the identity.*

Proof Again, it suffices to prove the first statement. Let $\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3}, \mathbf{v_4} \in F^3$ be nonzero vectors giving the points A, B, C, D, respectively, and $\mathbf{w_1}, \mathbf{w_2}, \mathbf{w_3}, \mathbf{w_4} \in F^3$ nonzero vectors giving the points A', B', C', D', respectively. Since no three of A, B, C, D are collinear, no three of $\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3}, \mathbf{v_4}$ are linearly dependent; similarly, since no three of A', B', C', D' are collinear, no three of $\mathbf{w_1}, \mathbf{w_2}, \mathbf{w_3}, \mathbf{w_4}$ are linearly dependent. Considering $\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3}$ as an ordered basis for F^3, we write

$$\mathbf{v_4} = \lambda_1 \mathbf{v_1} + \lambda_2 \mathbf{v_2} + \lambda_3 \mathbf{v_3},$$

where $\lambda_1, \lambda_2, \lambda_3 \neq 0$, since otherwise three of $\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3}, \mathbf{v_4}$ would be linearly dependent. By replacing $\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3}$ with nonzero scalar multiples of themselves if necessary, we can assume, without loss of generality, that $\mathbf{v_4} = \mathbf{v_1} + \mathbf{v_2} + \mathbf{v_3}$. Similarly, we can assume, without loss of generality, that $\mathbf{w_4} = \mathbf{w_1} + \mathbf{w_2} + \mathbf{w_3}$. Now, the element $T \in \mathrm{GL}_3(F)$ defined on the basis elements as $T(\mathbf{v_i}) = \mathbf{w_i}$ for $i = 1, 2, 3$ satisfies $T(\mathbf{v_4}) = \mathbf{w_4}$. Hence $\phi_T(A) = A'$, $\phi_T(B) = B'$, $\phi_T(C) = C'$, $\phi_T(D) = D'$.

To prove uniqueness, suppose that there are two elements $T, T' \in \mathrm{GL}_3(F)$ such that both ϕ_T and $\phi_{T'}$ have the required property. Then if we define $\psi = \phi_{T'}^{-1}\phi_T = \phi_{(T')^{-1}T}$, ψ fixes A, B, C, and D. We again assume that these four points are given by nonzero vectors $\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3}, \mathbf{v_4} \in F^3$ such that $\mathbf{v_4} = \mathbf{v_1} + \mathbf{v_2} + \mathbf{v_3}$. Now an argument similar to that used in the proof of Proposition 1.4.5 shows that $(T')^{-1}T$ is a scalar multiple of identity; in other words, T and T' are scalar multiples of each other. Hence $T = T'$ in $\mathrm{PGL}_3(F)$. □

We now have a nontrivial subgroup of $\mathrm{Aut}(\mathbb{P}^2(F))$, namely $\mathrm{PGL}_3(F)$.

Remark 1.4.7 It can be shown that if F is a finite field with N elements, the number of elements in $\mathrm{GL}_3(F)$ is

$$(N^3 - 1)(N^3 - N)(N^3 - N^2),$$

and hence the number of elements in $\mathrm{PGL}_3(F)$ is

$$\frac{(N^3 - 1)(N^3 - N)(N^3 - N^2)}{(N - 1)} = N^2(N^3 - 1)(N^3 - N).$$

Of course, if F is infinite, then $\mathrm{PGL}_3(F)$ is also infinite. Hence, there are many nontrivial automorphisms in $\mathrm{Aut}(\mathbb{P}^2(F))$.

The question now arises: is $\mathrm{Aut}(\mathbb{P}^2(F))$ actually equal to $\mathrm{PGL}_3(F)$? The answer is generally "no," and to understand why, we must investigate the automorphisms of $\mathbb{P}^2(F)$ induced by the automorphisms of the base field F. Recall that a *field automorphism* of F is a bijective map $\sigma : F \to F$ such that $\sigma(x+y) = \sigma(x)+\sigma(y)$ and $\sigma(xy) = \sigma(x)\sigma(y)$ for all $x, y \in F$. This implies, in particular, that $\sigma(0) = 0$, $\sigma(1) = 1$, $\sigma(-x) = -\sigma(x)$ for all $x \in F$, and $\sigma(x^{-1}) = \sigma(x)^{-1}$ for all nonzero $x \in F$.

Proposition 1.4.8 *The set $\mathrm{Aut}(F)$ of field automorphisms of F is a group under composition.*

A given field automorphism $\sigma \in \mathrm{Aut}(F)$ can be used to define a map $\chi_\sigma : \mathbb{P}^2(F) \to \mathbb{P}^2(F)$. Let $P = [x : y : z] \in \mathbb{P}^2(F)$. We define $\chi_\sigma(P) = [\sigma(x) : \sigma(y) : \sigma(z)]$. It is clear that χ_σ is well-defined.

Proposition 1.4.9 *Given a field automorphism $\sigma \in Aut(F)$, χ_σ is an automorphism of $\mathbb{P}^2(F)$. The map $X : Aut(F) \to Aut(\mathbb{P}^2(F))$ defined as $\sigma \mapsto \chi_\sigma$ for all $\sigma \in Aut(F)$ is an injective group homomorphism.*

Proof The proof is similar to that of Proposition 1.4.4. From the definition of χ_σ above, it follows that $\chi_{id} \in \mathrm{Aut}(\mathbb{P}^2(F))$ is the identity, and given $\sigma, \tau \in \mathrm{Aut}(F)$, $\chi_{\sigma\tau} = \chi_\sigma \chi_\tau$. Using these facts, it is easy to prove that $\chi_{\sigma^{-1}}$ is the inverse of χ_σ. Hence, χ_σ is bijective for any $\sigma \in \mathrm{Aut}(F)$. It is also clear that X is a group homomorphism.

Let $\ell : ax + by + cz = 0$ be a line in $\mathbb{P}^2(F)$. Define another line $\ell^\sigma : \sigma(a)x + \sigma(b)y + \sigma(c)z = 0$. Now it is easy to check that $\chi_\sigma(\ell) = \ell^\sigma$.

Finally, we must prove that X is injective. Suppose that χ_σ is the identity automorphism for some $\sigma \in \mathrm{Aut}(F)$. Let $x \in F$ be arbitrary. Then $[x : 0 : 1] = \chi_\sigma([x : 0 : 1]) = [\sigma(x) : 0 : 1]$, from which it follows that $\sigma(x) = x$. □

We will consider $\mathrm{Aut}(F)$ to be a subgroup of $\mathrm{Aut}(\mathbb{P}^2(F))$ via X.

Remark 1.4.10 Unlike $\mathrm{PGL}_3(F)$, the contribution of $\mathrm{Aut}(F)$ to $\mathrm{Aut}(\mathbb{P}^2(F))$ can be quite small and is not obviously related to the number of elements of F. For example, $\mathrm{Aut}(\mathbb{Q})$ and $\mathrm{Aut}(\mathbb{R})$ are both trivial (for the latter, see Exercise 3). However, the automorphism group $\mathrm{Aut}(\overline{\mathbb{Q}})$ of the field $\overline{\mathbb{Q}}$ of algebraic numbers is very large.

Lemma 1.4.11 *Any element $\phi \in PGL_3(F) \cap Aut(F) \subset Aut(\mathbb{P}^2(F))$ is the identity.*

Proof Suppose $\phi = \chi_\sigma$. Then it is trivially easy to see that χ_σ fixes P_1, P_2, P_3, Q. By Proposition 1.4.6, ϕ is the identity. □

Proposition 1.4.12 below shows that an automorphism of $\mathbb{P}^2(F)$ that fixes P_1, P_2, P_3, Q is of the form χ_σ for a unique $\sigma \in \mathrm{Aut}(F)$. Before we can prove it, we discuss how the addition and multiplication of elements of F can be performed geometrically in the affine plane $\mathbb{A}^2(F)$.[3] We impose some restrictions on these procedures, the reasons for which shall be clear during the proof of Proposition 1.4.12. The restrictions are as follows:

- We consider the elements $0, 1 \in F$ to be given. Other elements of F may also be given. The points on the x- and y-axes corresponding to given elements of F are also considered to be given.
- We can draw horizontal and vertical lines from any given point.
- We can take the intersection of two given lines that are not parallel.
- We can take the join of two distinct given points.

[3] The discussion below, up to and including the proof of Proposition 1.4.12, is based on the proof of [6, Proposition 3.11].

- We can draw the parallel line to a given line through a given point.
- As a consequence of the above items, the point $(0, 0)$ is given. Since the lines $x = 1$ and $y = 1$ are given, their intersection $(1, 1)$ is given. Hence, the line $y = x$ is given.

Given two elements $a, b \in F$ with $b \neq 0$ (the case $b = 0$ being trivial), their sum $a + b \in F$ can be constructed geometrically, subject to the restrictions listed above, as follows: since the lines $\ell_1 : y = 1$ and $\ell_2 : x = a$ are given, their intersection $\ell_1 \cdot \ell_2 = (a, 1)$ is given. Since the points $(0, 1)$ and $(b, 0)$ are given, their join $\ell_3 : y = -\frac{1}{b}x + 1$ is given. Then the line ℓ_4 through $(a, 1)$ parallel to ℓ_3 is given; its equation is $\ell_4 : y = -\frac{1}{b}x + \frac{a+b}{b}$. Finally, the intersection $\ell_4 \cdot \ell_x = (a + b, 0)$ is given.

Given two elements $a, b \in F$ with $b \neq 1$ (the case $b = 1$ being trivial), their product $ab \in F$ can be constructed geometrically, subject to the restrictions listed above, as follows: since the lines $\ell_1 : x = a$ and $\ell_2 : y = x$ are given, their intersection $\ell_1 \cdot \ell_2 = (a, a)$ is given. Since the points $(1, 1)$ and $(b, 0)$ are given, their join $\ell_3 : y = -\frac{1}{b-1}x + \frac{b}{b-1}$ is given. Then the line ℓ_4 through (a, a) parallel to ℓ_3 is given; its equation is $\ell_4 : y = -\frac{1}{b-1}x + \frac{ab}{b-1}$. Finally, the intersection $\ell_4 \cdot \ell_x = (ab, 0)$ is given.

Proposition 1.4.12 (Cf. [6, Proposition 3.11]) *Let $\phi \in Aut(\mathbb{P}^2(F))$ be an automorphism that fixes $P_1, P_2, P_3, Q \in \mathbb{P}^2(F)$. Then there exists a unique field automorphism $\sigma \in Aut(F)$ such that $\phi = \chi_\sigma$.*

Proof Uniqueness follows from Proposition 1.4.9.

By Definition 1.4.1, ϕ maps lines into lines. Since $P_1, P_2 \in \ell_\infty$, ϕ maps the line at infinity ℓ_∞ to itself, and hence it maps $\mathbb{A}^2(F)$ to itself. Note that the restrictions of ϕ to ℓ_∞ and $\mathbb{A}^2(F)$ are both bijective. Focusing on the restriction of ϕ to $\mathbb{A}^2(F)$, we now note some of its properties:

- Since P_1 and P_2 are ideal points of horizontal and vertical lines in $\mathbb{A}^2(F)$, respectively, and these points are fixed by ϕ, ϕ maps horizontal lines to horizontal lines and vertical lines to vertical lines.
- Since ϕ fixes P_3, it maps ℓ_x to itself and ℓ_y to itself.
- Since P_3 and Q lie on $\ell_1 : y = x$, ϕ maps ℓ_1 to itself.
- Since ϕ maps vertical lines to vertical lines, and Q lies on the vertical line $x = 1$, ϕ maps this line to itself.
- Since ϕ maps ℓ_x to itself and the line $x = 1$ to itself, it fixes their intersection point $(1, 0)$.

We define a map $\sigma : F \to F$ as follows: Given $a \in F$, $\phi(a, 0) \in \ell_x$ since ϕ maps ℓ_x to itself. We define $\sigma(a)$ to be the x-coordinate of $\phi(a, 0)$:

$$\phi(a, 0) = (\sigma(a), 0).$$

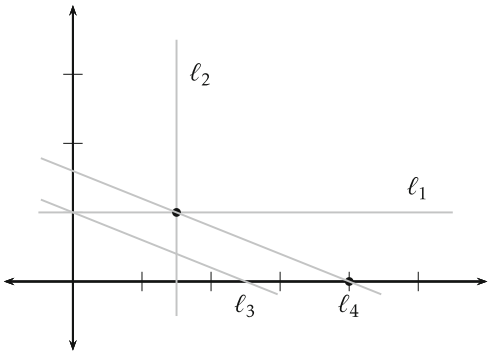


Fig. 1.8 Addition of $a, b \in F$ by geometrical operations, where $a = \frac{3}{2}$ and $b = \frac{5}{2}$, and $\text{char}(F) \neq 2$

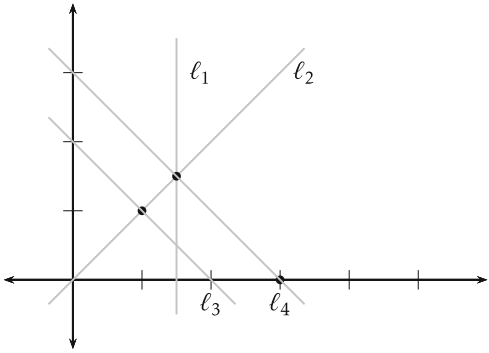


Fig. 1.9 Multiplication of $a, b \in F$ by geometrical operations, where $a = \frac{3}{2}$ and $b = 2$, and $\text{char}(F) \neq 2$

Since ϕ is bijective on $\mathbb{A}^2(F)$, σ is bijective. Since $\phi(0,0) = (0,0)$ and $\phi(1,0) = (1,0)$, we see that $\sigma(0) = 0$ and $\sigma(1) = 1$.

Since ϕ maps vertical lines to vertical lines, it maps $\ell_2 : x = a$ to $\ell_3 : x = \sigma(a)$. Therefore, $\phi(a,a) = (\sigma(a), \sigma(a))$ since $\ell_1 \cdot \ell_2 = (a,a)$ and $\ell_1 \cdot \ell_3 = (\sigma(a), \sigma(a))$.

Let $\ell_4 : y = a$. Since (a,a) lies on ℓ_4 and ϕ maps horizontal lines to horizontal lines, ϕ maps ℓ_4 to $\ell_5 : y = \sigma(a)$. Since $\ell_y \cdot \ell_4 = (0,a)$, $\ell_y \cdot \ell_5 = (0, \sigma(a))$, and ϕ maps ℓ_y to itself, $\phi(0,a) = (0, \sigma(a))$.

Given any point $(a,b) \in \mathbb{A}^2(F)$, let $\ell_6 : y = b$. Since ϕ maps $\ell_2 : x = a$ to $\ell_3 : x = \sigma(a)$, $\ell_6 : y = b$ to $\ell_7 : y = \sigma(b)$, $\ell_2 \cdot \ell_6 = (a,b)$, and $\ell_3 \cdot \ell_7 = (\sigma(a), \sigma(b))$, we have $\phi(a,b) = (\sigma(a), \sigma(b))$.

We must now show that for any $a, b \in F$, $\sigma(a+b) = \sigma(a) + \sigma(b)$ and $\sigma(ab) = \sigma(a)\sigma(b)$. Recall that addition and multiplication can be described using geometrically, as shown in Figs. 1.8 and 1.9. Applying χ_σ to Fig. 1.8, for example, we see that the same geometric procedure gives both $\sigma(a+b)$ and $\sigma(a) + \sigma(b)$. Therefore, $\sigma(a+b) = \sigma(a) + \sigma(b)$. Similarly, $\sigma(ab) = \sigma(a)\sigma(b)$. Hence, $\sigma \in \text{Aut}(F)$.

It remains to consider the image under ϕ of a point $[x_0 : y_0 : z_0] \in \mathbb{P}^2(F)$. There are two cases to consider. If $z_0 \neq 0$, $[x_0 : y_0 : z_0]$ corresponds to the affine point $\left(\frac{x_0}{z_0}, \frac{y_0}{z_0}\right)$. Hence,

$$\begin{aligned}\phi([x_0 : y_0 : z_0]) &= \phi\left(\frac{x_0}{z_0}, \frac{y_0}{z_0}\right)\\ &= \left(\sigma\left(\frac{x_0}{z_0}\right), \sigma\left(\frac{y_0}{z_0}\right)\right)\\ &= \left(\frac{\sigma(x_0)}{\sigma(z_0)}, \frac{\sigma(y_0)}{\sigma(z_0)}\right)\\ &= [\sigma(x_0) : \sigma(y_0) : \sigma(z_0)]\\ &= \chi_\sigma([x_0 : y_0 : z_0]).\end{aligned}$$

Suppose now that $z_0 = 0$. Let $m : y_0x - x_0y = 0$. Then $[x_0 : y_0 : 0] = m \cdot \ell_\infty$. Note that m passes through $[0 : 0 : 1]$ and $[x_0 : y_0 : 1]$, both of which are affine points. We have

$$\begin{aligned}\phi(0, 0) &= (0, 0),\\ \phi(x_0, y_0) &= (\sigma(x_0), \sigma(y_0)).\end{aligned}$$

Therefore, ϕ maps m to the line $n : \sigma(y_0)x - \sigma(x_0)y = 0$, which passes through $[0 : 0 : 1]$ and $[\sigma(x_0) : \sigma(y_0) : 1]$. Hence, $\phi([x_0 : y_0 : 0]) = n \cdot \ell_\infty = [\sigma(x_0) : \sigma(y_0) : 0] = \chi_\sigma([x_0 : y_0 : 0])$. □

We finally arrive at the main result of this section, which concerns the structure of $\mathrm{Aut}(\mathbb{P}^2(F))$. However, before we can state it, let us recall that we have found two types of automorphisms of $\mathbb{P}^2(F)$. One type is given by elements of $\mathrm{PGL}_3(F)$; the other type is given by elements of $\mathrm{Aut}(F)$. These two types of automorphisms are related to each other in the following way: given $\sigma \in \mathrm{Aut}(F)$, and

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix} \in \mathrm{PGL}_3(F),$$

applying σ to the entries of A gives the matrix

$$\omega_\sigma(A) = \begin{bmatrix} \sigma(a_{11}) & \sigma(a_{12}) & \sigma(a_{13})\\ \sigma(a_{21}) & \sigma(a_{22}) & \sigma(a_{23})\\ \sigma(a_{31}) & \sigma(a_{32}) & \sigma(a_{33})\end{bmatrix} \in \mathrm{PGL}_3(F).$$

This defines a function $\omega_\sigma : \mathrm{PGL}_3(F) \to \mathrm{PGL}_3(F)$, which can be easily checked to be an automorphism of the group $\mathrm{PGL}_3(F)$. We then get another function

$$\Omega : \mathrm{Aut}(F) \to \mathrm{Aut}(\mathrm{PGL}_3(F)),$$

defined by $\Omega(\sigma) = \omega_\sigma$. Ω is a group homomorphism, the verification of which we leave to the reader.

Theorem 1.4.13 *$Aut(\mathbb{P}^2(F))$ is isomorphic as a group to $PGL_3(F) \rtimes_\Omega Aut(F)$.*

Proof We claim that $\mathrm{PGL}_3(F)$ is a normal subgroup of $\mathrm{Aut}(\mathbb{P}^2(F))$ and every element $\phi \in \mathrm{Aut}(\mathbb{P}^2(F))$ can be written uniquely as $\phi = \phi_T \chi_\sigma$, where $T \in \mathrm{PGL}_3(F)$ and $\sigma \in \mathrm{Aut}(F)$.

We first prove the latter statement. Let $\phi \in \mathrm{Aut}(\mathbb{P}^2(F))$. Then no three of the four points $A = \phi(P_1)$, $B = \phi(P_2)$, $C = \phi(P_3)$, $D = \phi(Q)$ are collinear. Hence, by Proposition 1.4.6, there exists an element $T \in \mathrm{PGL}_3(F)$ such that $\phi_T(P_1) = A$, $\phi_T(P_2) = B$, $\phi_T(P_3) = C$, $\phi_T(P_4) = D$. It follows that $\phi_T^{-1}\phi$ fixes P_1, P_2, P_3, Q. By Proposition 1.4.12, there exists a $\sigma \in \mathrm{Aut}(F)$ such that $\phi_T^{-1}\phi = \chi_\sigma$. Hence, $\phi = \phi_T \chi_\sigma$. To prove the uniqueness of this decomposition, suppose that $\phi = \phi_{T_1}\chi_{\sigma_1} = \phi_{T_2}\chi_{\sigma_2}$ with $T_1, T_2 \in \mathrm{PGL}_3(F)$ and $\sigma_1, \sigma_2 \in \mathrm{Aut}(F)$. Hence $\phi_{T_2}^{-1}\phi_{T_1} = \chi_{\sigma_2}\chi_{\sigma_1}^{-1}$ and this element is the identity by Lemma 1.4.11.

To show that $\mathrm{PGL}_3(F)$ is a normal subgroup of $\mathrm{Aut}(\mathbb{P}^2(F))$, it suffices to prove that, for any $A \in \mathrm{PGL}_3(F)$ and $\sigma \in \mathrm{Aut}(F)$, the automorphism $\chi_\sigma \phi_A \chi_\sigma^{-1}$ is in $\mathrm{PGL}_3(F)$. By a simple calculation, it can be seen that if

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix},$$

then the automorphism $\chi_\sigma \phi_A \chi_\sigma^{-1}$ can be given as a 3×3 matrix of the form

$$\omega_\sigma(A) = \begin{bmatrix} \sigma(a_{11}) & \sigma(a_{12}) & \sigma(a_{13}) \\ \sigma(a_{21}) & \sigma(a_{22}) & \sigma(a_{23}) \\ \sigma(a_{31}) & \sigma(a_{32}) & \sigma(a_{33}) \end{bmatrix},$$

as given above. Hence, $\mathrm{PGL}_3(F)$ is a normal subgroup of $\mathrm{Aut}(\mathbb{P}^2(F))$.

Finally, we note that the homomorphism

$$\mathrm{Aut}(F) \to \mathrm{Aut}(\mathrm{PGL}_3(F))$$

given by mapping $\sigma \in \mathrm{Aut}(F)$ to the automorphism of $\mathrm{PGL}_3(F)$ that maps ϕ_A to $\chi_\sigma \phi_A \chi_\sigma^{-1}$ is precisely Ω. □

Exercises

1. Prove Propositions 1.4.3 and 1.4.8.
2. Prove that if F is a finite field with N elements, the number of elements in $\mathrm{GL}_n(F)$ is $(N^n - 1)(N^n - N)\cdots(N^n - N^{n-1})$. Prove also that if F is an infinite field, then $\mathrm{PGL}_n(F)$ is also infinite for $n \geq 2$.

3. Prove that the only automorphism of $\mathbb{R}$ is the trivial one. Hence, $\mathrm{Aut}(\mathbb{P}^2(\mathbb{R})) = \mathrm{PGL}_3(\mathbb{R})$. (Hint: Prove that an automorphism of $\mathbb{R}$ must be order-preserving.)
4. Let p be a prime number. Prove that the only automorphism of $\mathbb{F}_p$ is the trivial one. Hence, $\mathrm{Aut}(\mathbb{P}^2(\mathbb{F}_p)) = \mathrm{PGL}_3(\mathbb{F}_p)$.
5. Prove that ω_σ is well-defined, and an automorphism of $\mathrm{PGL}_3(F)$. Prove also that Ω is a group homomorphism.

1.5 Digression: Automorphisms of Affine Planes over a Field

In this section, we define the automorphisms of $\mathbb{A}^2(F)$ and investigate some of their properties. This section comes immediately after Sect. 1.4 because it makes use of the ideas developed there. It is also intended as a digression that can be skipped safely by the reader.

Definition 1.5.1 An *automorphism* of $\mathbb{A}^2(F)$ is a bijective map $\phi : \mathbb{A}^2(F) \to \mathbb{A}^2(F)$ that maps lines to lines. The set of automorphisms of $\mathbb{A}^2(F)$ is denoted by $\mathrm{Aut}(\mathbb{A}^2(F))$.

Proposition 1.5.2 *$Aut(\mathbb{A}^2(F))$ is a group under composition.*

Proof The proof is similar to that of Proposition 1.4.2. We omit the details. □

Remark 1.5.3 For ease of notation, we identify elements of $\mathbb{A}^2(F)$ with vectors in F^2, which we write as column vectors, as before.

In Sect. 1.4, we found nontrivial automorphisms of $\mathbb{P}^2(F)$ by using 3×3 matrices over F with nonzero determinant, which form a group, denoted $\mathrm{GL}_3(F)$, under matrix multiplication. We follow a similar approach here, using 2×2 matrices over F with nonzero determinant. However, there is another kind of automorphism that can be combined with them.

Definition 1.5.4 Let

$$\mathbf{v} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \in F^2$$

be a given vector. For an arbitrary

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix} \in F^2,$$

we define the *translation* by $\mathbf{v}$ as the map $T_\mathbf{v} : \mathbb{A}^2(F) \to \mathbb{A}^2(F)$ defined by

$$T_{\mathbf{v}}(\mathbf{x}) = \mathbf{x} + \mathbf{v} = \begin{bmatrix} x + x_0 \\ y + y_0 \end{bmatrix}.$$

Proposition 1.5.5 *Given a vector*

$$\mathbf{v} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \in F^2,$$

the translation $T_{\mathbf{v}}$ is an automorphism of $\mathbb{A}^2(F)$.

Proof We note that $T_{-\mathbf{v}}$ is the inverse of $T_{\mathbf{v}}$; therefore, $T_{\mathbf{v}}$ is bijective.

Let $\ell : ax + by + c = 0$ be an affine line. Then it is easy to check that $T_{\mathbf{v}}(\ell)$ is the line $\ell' : ax + by + c - ax_0 - by_0 = 0$. □

Proposition 1.5.6 *Let there be given $A \in GL_2(F)$ and a vector*

$$\mathbf{v} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \in F^2.$$

For arbitrary

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix} \in F^2,$$

we define the map $T_{A,\mathbf{v}} : \mathbb{A}^2(F) \to \mathbb{A}^2(F)$ by

$$T_{A,\mathbf{v}}(\mathbf{x}) = A\mathbf{x} + \mathbf{v}.$$

$T_{A,\mathbf{v}}$ is an automorphism of $\mathbb{A}^2(F)$. The set of all such automorphisms forms a subgroup of Aut($\mathbb{A}^2(F)$), denoted by G.

Proof Since $T_{A,\mathbf{v}}$ is the composition $T_{\mathbf{v}}T_{A,\mathbf{0}}$, and $T_{\mathbf{v}}$ is an automorphism by Proposition 1.5.5, it suffices to prove that $T_{A,\mathbf{0}}$ is an automorphism. Since the matrix A has nonzero determinant, it is clear that $T_{A,\mathbf{0}}$ is bijective since $T_{A^{-1},\mathbf{0}}$ is its inverse.

Let $\ell : ax + by + c = 0$ be an affine line. Write

$$A^{-1} = \begin{bmatrix} p & q \\ r & s \end{bmatrix},$$

and define another affine line by $\ell' : (ap + br)x + (aq + bs)y + c = 0$ (why is this an affine line?). Then it can be checked that $T_{A,\mathbf{0}}(\ell) = \ell'$.

In order to prove that automorphisms of the form $T_{A,\mathbf{v}}$ for $A \in \mathrm{GL}_2(F)$ and $\mathbf{v} \in F^2$ form a subgroup of Aut($\mathbb{A}^2(F)$), it suffices to note the following:

- $T_{I,\mathbf{0}}$ is the identity automorphism,
- $T_{B,\mathbf{w}}T_{A,\mathbf{v}} = T_{BA,B\mathbf{v}+\mathbf{w}}$ for $A, B \in \mathrm{GL}_2(F)$ and $\mathbf{v}, \mathbf{w} \in F^2$,
- $T_{A,\mathbf{v}}^{-1} = T_{A^{-1},-A^{-1}\mathbf{v}}$ for $A \in \mathrm{GL}_2(F)$ and $\mathbf{v} \in F^2$.

□

Proposition 1.5.7 *The group G is isomorphic to the multiplicative group of 3×3 matrices with nonzero determinant of the form*

$$\begin{bmatrix} & A & x_0 \\ & & y_0 \\ 0 & 0 & 1 \end{bmatrix}$$

over F.

Proof Given

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{GL}_2(F)$$

and

$$\mathbf{v} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \in F^2,$$

we form the matrix

$$T_{A,\mathbf{v}} = \begin{bmatrix} a & b & x_0 \\ c & d & y_0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} & \mathrm{A} & x_0 \\ & & y_0 \\ 0 & 0 & 1 \end{bmatrix}.$$

This establishes a one-to-one correspondence between 3×3 matrices with nonzero determinant of the form

$$\begin{bmatrix} * & * & * \\ * & * & * \\ 0 & 0 & 1 \end{bmatrix}$$

over F, and automorphisms of $\mathbb{A}^2(F)$ of the form $T_{A,\mathbf{v}}$ with $A \in \mathrm{GL}_2(F)$ and $\mathbf{v} \in F^2$. All that is needed now is to observe that, given

$$B = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \in \mathrm{GL}_2(F),$$

and

$$\mathbf{w} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \in F^2,$$

one has

$$\begin{aligned} T_{B,\mathbf{w}} T_{A,\mathbf{v}} &= \begin{bmatrix} & \mathrm{B} & \\ & & \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} & \mathrm{A} & \\ & & \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} x_1 \\ y_1 \\ \end{matrix} \begin{matrix} x_0 \\ y_0 \\ \end{matrix} \\ &= \begin{bmatrix} & \mathrm{BA} & \\ & & \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} px_0 + qy_0 + x_1 \\ rx_0 + sy_0 + y_1 \\ \end{matrix} \\ &= T_{BA, B\mathbf{v}+\mathbf{w}}. \end{aligned}$$

□

Until the end of the section, we define the three points

$$X_1 = (0, 0), X_2 = (1, 0), X_3 = (0, 1) \in \mathbb{A}^2(F).$$

These points are noncollinear by Exercise 10 of Sect. 1.1.

Proposition 1.5.8 *Given three noncollinear points P, Q, $R \in \mathbb{A}^2(F)$, there exists a unique automorphism $T_{A,\mathbf{v}} \in G$ with $A \in GL_2(F)$ and $\mathbf{v} \in F^2$ such that $T_{A,\mathbf{v}}(X_1) = P$, $T_{A,\mathbf{v}}(X_2) = Q$, $T_{A,\mathbf{v}}(X_3) = R$. Any automorphism $T_{A,\mathbf{v}} \in G$ that fixes P, Q, R is the identity.*

Proof Consider the vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in F^2$ corresponding to P, Q, R, respectively. Since P, Q, R are noncollinear, $\mathbf{v} - \mathbf{u}$ and $\mathbf{w} - \mathbf{u}$ are linearly independent. Hence, there exists a unique $A \in \mathrm{GL}_2(F)$ that maps the unit vectors $\mathbf{i}$ and $\mathbf{j}$ to $\mathbf{v} - \mathbf{u}$ and $\mathbf{w} - \mathbf{u}$, respectively. It is now clear that the automorphism $T_{A,\mathbf{u}}$ is the unique automorphism of this form with the required property.

To prove the second part, let $T_{B,\mathbf{w}}$ with $B \in \mathrm{GL}_2(F)$ and $\mathbf{w} \in F^2$ be the unique automorphism of this form such that $T_{B,\mathbf{w}}(X_1) = P$, $T_{B,\mathbf{w}}(X_2) = Q$, $T_{B,\mathbf{w}}(X_3) = R$. Then the composition

$$T_{B,\mathbf{w}}^{-1} T_{A,\mathbf{v}} T_{B,\mathbf{w}}$$

fixes X_1, X_2, X_3. Hence, by the first part, it is the identity automorphism. This implies that $T_{A,\mathbf{v}}$ is the identity automorphism. □

Proposition 1.5.9 *The maps $A \mapsto T_{A,\mathbf{0}}$ and $\mathbf{v} \mapsto T_{I,\mathbf{v}}$ define injective group homomorphisms $GL_2(F) \to G$ and $F^2 \to G$, respectively, where F^2 is considered as a group under vector addition. G is isomorphic as a group to $F^2 \rtimes GL_2(F)$,*

where the factors in the semidirect product are considered as subgroups of G via the injective maps above.

Proof It is clear that the two maps are one-to-one; that they are group homomorphisms can be verified with the description, given in Proposition 1.5.7, of the elements of G as matrices, or alternatively with the rules of composition given in the proof of Proposition 1.5.6.

To prove the second claim, we must prove that F^2 is a normal subgroup of G and that every element $T \in G$ can be written uniquely as $T_{I,\mathbf{v}}T_{A,\mathbf{0}}$ for some $\mathbf{v} \in F^2$ and $A \in \mathrm{GL}_2(F)$.

Given $T = T_{A,\mathbf{v}} \in G$ for some $\mathbf{v} \in F^2$ and $A \in \mathrm{GL}_2(F)$, it is clear that $T = T_{I,\mathbf{v}}T_{A,\mathbf{0}}$. To prove uniqueness, suppose that $T_{I,\mathbf{v}}T_{A,\mathbf{0}} = T_{I,\mathbf{w}}T_{B,\mathbf{0}}$ for some $B \in \mathrm{GL}_2(F)$ and $\mathbf{w} \in F^2$. Then $T_{I,\mathbf{w}}^{-1}T_{I,\mathbf{v}} = T_{B,\mathbf{0}}T_{A,\mathbf{0}}^{-1}$. This implies $T_{I,\mathbf{v}-\mathbf{w}} = T_{BA^{-1},\mathbf{0}}$. Since $T_{BA^{-1},\mathbf{0}}$ maps $\mathbf{0}$ to $\mathbf{0}$, we have $\mathbf{v} = \mathbf{w}$ and therefore $A = B$.

To prove that F^2 is a normal subgroup of G, it suffices to prove that, for any $\mathbf{v} \in F^2$ and $A \in \mathrm{GL}_2(F)$, $T_{A,\mathbf{0}}T_{I,\mathbf{v}}T_{A,\mathbf{0}}^{-1}$ is in F^2. This can be easily calculated to be $T_{I,A\mathbf{v}}$ and therefore in F^2. □

A given field automorphism $\sigma \in \mathrm{Aut}(F)$ can be used to define a map $\chi_\sigma : \mathbb{A}^2(F) \to \mathbb{A}^2(F)$ as

$$\chi_\sigma(x, y) = (\sigma(x), \sigma(y)).$$

Proposition 1.5.10 *Given $\sigma \in Aut(F)$, χ_σ is an automorphism of $\mathbb{A}^2(F)$. The map $X : Aut(F) \to Aut(\mathbb{A}^2(F))$ defined as $\sigma \mapsto \chi_\sigma$ for all $\sigma \in Aut(F)$ is an injective group homomorphism.*

Proof The proof is similar to that of Proposition 1.4.9. We omit the details. □

We consider $\mathrm{Aut}(F)$ to be a subgroup of $\mathrm{Aut}(\mathbb{P}^2(F))$ via X.

Lemma 1.5.11 *Any element $\phi \in G \cap Aut(F) \subset Aut(\mathbb{A}^2(F))$ is the identity.*

Proof Suppose $\phi = \chi_\sigma$. Then it is trivially easy to see that χ_σ fixes X_1, X_2, X_3. By Proposition 1.5.8, ϕ is the identity. □

Proposition 1.5.12 *Let $\phi \in Aut(\mathbb{A}^2(F))$ be an automorphism that fixes $X_1, X_2, X_3 \in \mathbb{A}^2(F)$. Then there exists a unique field automorphism $\sigma \in Aut(F)$ such that $\phi = \chi_\sigma$.*

Proof The uniqueness follows from Proposition 1.5.10.

Recall that ϕ maps lines to lines by Definition 1.5.1; it is also easy to see that it maps parallel lines to parallel lines. As in the proof of Proposition 1.4.12, let us note some properties of ϕ:

- Since ϕ fixes X_1, X_2, X_3, it maps ℓ_x to itself and ℓ_y to itself.
- Since ϕ maps parallel lines to parallel lines, it maps horizontal lines to horizontal lines and vertical lines to vertical lines.
- Hence, ϕ maps the line $x = 1$ to itself and $y = 1$ to itself. Therefore, ϕ fixes the point $(1, 1)$.
- Since ϕ fixes the points $X_1 = (0, 0)$ and $(1, 1)$, it maps the line $y = x$ to itself.

The rest of the proof is identical to that of Proposition 1.4.12. □

Theorem 1.5.13 *$Aut(\mathbb{A}^2(F))$ is isomorphic as a group to $G \rtimes Aut(F)$.*

Proof We claim that G is a normal subgroup of $\mathrm{Aut}(\mathbb{A}^2(F))$ and that every element $\phi \in \mathrm{Aut}(\mathbb{A}^2(F))$ can be written uniquely as $\phi = T_{A,\mathbf{v}}\chi_\sigma$, where $T_{A,\mathbf{v}} \in G$, for some $A \in \mathrm{GL}_2(F)$ and $\mathbf{v} \in F^2$, and $\sigma \in \mathrm{Aut}(F)$.

We first prove the latter statement. Let $\phi \in \mathrm{Aut}(\mathbb{A}^2(F))$. Then the three points $A = \phi(X_1)$, $B = \phi(X_2)$, $C = \phi(X_3)$ are not collinear. Hence, by Proposition 1.5.8, there exists an automorphism $T_{A,\mathbf{v}} \in G$, for some $A \in \mathrm{GL}_2(F)$ and $\mathbf{v} \in F^2$, such that $T_{A,\mathbf{v}}(X_1) = A$, $T_{A,\mathbf{v}}(X_2) = B$, $T_{A,\mathbf{v}}(X_3) = C$. It follows that $T_{A,\mathbf{v}}^{-1}\phi$ fixes X_1, X_2, X_3. By Proposition 1.5.12, there exists a $\sigma \in \mathrm{Aut}(F)$ such that $T_{A,\mathbf{v}}^{-1}\phi = \chi_\sigma$. Hence, $\phi = T_{A,\mathbf{v}}\chi_\sigma$.

To prove the uniqueness of this decomposition, suppose that $\phi = T_{A,\mathbf{v}}\chi_\sigma = T_{B,\mathbf{w}}\chi_\tau$ with $A, B \in \mathrm{GL}_2(F)$, $\mathbf{v}, \mathbf{w} \in F^2$, and $\sigma, \tau \in \mathrm{Aut}(F)$. Hence $T_{B,\mathbf{w}}^{-1}T_{A,\mathbf{v}} = \chi_\tau\chi_\sigma^{-1}$ and this element is the identity by Lemma 1.5.11.

To show that G is a normal subgroup of $\mathrm{Aut}(\mathbb{A}^2(F))$, it suffices to prove that, given $T_{A,\mathbf{v}} \in G$, for some $A \in \mathrm{GL}_2(F)$, $\mathbf{v} \in F^2$, and $\sigma \in \mathrm{Aut}(F)$, the automorphism $\chi_\sigma T_{A,\mathbf{v}}\chi_\sigma^{-1}$ is in G. Write

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

and

$$\mathbf{v} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}.$$

Since $T_{A,\mathbf{v}}$ can be represented by the matrix

$$A = \begin{bmatrix} a & b & x_0 \\ c & d & y_0 \\ 0 & 0 & 1 \end{bmatrix},$$

the automorphism $\chi_\sigma T_{A,\mathbf{v}}\chi_\sigma^{-1}$ can be represented by the matrix

$$\begin{bmatrix} \sigma(a) & \sigma(b) & \sigma(x_0) \\ \sigma(c) & \sigma(d) & \sigma(y_0) \\ 0 & 0 & 1 \end{bmatrix},$$

which implies that it is an element of G. Hence, G is a normal subgroup of $\mathrm{Aut}(\mathbb{P}^2(F))$. □

Exercise

1. An element $\phi \in \mathrm{Aut}(\mathbb{A}^2(F))$ is called a *dilation* if $\phi(\ell) \parallel \ell$ for every line $\ell \subset \mathbb{A}^2(F)$.
   (a) Prove that a dilation corresponds uniquely to an automorphism of $\mathbb{P}^2(F)$ that leaves invariant the points on the line at infinity $\ell_\infty : z = 0$.
   (b) Prove that the dilations form a subgroup of $\mathrm{Aut}(\mathbb{A}^2(F))$. We denote this subgroup by $\mathrm{Dil}(\mathbb{A}^2(F))$.
   (c) Prove that an element of $\mathrm{Aut}(F) \subset \mathrm{Aut}(\mathbb{A}^2(F))$ is a dilation if and only if it is the identity. Hence, $\mathrm{Dil}(\mathbb{A}^2(F))$ is, in fact, a subgroup of G.
   (d) Prove that a translation is a dilation.
   (e) Determine the matrices $A \in \mathrm{GL}_2(F)$ for which $T_{A,\mathbf{0}}$ is a dilation.
   (f) Prove that a dilation that is not the identity can have at most one fixed point.
   (g) Prove that a dilation that has no fixed points is a translation.
   (h) Determine the matrices $A \in \mathrm{GL}_2(F)$ for which $T_{A,\mathbf{0}}$ is a dilation with exactly one fixed point. Such dilations are called *scalings*, and we shall use them extensively in Sect. 2.5.
   (i) Determine the structure of the group $\mathrm{Dil}(\mathbb{A}^2(F))$. In particular, note that the translations form a normal subgroup of $\mathrm{Dil}(\mathbb{A}^2(F))$. We denote the group of translations by $\mathrm{Tran}(\mathbb{A}^2(F))$.

1.6 The Theorems of Desargues and Pappus

In this section, we use the tools developed in Sect. 1.4 to prove two famous theorems in projective geometry, namely Desargues' and Pappus' theorems (Theorems 1.6.5 and 1.6.6).

The definition of a triangle in $\mathbb{P}^2(F)$ is identical to Definition 1.1.15, which was used for $\mathbb{A}^2(F)$. Note that this definition is self-dual.

Definition 1.6.1 A *triangle* in $\mathbb{P}^2(F)$ is a collection of three noncollinear points, called the *vertices* of the triangle, together with the joins of pairs of these points, called the *sides* of the triangle, which are then three nonconcurrent lines. The triangle with vertices A, B, C is denoted by ABC, or by any permutation of these letters. The sides of the triangle ABC are the lines AB, AC, BC. The triangle with sides a, b, c is denoted by abc, or by any permutation of these letters. The vertices of the triangle abc are the points $a \cdot b$, $a \cdot c$, $b \cdot c$. By convention, we will label the

points $a \cdot b = C$, $a \cdot c = B$, $b \cdot c = A$, and we will label the lines $AB = c$, $AC = b$, $BC = a$.

We point out that, in contrast to Euclidean geometry, there is no concept of angle or side length in projective geometry. Thus, the sides of a triangle are entire lines.

Definition 1.6.2 Two triangles ABC and $A'B'C'$ in $\mathbb{P}^2(F)$ are *perspective from point* O if AA', BB', CC' pass through O.

Dually, we have the following definition:

Definition 1.6.3 Two triangles ABC and $A'B'C'$ in $\mathbb{P}^2(F)$ are *perspective from line* ℓ if $AB \cdot A'B'$, $AC \cdot A'C'$, $BC \cdot B'C'$ lie on ℓ.

Definition 1.6.4 A *Desargues configuration* in $\mathbb{P}^2(F)$ consists of two triangles ABC and $A'B'C'$ that are perspective from a point O, such that the lines OAA', OBB', OCC' are distinct; the points O, A, A' are distinct; the points O, B, B' are distinct; and the points O, C, C' are distinct.

Theorem 1.6.5 (Desargues' Theorem) *Two triangles in a Desargues configuration are perspective from a line.*

Proof Let the two triangles be ABC and $A'B'C'$, and suppose that they are perspective from point O. We must prove that the intersection points $P = AB \cdot A'B'$, $Q = AC \cdot A'C'$, $R = BC \cdot B'C'$ are collinear.

The hypotheses of the theorem imply that no three of A, B, C, O are collinear. Hence, by Proposition 1.4.6, there exists an automorphism ϕ of $\mathbb{P}^2(F)$ such that $\phi(A) = [1 : 0 : 0]$, $\phi(B) = [0 : 1 : 0]$, $\phi(C) = [0 : 0 : 1]$, $\phi(O) = [1 : 1 : 1]$. Since ϕ maps lines into lines, it suffices to prove the theorem for the case where $A = [1 : 0 : 0]$, $B = [0 : 1 : 0]$, $C = [0 : 0 : 1]$, $O = [1 : 1 : 1]$. With this assumption, we have the following equations of lines:

$$OA : y = z,\ AB : z = 0,$$
$$OB : x = z,\ AC : y = 0,$$
$$OC : x = y,\ BC : x = 0.$$

We write $A' = [a : 1 : 1]$, $B' = [1 : b : 1]$, $C' = [1 : 1 : c]$, where $a, b, c \neq 1$, and we find the equations of the sides of the triangle $A'B'C'$ as

$$A'B' : (1-b)x + (1-a)y + (ab-1)z = 0,$$
$$A'C' : (c-1)x + (1-ac)y + (a-1)z = 0,$$
$$B'C' : (bc-1)x + (1-c)y + (1-b)z = 0.$$

Taking intersections, we obtain the coordinates of P, Q, R as

$$P = [1 - a : b - 1 : 0],$$
$$Q = [a - 1 : 0 : 1 - c],$$
$$R = [0 : 1 - b : c - 1].$$

Using Proposition 1.2.12, one can now easily check that P, Q, R are collinear. □

We come to the second main result of this section.

Theorem 1.6.6 (Pappus' Theorem) *Let six distinct points $A, B, C, D, E, F \in \mathbb{P}^2(F)$ be given such that A, B, C lie on one line, D, E, F lie on a second, different line, and suppose that none of the six points is the point of intersection of the two lines. Then the intersection points $X = AE \cdot BD$, $Y = AF \cdot CD$, $Z = BF \cdot CE$ are collinear (Fig. 1.11).*

Proof The hypotheses of the theorem imply that no three of A, B, D, E are collinear. Therefore, using Proposition 1.4.6 we can assume, without loss of generality, that $A = [1 : 0 : 0]$, $B = [0 : 1 : 0]$, $D = [0 : 0 : 1]$, $E = [1 : 1 : 1]$. Then $AE : y = z$, $BD : x = 0$, and hence $X = [0 : 1 : 1]$. Note that $AB : z = 0$, $DE : x = y$, and hence $ABC \cdot DEF = [1 : 1 : 0]$. We write $C = [\lambda : \mu : 0]$, where $\lambda \neq 0$, $\mu \neq 0$, and $\lambda \neq \mu$. Similarly, we write $F = [1 : 1 : \nu]$, where $\nu \neq 0, 1$.

Then $BF : \nu x = z$, $CE : \mu x - \lambda y + (\lambda - \mu)z = 0$, and hence $Z = [\lambda : \mu + (\lambda - \mu)\nu : \lambda\nu]$. Finally, $AF : \nu y = z$, $CD : \mu x = \lambda y$, and hence $Y = [\lambda : \mu : \mu\nu]$. Proposition 1.2.12 now implies that X, Y, Z are collinear. □

Exercises

1. Write the duals of Definition 1.6.4 (defining a *dual Desargues configuration*) and Theorems 1.6.5 and 1.6.6.
2. Try to prove Theorems 1.6.5 and 1.6.6 by using automorphisms that map *other* groups of four points, no three of which are collinear, to $[1 : 0 : 0]$, $[0 : 1 : 0]$, $[0 : 0 : 1]$, $[1 : 1 : 1]$.
3. Prove that a Desargues configuration as shown in Fig. 1.10 consists of 10 points and 10 lines, each point of which lies on exactly 3 lines in the configuration, and every line of which contains exactly 3 points in the configuration. Thus, there exists no Desargues configuration on the Fano plane $\mathbb{P}^2(\mathbb{F}_2)$. Are there other projective planes $\mathbb{P}^2(F)$ on which there exists no Desargues configuration?
4. What happens if, in the hypothesis of Pappus' Theorem, we remove the requirement that none of the six distinct points be the point of intersection of the two lines?

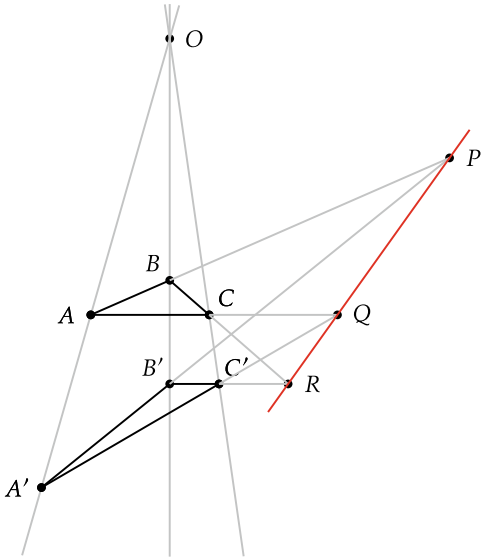


Fig. 1.10 Desargues' Theorem

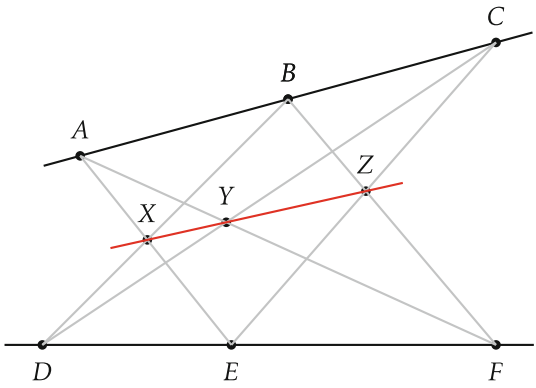


Fig. 1.11 Pappus' Theorem

5. Investigate whether Desargues' and Pappus' theorems remain valid for *degenerate* cases, that is, when objects that were assumed to be distinct are no longer necessarily so.

1.7 Projectivities

We begin with the following pair of dual definitions.

Definition 1.7.1 A *range* of points in $\mathbb{P}^2(F)$ is the set of points on a given line. A *pencil* of lines in $\mathbb{P}^2(F)$ is the set of lines passing through a given point.

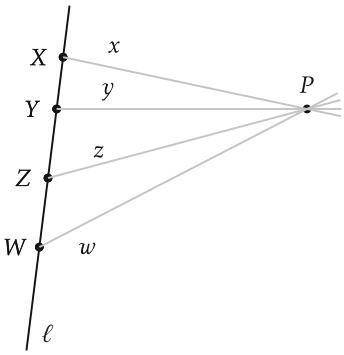


Fig. 1.12 An elementary correspondence, $XYZW \mathrel{\overline{\wedge}} xyzw$

In this section, we will be concerned with one-to-one correspondences between ranges and pencils. The building blocks of such correspondences are the elementary correspondences.

Definition 1.7.2 Let $\ell \subset \mathbb{P}^2(F)$ be a line and $P \in \mathbb{P}^2(F)$ be a point not on ℓ. The *elementary correspondence* between the range of points on ℓ and the pencil of lines passing through P is constructed by mapping an arbitrary point $X \in \ell$ to the line $x = PX$. This correspondence is denoted by $X \mathrel{\overline{\wedge}} x$. The range is called a *section* of the pencil by ℓ and the pencil is said to *project* the range from P (Fig. 1.12).

Proposition 1.7.3 *An elementary correspondence is a one-to-one correspondence. The inverse of an elementary correspondence is also an elementary correspondence.*

The proof is left to the reader.

Most frequently, we will be interested in one-to-one correspondences between two ranges. The simplest type of such correspondence is the perspectivity.

Definition 1.7.4 Let ℓ_1 and ℓ_2 be two distinct lines in $\mathbb{P}^2(F)$, and let $P \in \mathbb{P}^2(F)$ be a point not on ℓ_1 or ℓ_2. A *perspectivity* between ℓ_1 and ℓ_2 with *center* P is a one-to-one correspondence between the ranges of points on these lines defined as the composition of the elementary correspondence between the range of points on ℓ_1 and the pencil of lines passing through P followed by the elementary correspondence between the pencil of lines passing through P and the range of points on ℓ_2. If the points $X \in \ell_1$ and $X' \in \ell_2$ correspond to each other, this is denoted by $X \mathrel{\overline{\overline{\wedge}}} X'$, or $X \overset{P}{\mathrel{\overline{\overline{\wedge}}}} X'$ if we want to emphasize the center P (Fig. 1.13).

Definition 1.7.5 Let ℓ_1 and ℓ_2 be two lines in $\mathbb{P}^2(F)$, not necessarily distinct. A *projectivity* between the ranges of points on ℓ_1 and ℓ_2 is a composition of perspectivities such that the source of the first perspectivity is ℓ_1 and the target of the last one is ℓ_2 (Fig. 1.14).

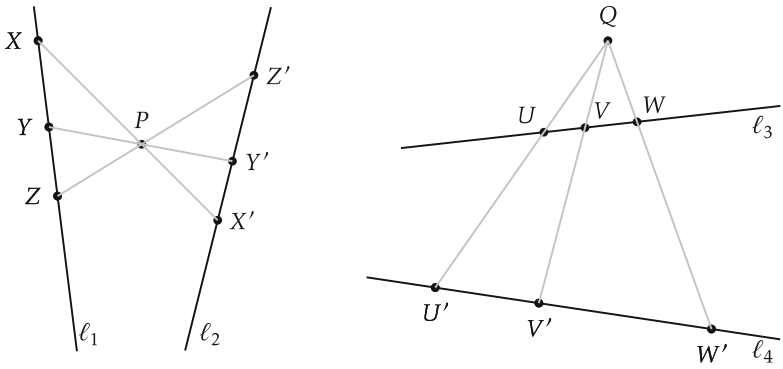


Fig. 1.13 Two perspectivities, with centers P and Q

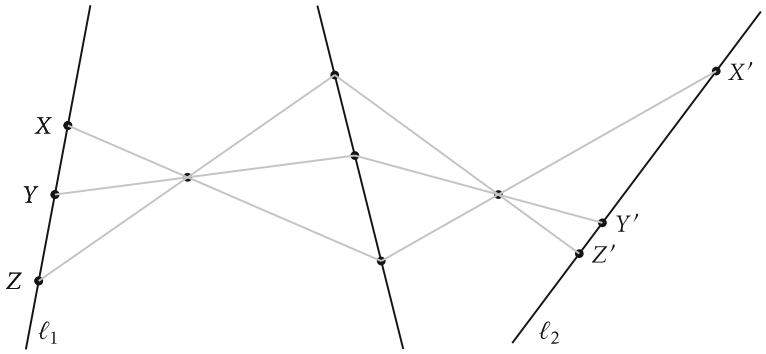


Fig. 1.14 A projectivity that is the composition of two perspectivities

Proposition 1.7.6 *The inverse of a perspectivity is also a perspectivity. A projectivity is a one-to-one correspondence. The identity map from a range or pencil to itself is a projectivity. The composition of two projectivities and the inverse of a projectivity are also projectivities.*

Again, we leave the proof to the reader. We note that the projectivities from a given line to itself form a group under composition.

Definition 1.7.7 Let $\ell \subset \mathbb{P}^2(F)$ be a line. The *projectivity group* $\mathrm{PJ}(\ell)$ is the group of projectivities of ℓ to itself, under composition.

Remark 1.7.8 Since perspectivities and projectivities are one-to-one correspondences, we can consider them as bijective functions from the range defined by one line to the range defined by a second (possibly the same in the case of projectivities) line.

Proposition 1.7.9 *Let $\ell, \ell' \subset \mathbb{P}^2(F)$ be two lines. Then $PJ(\ell) \cong PJ(\ell')$.*

Proof Using Proposition 1.4.6, we can find an automorphism ϕ of $\mathbb{P}^2(F)$ such that $\phi(\ell) = \ell'$. Let there be a projectivity $f : \ell \to \ell$ given by the composition of a chain of perspectivities such that the source of the first perspectivity and the target of the last are both ℓ, and the centers of the perspectivities are, in order, $P_1, \ldots, P_n$. Applying ϕ to this chain of perspectivities gives another chain of perspectivities such that the source of the first perspectivity and the target of the last one are both $\ell' = \phi(\ell)$, and the centers are $\phi(P_1), \ldots, \phi(P_n)$. This gives a mapping $\mathrm{PJ}(\ell) \to \mathrm{PJ}(\ell')$ that is a group homomorphism whose inverse can be constructed similarly using ϕ^{-1}. □

We now turn to the structure of the projectivity group $\mathrm{PJ}(\ell)$ of a line $\ell \subset \mathbb{P}^2(F)$. Since it is clear that a projectivity of ℓ to itself is a bijective mapping from ℓ to itself, it makes sense to start by investigating the latter. We recall that we have started the study of automorphisms of $\mathbb{P}^2(F)$ by considering vector space automorphisms of F^3. Now, by Remark 1.2.8, a line in $\mathbb{P}^2(F)$ can be considered to correspond to a 2-dimensional vector subspace of F^3. For this reason, we will consider, similarly to what we did in Sect. 1.4, bijective mappings from ℓ to itself induced by vector space automorphisms of F^2. We remind the reader that the set of 2×2 invertible matrices over F, denoted $\mathrm{GL}_2(F)$, is a group by Proposition 1.4.3, and it is isomorphic to the group of vector space automorphisms of F^2.

By Proposition 1.7.9, we are free to choose the line $\ell \subset \mathbb{P}^2(F)$ as we wish. We choose the line at infinity $\ell_\infty : z = 0$ for convenience of notation and calculation. Suppose that we are given $A \in \mathrm{GL}_2(F)$. We define a map $\phi_A : \ell_\infty \to \ell_\infty$ as follows, which is very similar to the way (some) automorphisms of $\mathbb{P}^2(F)$ were defined in Sect. 1.4: Let $P = [x : y : 0] \in \ell_\infty$. Define the vector

$$\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix} \in F^2.$$

The matrix product $A\mathbf{v}$ is then another vector in F^2, which is nonzero since $\mathbf{v}$ is nonzero and A is invertible. Hence, the components of $A\mathbf{v}$ give a point, denoted by $[A\mathbf{v}] \in \ell_\infty$, where the last coordinate is 0. We define $\phi_A(P) = [A\mathbf{v}]$.

Proposition 1.7.10 *Given $A \in GL_2(F)$, ϕ_A is a bijective mapping from ℓ_∞ to itself. The map $\Phi : GL_2(F) \to Sym(\ell_\infty)$ defined as $A \mapsto \phi_A$ for all $A \in GL_2(F)$ is a group homomorphism.*

Similarly as in Sect. 1.4, different matrices in $\mathrm{GL}_2(F)$ can give rise to the same bijective mapping from ℓ_∞ to itself.

Proposition 1.7.11 *Let $A, B \in GL_2(F)$. Then $\phi_A = \phi_B$ if and only if $B = \lambda A$ for some nonzero $\lambda \in F$.*

Again, as in Sect. 1.4, we define *projective general linear group of degree 2* as $\mathrm{PGL}_2(F) := \mathrm{GL}_2(F)/F^*$ and, by an abuse of notation, we will represent elements of $\mathrm{PGL}_2(F)$ as matrices. The map Φ descends to a map, also denoted Φ, from $\mathrm{PGL}_2(F)$ to $\mathrm{Sym}(\ell_\infty)$; this map is an injective group homomorphism.

On the line ℓ_∞, every point can be written as $[x : y : 0]$, where $x, y \in F$, not both zero. We consider their ratio

$$t = \frac{x}{y},$$

to be a *linear coordinate* on ℓ, since it is well-defined. In case $y = 0$—in other words, for the point $[1 : 0 : 0]$—we define $t = \infty$. It is easy to check that the linear coordinate t defines a bijection between ℓ_∞ and $F \cup \{\infty\}$. We will give a more general definition of a linear coordinate below (Definition 1.7.18) and investigate some of their properties.

Consider

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{PGL}_2(F).$$

Then for any point $[x : y : 0] \in \ell_\infty$, we have $\phi_A([x : y : 0]) = [ax + by : cx + dy : 0]$. Using the linear coordinate t for both the source and the target of ϕ_A, and a small abuse of notation, we can write

$$\phi_A(t) = \frac{at + b}{ct + d}.$$

Definition 1.7.12 A *fractional linear transformation* is a function $\phi : F \cup \{\infty\} \to F \cup \{\infty\}$ of the form

$$\phi(t) = \frac{at + b}{ct + d},$$

where $a, b, c, d \in F$ and $ad - bc \neq 0$.

Remark 1.7.13 In Definition 1.7.12, if $c = 0$ we define $\phi(\infty) = \infty$, and if $c \neq 0$ we define $\phi(-\frac{d}{c}) = \infty$ and $\phi(\infty) = \frac{a}{c}$. It can be checked that the situation $\frac{0}{0}$ never arises since $ad - bc \neq 0$. With these conventions, fractional linear transformations are invertible, and inverses of fractional linear transformations are also fractional linear transformations. Hence, fractional linear transformations form a group under composition. In fact, this group is isomorphic to $\mathrm{PGL}_2(F)$ with the isomorphism given by $A \mapsto \phi_A$. We leave the verification of the details to the reader. We will identify an element $A \in \mathrm{PGL}_2(F)$ and the corresponding fractional linear transformation ϕ_A from now on.

Let us now investigate the structure of $\mathrm{PGL}_2(F)$. The key fact is that $\mathrm{PGL}_2(F)$ is generated by three types of fractional linear transformations, each of which has a particularly simple form.

Theorem 1.7.14 (Cf. [6, Proposition 8.2]) *$PGL_2(F)$ is generated by fractional linear transformations of the types (i) $\phi(t) = t + a$, (ii) $\phi(t) = at$ with $a \neq 0$, and (iii) $\phi(t) = \frac{1}{t}$.*

Proof Let

$$\phi(t) = \frac{at + b}{ct + d}$$

be an arbitrary fractional linear transformation. If $c = 0$, then the condition $ad - bc \neq 0$ forces $a \neq 0$ and $d \neq 0$, and the given transformation, which is

$$\phi(t) = \frac{a}{d}t + \frac{b}{d},$$

can be written as the composition of two transformations of types *(i)* and *(ii)*.

Suppose now that $c \neq 0$. Then the given transformation is

$$\phi(t) = \frac{b - \frac{ad}{c}}{ct + d} + \frac{a}{c},$$

which is a composition of $\phi_1(t) = ct + d$, itself a composition of transformations of types *(i)* and *(ii)*, $\phi_2(t) = \frac{1}{t}$, $\phi_3(t) = (b - \frac{ad}{c})t$, and $\phi_4(t) = t + \frac{a}{c}$. □

Proposition 1.7.15 (Cf. [6, Proposition 8.1]) *Given three distinct points $A, B, C \in \ell_\infty$, there exists a unique $\phi \in PGL_2(F)$ such that $\phi(0) = A$, $\phi(1) = B$, $\phi(\infty) = C$.*

Proof We suppose $A, B, C \in F$ and leave the case where one of them is ∞ to the reader.

We have to solve the system of equations

$$A = \frac{b}{d},$$
$$B = \frac{a + b}{c + d},$$
$$C = \frac{a}{c}.$$

Since $A, B, C \in F$, neither c nor d can be 0. Since multiplication of a, b, c, d by a nonzero scalar will yield the same fractional linear transformation, we can assume

$c = 1$. Then the above system can be solved to give the unique solutions

$$\begin{aligned} a &= C, \\ b &= \frac{A(C-B)}{B-A}, \\ c &= 1, \\ d &= \frac{(C-B)}{B-A}. \end{aligned}$$

We also note that

$$ad - bc = \frac{(C-A)(C-B)}{B-A} \neq 0,$$

since $A, B, C \in F$ were assumed to be distinct. □

Corollary 1.7.16 *Given two sets of distinct points $A, B, C \in \ell_\infty$ and $A', B', C' \in \ell_\infty$, there exists a unique $\phi \in PGL_2(F)$ such that $\phi(A) = A'$, $\phi(B) = B'$, $\phi(C) = C'$.*

Corollary 1.7.16 can also be stated for two sets of distinct points in $F \cup \{\infty\}$.

Proposition 1.7.17 (Cf. [6, Proposition 8.3]) *The three types of fractional linear transformations in Theorem 1.7.14 are projectivities of ℓ_∞ to itself.*

Proof In each case, we will describe the projectivities as a composition of perspectivities and perform the required calculations.

Case (i). The transformation $\phi(t) = t + a$ is obtained as the composition of the perspectivity from ℓ_∞ to $\ell : y = 0$ with center $P = [0 : 1 : 1]$ and the perspectivity from ℓ to ℓ_∞ with center $Q = [a : 1 : 1]$.

Given $t \in F$, the corresponding point on ℓ_∞ is $X = [t : 1 : 0]$. We compute $PX : x - ty + tz = 0$, $X' = PX \cdot \ell = [-t : 0 : 1]$, $QX' = x - (a+t)y + tz = 0$, $X'' = QX' \cdot \ell_\infty = [t + a : 1 : 0]$. Hence, $X \overset{P}{\barwedge} X' \overset{Q}{\barwedge} X''$; in other words, the given projectivity maps t to $t + a$. It can be checked that it also maps ∞ to ∞.

Case (ii). The transformation $\phi(t) = at$ with $a \neq 0$ is obtained as the composition of the perspectivity from ℓ_∞ to $\ell : x = z$ with center $P = [0 : 0 : 1]$ and the perspectivity from ℓ to ℓ_∞ with center $Q = [1 - a : 0 : 1]$.

Given $t \in F$, the corresponding point on ℓ_∞ is $X = [t : 1 : 0]$. We compute $PX : -x + ty = 0$, $X' = PX \cdot \ell = [t : 1 : t]$, $QX' = -x + aty + (1-a)z = 0$, $X'' = QX' \cdot \ell_\infty = [at : 1 : 0]$. Hence, $X \overset{P}{\barwedge} X' \overset{Q}{\barwedge} X''$; in other words, the given projectivity maps t to at. It can be checked that it also maps ∞ to ∞.

Case (iii). The transformation $\phi(t) = \frac{1}{t}$ is obtained as the composition of the perspectivity from ℓ_∞ to $\ell_1 : y = 0$ with center $P = [1 : 1 : 1]$, followed by the perspectivity from ℓ_1 to $\ell_2 : x = z$ with center $Q = [1 : 1 : 0]$, followed by the perspectivity from ℓ_2 to ℓ_∞ with center $R = [0 : 0 : 1]$.

Given $t \in F$, $t \neq 0$, the corresponding point on ℓ_∞ is $X = [t : 1 : 0]$. We compute $PX : x - ty + (t-1)z = 0$, $X' = PX \cdot \ell_1 = [1 - t : 0 : 1]$, $QX' = x - y + (t-1)z = 0$, $X'' = QX' \cdot \ell_2 = [1 : t : 1]$, $RX'' : -tx + y = 0$, $X''' = RX'' \cdot \ell_\infty = [1 : t : 0]$. Hence, $X \overset{P}{\wedge} X' \overset{Q}{\wedge} X'' \overset{R}{\wedge} X'''$; in other words, the given projectivity maps t to $\frac{1}{t}$. It can be checked that it also maps 0 to ∞ and ∞ to 0. □

Consequently, $\mathrm{PGL}_2(F) \subset \mathrm{PJ}(\ell_\infty)$, if these two groups are considered as subsets of $\mathrm{Sym}(\ell_\infty)$.

In fact, we can also prove that $\mathrm{PGL}_2(F) = \mathrm{PJ}(\ell_\infty)$, but this will require us to develop the concept of a linear coordinate in more detail. Recall that the ratio

$$t = \frac{x}{y}$$

was defined to be a linear coordinate on ℓ_∞ and it provided a bijection between ℓ_∞ and $F \cup \{\infty\}$. The numerator and denominator of this ratio are not well-defined on ℓ_∞ as a given point $P = [x_0 : y_0 : 0]$ on ℓ_∞ can also be represented as $P = [\lambda x_0 : \lambda y_0 : 0]$ for any $\lambda \in F^*$. However, the ratio $\frac{x}{y}$ itself *is* well-defined. Note also that x and y *are* well-defined on the 2-dimensional subspace $V \subset F^3$ that corresponds to ℓ_∞ by Remark 1.2.8, that they are linear forms on V, and that they are linearly independent. It is precisely these properties that we generalize to define a linear coordinate on a line in $\mathbb{P}^2(F)$.

Definition 1.7.18 Let $\ell \subset \mathbb{P}^2(F)$ be a line and V the 2-dimensional subspace of F^3 that corresponds to ℓ by Remark 1.2.8. A *linear coordinate* on ℓ is a function $t : \ell \to F \cup \{\infty\}$ of the form

$$P = [\mathbf{v}] \mapsto \frac{\phi(\mathbf{v})}{\psi(\mathbf{v})},$$

where ϕ and ψ are two linearly independent elements of the dual vector space $V^\vee = \mathrm{Hom}_F(V, F)$.

A linear coordinate on a line $\ell \subset \mathbb{P}^2(F)$ is well-defined since

$$\frac{\phi(\lambda\mathbf{v})}{\psi(\lambda\mathbf{v})} = \frac{\lambda\phi(\mathbf{v})}{\lambda\psi(\mathbf{v})} = \frac{\phi(\mathbf{v})}{\psi(\mathbf{v})}$$

for $\lambda \in F^*$ and $\mathbf{v} \in V$.

Proposition 1.7.19 *A linear coordinate on a line $\ell \subset \mathbb{P}^2(F)$ provides a bijection between ℓ and $F \cup \{\infty\}$.*

Proof Let $V \subset F^3$ be the 2-dimensional subspace that corresponds to ℓ, and let a linear coordinate t on ℓ be defined by two linearly independent elements $\phi, \psi \in V^\vee$ as in Definition 1.7.18. Note that since ϕ and ψ are nonzero elements of $V^\vee = \mathrm{Hom}_F(V, F)$, they are surjective and have 1-dimensional kernels. Also, since ϕ and ψ are linearly independent, it can easily be seen that they cannot have the same kernel, since otherwise they would be scalar multiples of each other.

We first prove surjectivity. Since ψ has a nontrivial kernel, there exists a nonzero vector $\mathbf{v} \in V$ such that $\psi(\mathbf{v}) = 0$; we then have $t([\mathbf{v}]) = \infty$. Next, given $k \in F$, consider $\phi - k\psi \in V^\vee$. There exists a nonzero vector $\mathbf{v} \in V$ such that $(\phi - k\psi)(\mathbf{v}) = 0$. Note that $\psi(\mathbf{v})$ is nonzero since, if it were 0, then $\phi(\mathbf{v})$ would also be 0 and ϕ and ψ would have the same kernel, which is impossible. Hence, $t([\mathbf{v}]) = k$.

For injectivity, suppose there exist two linearly independent vectors $\mathbf{v}, \mathbf{w} \in V$ such that

$$\frac{\phi(\mathbf{v})}{\psi(\mathbf{v})} = \frac{\phi(\mathbf{w})}{\psi(\mathbf{w})}.$$

If both ratios are equal to ∞, then $\mathbf{v}$ and $\mathbf{w}$ both lie in the kernel of ψ, which is impossible since the kernel of ψ is 1-dimensional. Suppose now that

$$\frac{\phi(\mathbf{v})}{\psi(\mathbf{v})} = \frac{\phi(\mathbf{w})}{\psi(\mathbf{w})} = k,$$

for some $k \in F$. Then since ϕ and ψ are linearly independent, the element $\phi - k\psi \in V^\vee$ is nonzero and therefore has 1-dimensional kernel, and it has both $\mathbf{v}$ and $\mathbf{w}$ in its kernel, a contradiction. □

Proposition 1.7.20 *Let $\ell \subset \mathbb{P}^2(F)$ be a line, and $A, B, C \in \ell$ three distinct points. Then there exists a unique linear coordinate t on ℓ such that $t(A) = 0$, $t(B) = 1$, $t(C) = \infty$.*

Proof Let $V \subset F^3$ be the 2-dimensional vector subspace corresponding to ℓ. We choose nonzero vectors $\mathbf{u}, \mathbf{v} \in V$ such that $A = [\mathbf{u}]$ and $B = [\mathbf{v}]$. Since A and B are distinct, $\mathbf{u}$ and $\mathbf{v}$ are linearly independent in V. Since C is different from A and B, we can write $\mathbf{w}$ as a linear combination of $\mathbf{u}$ and $\mathbf{v}$ with nonzero coefficients; after a suitable scaling, we can also assume that $\mathbf{w} = \mathbf{u} + \mathbf{v}$ without loss of generality.

We now have to find two linearly independent elements $\phi, \psi \in V^\vee$ such that

$$\frac{\phi(\mathbf{u})}{\psi(\mathbf{u})} = 0,$$

$$\frac{\phi(\mathbf{v})}{\psi(\mathbf{v})} = 1,$$
$$\frac{\phi(\mathbf{w})}{\psi(\mathbf{w})} = \frac{\phi(\mathbf{u}) + \phi(\mathbf{v})}{\psi(\mathbf{u}) + \psi(\mathbf{v})} = \infty.$$

The first condition imposes $\phi(\mathbf{u}) = 0$; the second condition imposes $\phi(\mathbf{v}) = \psi(\mathbf{v})$; the third condition imposes $\psi(\mathbf{u}) = -\psi(\mathbf{v})$ while $\phi(\mathbf{u}) + \phi(\mathbf{v}) \neq 0$. Let now $k \in F$ be an arbitrary nonzero element, and define $\phi, \psi \in V^\vee$ by

$$\phi(\mathbf{u}) = 0,$$
$$\phi(\mathbf{v}) = k,$$
$$\psi(\mathbf{u}) = -k,$$
$$\psi(\mathbf{v}) = k.$$

It is clear that the linear coordinate t on ℓ defined by

$$t([\mathbf{v}]) = \frac{\phi(\mathbf{v})}{\psi(\mathbf{v})}$$

satisfies $t(A) = 0$, $t(B) = 1$, $t(C) = \infty$, and it is unique with these properties. □

Proposition 1.7.21 *Let $\ell \subset \mathbb{P}^2(F)$ be a line, and let t and u be two linear coordinates on ℓ. Then there exist $a, b, c, d \in F$, with $ad - bc \neq 0$, such that*

$$t(P) = \frac{au(P) + b}{cu(P) + d}$$

for all $P \in \ell$.

Proof Let $V \subset F^3$ be the 2-dimensional vector subspace corresponding to ℓ. Let t be defined by two linearly independent elements $\phi_1, \psi_1 \in V^\vee$ as in Definition 1.7.18, and likewise let u be defined by two linearly independent elements $\phi_2, \psi_2 \in V^\vee$. Since $V^\vee$ is 2-dimensional, there exist $a, b, c, d \in F$, with $ad - bc \neq 0$, such that

$$\phi_1 = a\phi_2 + b\psi_2,$$
$$\psi_1 = c\phi_2 + d\psi_2.$$

These equations imply the desired result. □

Let $A \in \mathrm{PGL}_3(F)$, $\ell \subset \mathbb{P}^2(F)$ a line, and t a linear coordinate on ℓ. Suppose t is defined by two linearly independent elements $\phi, \psi \in V^\vee$, where $V \subset F^3$ is the 2-dimensional vector subspace corresponding to ℓ. Then $\ell' = \phi_A(\ell)$ is a line

in $\mathbb{P}^2(F)$ and the 2-dimensional vector subspace $V' \subset F^3$ corresponding to ℓ' is simply $V' = A(V)$. We obtain two linear maps $\phi A^{-1} : V' \to F$ and $\psi A^{-1} : V' \to F$. It is clear that these two linear maps are linearly independent as elements of $(V')^\vee$ and therefore define a linear coordinate on ℓ', which is denoted by t^A and $t^A(\phi_A(P)) = t(P)$ for all $P \in \ell$.

We can now use the properties of linear coordinates developed so far to prove the following theorem with important corollaries.

Theorem 1.7.22 *Given two lines $\ell_1, \ell_2 \subset \mathbb{P}^2(F)$ and a point P not on either of them, and linear coordinates t_1 and t_2 on ℓ_1 and ℓ_2, respectively, the perspectivity from ℓ_1 to ℓ_2 with center P can be written as a fractional linear transformation in t_1.*

Proof We first prove a special case. Suppose $\ell_1 : z = 0$, $\ell_2 : y = 0$, and $P = [1 : 1 : 1]$. On ℓ_1, let $t_1 = \frac{x}{y}$; on ℓ_2, let $t_2 = \frac{x}{z}$. Given $t_1 \in F$ (we leave the case $t_1 = \infty$ to the reader), the corresponding point on ℓ_1 is $X = [t_1 : 1 : 0]$. We compute $PX : -x + t_1 y + (1 - t_1)z = 0$ and $X' = PX \cdot \ell_2 = [1 - t_1 : 0 : 1]$. Hence, the perspectivity from ℓ_1 to ℓ_2 with center P maps t_1 to $t_2 = 1 - t_1$, a fractional linear transformation.

For the general case, suppose that we are given $\ell_1, \ell_2 \subset \mathbb{P}^2(F)$ and a point P not on either of them, and linear coordinates t_1 and t_2 on ℓ_1 and ℓ_2, respectively. Using Proposition 1.4.6, we choose $A \in \mathrm{PGL}_3(F)$ such that ϕ_A maps ℓ_1 and ℓ_2 to the lines $z = 0$ and $y = 0$, respectively, and P to $[1 : 1 : 1]$. Let ϕ_A convert t_1 and t_2 to t_1^A and t_2^A, respectively. Converting these linear coordinates to $\frac{x}{y}$ and $\frac{x}{z}$, which is done by fractional linear transformations by Proposition 1.7.21, we reduce the problem to the special case proved above. □

Corollary 1.7.23 *Given two lines $\ell_1, \ell_2 \subset \mathbb{P}^2(F)$ and linear coordinates t_1 and t_2 on ℓ_1 and ℓ_2, respectively, a projectivity from ℓ_1 to ℓ_2 can be written as a fractional linear transformation in t_1.*

Corollary 1.7.24 *The projectivity group $PJ(\ell)$ of a line $\ell \subset \mathbb{P}^2(F)$ is isomorphic to $PGL_2(F)$.*

Corollary 1.7.25 (Fundamental Theorem of Projective Geometry) *Let $\ell, \ell' \subset \mathbb{P}^2(F)$ be two lines, not necessarily distinct, A, B, C distinct points on ℓ, and A', B', C' distinct points on ℓ'. Then there exists a unique projectivity from ℓ to ℓ' that maps A to A', B to B', and C to C'.*

Proof This follows from Corollaries 1.7.16 and 1.7.25. □

The cross-ratio of four distinct points on a line is an important quantity in projective geometry. Until the end of the section, we assume that $\mathrm{char}(F) \neq 2$.

Definition 1.7.26 Given four distinct points $a, b, c, d \in F \cup \{\infty\}$, their *cross-ratio* is defined as

$$(a, b; c, d) = \frac{a-c}{b-c}\frac{b-d}{a-d},$$

with the understanding that if any one of a, b, c, or d is equal to ∞, the factors containing it are left out of the formula.

Remark 1.7.27 An easy way to remember what happens when one of a, b, c, or d is equal to ∞ is to consider the "limit" of the cross-ratio as that element approaches infinity. As an example, suppose $F = \mathbb{R}$ and $c = \infty$. Then it is clear that

$$(a, b; \infty, d) = \frac{b-d}{a-d} = \lim_{c' \to \infty} \frac{a-c'}{b-c'}\frac{b-d}{a-d}.$$

Theorem 1.7.28 (Cf. [6, Theorem 8.5]) *$PGL_2(F)$ consists of precisely those permutations of $F \cup \{\infty\}$ that preserve the cross-ratio.*

Proof Suppose first that $\phi \in \mathrm{Sym}(F \cup \{\infty\})$ preserves the cross-ratio. Let $a = \phi(0)$, $b = \phi(1)$, $c = \phi(\infty)$, and $t' = \phi(t)$ for any $t \in F \cup \{\infty\}$ not equal to 0, 1, ∞. Then, assuming $a, b, c \in F$ (the reader can easily check the remaining cases), we have

$$(0, 1; \infty, t) = (a, b; c, t'),$$

or equivalently,

$$\frac{t-1}{t} = \frac{a-c}{b-c}\frac{b-t'}{a-t'}.$$

Solving this equation for t' yields

$$t' = \frac{c(b-a)t + a(c-b)}{(b-a)t + (c-b)},$$

which is a fractional linear transformation since

$$c(b-a)(c-b) - a(c-b)(b-a) = (c-a)(b-a)(c-b) \neq 0.$$

Let us now consider the other direction. Since $\mathrm{PGL}_2(F)$ is generated by fractional linear transformations of types *(i)*, *(ii)*, *(iii)* by Theorem 1.7.14, it suffices to verify that these three types of fractional linear transformations preserve the cross-ratio. Below, we indicate the cases where the cross-ratios are those of four distinct elements of F and leave the cases where one of them is ∞ to the reader.

Let $t \mapsto t + k$ be a fractional linear transformation of type *(i)*. Then

$$(a+k, b+k; c+k, d+k) = \frac{(a+k)-(c+k)}{(b+k)-(c+k)}\frac{(b+k)-(d+k)}{(a+k)-(d+k)} = \frac{a-c}{b-c}\frac{b-d}{a-d}$$
$$= (a, b; c, d).$$

Let $t \mapsto kt$ with $k \neq 0$ be a fractional linear transformation of type *(ii)*. Then

$$(ka, kb; kc, kd) = \frac{ka-kc}{kb-kc}\frac{kb-kd}{ka-kd} = \frac{a-c}{b-c}\frac{b-d}{a-d} = (a, b; c, d).$$

Finally, let $t \mapsto \frac{1}{t}$ be the unique fractional linear transformation of type *(iii)*. Then

$$(\frac{1}{a}, \frac{1}{b}; \frac{1}{c}, \frac{1}{d}) = \frac{\frac{1}{a}-\frac{1}{c}}{\frac{1}{b}-\frac{1}{c}}\frac{\frac{1}{b}-\frac{1}{d}}{\frac{1}{a}-\frac{1}{d}} = \frac{c-a}{c-b}\frac{d-b}{d-a} = (a, b; c, d).$$

□

Definition 1.7.29 Let $\ell \subset \mathbb{P}^2(F)$ be a line. Given four distinct points $A, B, C, D \in \ell$, their *cross-ratio* is defined as

$$(A, B; C, D) = (t(A), t(B); t(C), t(D)),$$

where t is any linear coordinate on ℓ.

Remark 1.7.30 The cross-ratio of four distinct points on a line is well-defined due to Proposition 1.7.21 and Theorem 1.7.28.

Proposition 1.7.31 *Let $\ell \subset \mathbb{P}^2(F)$ be a line, and $T \in PGL_3(F)$. Given four distinct points $A, B, C, D \in \ell$, we have*

$$(A, B; C, D) = (\phi_T(A), \phi_T(B); \phi_T(C), \phi_T(D)).$$

Proof Let t be a linear coordinate on ℓ. Then we have seen that T transforms t to a linear coordinate t^T on $\phi_T(\ell)$ such that $t^T(\phi_T(P)) = t(P)$ for all $P \in \ell$. Then,

$$(\phi_T(A), \phi_T(B); \phi_T(C), \phi_T(D)) = (t^T(\phi_T(A)), t^T(\phi_T(B)); t^T(\phi_T(C)), t^T(\phi_T(D)))$$
$$= (t(A), t(B); t(C), t(D))$$
$$= (A, B; C, D).$$

□

Remark 1.7.32 In contrast to Proposition 1.7.31, automorphisms of $\mathbb{P}^2(F)$ induced by field automorphisms of F do not preserve the cross-ratio.

Proposition 1.7.33 *Let $\ell \subset \mathbb{P}^2(F)$ be a line and $A, B, C, D \in \ell$ four distinct points. Write $\lambda = (A, B; C, D)$. We have*

$$(C, D; A, B) = \lambda,$$
$$(A, B; D, C) = \frac{1}{\lambda},$$
$$(A, C; B, D) = 1 - \lambda.$$

These formulas can be used to compute the cross-ratio for all 24 permutations of A, B, C, D. Let

$$V = \{1, (12)(34), (13)(24), (14)(23)\} \subset S_4$$

be the *Klein four-group*, here considered as a (normal) subgroup of S_4. It can be checked that any $\sigma \in V$ leaves the cross-ratio of A, B, C, D invariant.

Since $S_4/V \cong S_3$, we can expect no more than six different values for the cross-ratio for the 24 permutations of A, B, C, D. Indeed, exactly six different values (depending on λ) occur:

$$(A, B; D, C) = \frac{1}{\lambda},$$
$$(A, C; B, D) = 1 - \lambda,$$
$$(A, C; D, B) = \frac{1}{1 - \lambda},$$
$$(A, D; B, C) = \frac{\lambda - 1}{\lambda},$$
$$(A, D; C, B) = \frac{\lambda}{\lambda - 1}.$$

Considered as linear fractional transformations, these expressions form a subgroup of $\mathrm{PGL}_2(F)$ isomorphic to S_3.

We finish this section with a discussion of a special kind of projectivity, the involution.

Definition 1.7.34 An *involution* is a projectivity of period 2.

We leave it to the reader to verify that an involution is a nontrivial projectivity that interchanges pairs of points.

Corollary 1.7.24 allows us to classify involutions.

Proposition 1.7.35 *Let $\ell \subset \mathbb{P}^2(F)$ be a line, and let*

$$\sigma = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in PGL_2(F)$$

be a projectivity on ℓ (with respect to a given linear coordinate on ℓ). Then σ is an involution on ℓ if and only if $d = -a$.

Proof We prove only the "only if" part, leaving the "if" part to the reader. Since σ is an involution,

$$\sigma^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2 + bc & b(a+d) \\ c(a+d) & d^2 + bc \end{bmatrix}$$

must be a scalar multiple of the identity. Hence, $a^2 = d^2$ and $d = a$ or $d = -a$. In the second case, the proof is complete.

Suppose $d = a$. Then $a + d = 0$ or $a + d \neq 0$. If $a + d \neq 0$, then $b = c = 0$. In this case, we would have σ equal to a scalar multiple of the identity matrix, which contradicts Definition 1.7.34. If $a + d = 0$, then the proof is complete. □

Proposition 1.7.36 *Any involution has 0 or 2 fixed points.*

Proof Let $\sigma \in \mathrm{PGL}_2(F)$ be an involution on a line ℓ. By Proposition 1.7.35, we can write

$$\sigma = \begin{bmatrix} a & b \\ c & -a \end{bmatrix},$$

where $a^2 + bc \neq 0$. Choosing a linear coordinate t on ℓ, σ can be expressed as a function $F \cup \{\infty\} \to F \cup \{\infty\}$ by the formula

$$\sigma(t) = \frac{at + b}{ct - a}.$$

The fixed points of σ correspond to the solutions of the equation

$$\frac{at + b}{ct - a} = t.$$

We distinguish two cases. If $c = 0$, the solutions to the equation are given by $t = -\frac{b}{2a}$ and $t = \infty$. If $c \neq 0$, the equation takes the form

$$ct^2 - 2at - b = 0,$$

which has nonzero discriminant, and hence has either 0 or 2 solutions. □

Proposition 1.7.37 *Let $\ell \subset \mathbb{P}^2(F)$ be a line, and $A, B \in \ell$ two distinct points. Then there is a unique involution on ℓ with fixed points A and B.*

Proof By Proposition 1.7.20, we can choose a linear coordinate t on ℓ such that A and B correspond to 0 and 1, respectively. Now, we must find an element $\sigma \in \mathrm{PGL}_2(F)$ that can be written as

$$\sigma(t) = \frac{at + b}{ct - a},$$

such that $\sigma(0) = 0$ and $\sigma(1) = 1$. These conditions give $b = 0$ and $2a = c$. Hence,

$$\sigma(t) = \frac{t}{2t - 1}.$$

□

Proposition 1.7.38 *A projectivity on a line ℓ that interchanges two distinct points is an involution.*

Proof By Proposition 1.7.20, we can choose a linear coordinate t on ℓ such that the two distinct points are 0 and 1. A projectivity $\sigma \in \mathrm{PGL}_2(F)$ that interchanges 0 and 1 has the form

$$\sigma(t) = \frac{at + b}{ct + d},$$

with $\sigma(0) = 1$ and $\sigma(1) = 0$. These conditions give $b = d$ and $a + b = 0$. Hence σ is an involution. □

Exercises

1. Prove Propositions 1.7.3, 1.7.6, 1.7.10, and 1.7.11.
2. Prove the assertions in Remark 1.7.13.
3. Prove Corollary 1.7.16.
4. Complete the cases in Theorem 1.7.28.
5. Prove Proposition 1.7.33.
6. Given four distinct points $a, b, c, d \in F \cup \{\infty\}$, prove that the cross-ratio $(a, b; c, d)$ is an element of $F \setminus \{0, 1\}$, and that it can take every value in that set.
7. Given four distinct points $a, b, c, d \in F \cup \{\infty\}$ and a field automorphism $\sigma \in \mathrm{Aut}(F)$, calculate $(\sigma(a), \sigma(b); \sigma(c), \sigma(d))$ in terms of $(a, b; c, d)$. (Assume $\sigma(\infty) = \infty$.) The result shows that the cross-ratio is not preserved by the automorphisms of $\mathbb{P}^2(F)$ induced by field automorphisms of F.
8. A projectivity from a line ℓ onto itself is called *elliptic*, *parabolic*, or *hyperbolic*, depending on the number of its fixed points being 0, 1, or 2, respectively. Prove

that parabolic and hyperbolic projectivities always exist from any line $\ell \subset \mathbb{P}^2(F)$ to itself. Do elliptic projectivities exist, and if so, under what conditions on F?

9. Prove that an involution is a nontrivial projectivity that interchanges pairs of points, that is, a projectivity $\phi : \ell \to \ell$ for some line $\ell \subset \mathbb{P}^2(F)$ such that if $\phi(A) = B$, then $\phi(B) = A$.

1.8 Harmonic Points

Definition 1.8.1 A *complete quadrangle* in $\mathbb{P}^2(F)$ is a collection of four points, no three of which are collinear. The four points are called the *vertices* of the complete quadrangle. The joins of pairs of vertices of a complete quadrangle are six distinct lines; they are called the *sides* of the complete quadrangle. Two sides are called *opposite* if their intersection is not a vertex. The intersections of the three pairs of opposite sides are called the *diagonal points*. If the vertices of the complete quadrangle are denoted by A, B, C, D, the complete quadrangle is denoted by $ABCD$, or by any permutation of these letters (Fig. 1.15).

Dually, we have the following:

Definition 1.8.2 A *complete quadrilateral* in $\mathbb{P}^2(F)$ is a collection of four lines, no three of which are concurrent. The four lines are called the *sides* of the complete quadrilateral. The intersections of pairs of sides of a complete quadrilateral are six distinct points; they are called the *vertices* of the complete quadrilateral. Two vertices are called *opposite* if their join is not a side. The joins of the three pairs of opposite vertices are called the *diagonal lines*. If the sides of the complete quadrilateral are denoted by a, b, c, d, the complete quadrilateral is denoted by $abcd$, or by any permutation of these letters (Fig. 1.16).

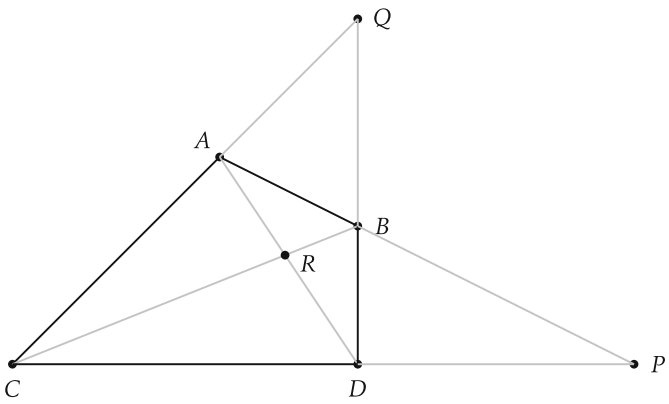


Fig. 1.15 A complete quadrangle and its diagonal points

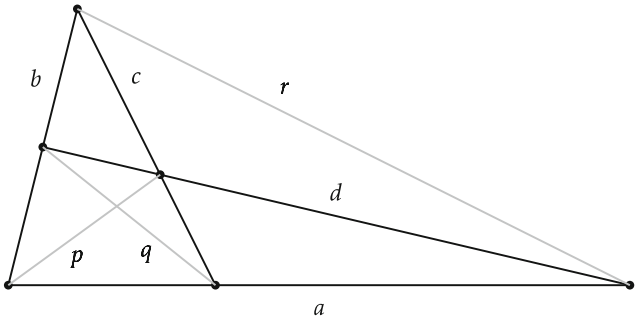


Fig. 1.16 A complete quadrilateral and its diagonal lines

Proposition 1.8.3 *The diagonal points of a complete quadrangle in* $\mathbb{P}^2(F)$ *are collinear if and only if* $char(F) = 2$.

Proof Let the given complete quadrangle be $ABCD$. Using Proposition 1.4.6 we can assume, without loss of generality, that $A = [1 : 0 : 0]$, $B = [0 : 1 : 0]$, $C = [0 : 0 : 1]$, $D = [1 : 1 : 1]$. We then calculate the coordinates of the diagonal points as

$$P = AB \cdot CD = [1 : 1 : 0],$$
$$Q = AC \cdot BD = [1 : 0 : 1],$$
$$R = AD \cdot BC = [0 : 1 : 1].$$

Hence, by Proposition 1.2.12, P, Q, R are collinear if and only if the determinant

$$\begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} = -2$$

vanishes, and that happens if and only if $\text{char}(F) = 2$. □

Dually, we have

Proposition 1.8.4 *The diagonal lines of a complete quadrilateral in* $\mathbb{P}^2(F)$ *are concurrent if and only if* $char(F) = 2$.

In order to define harmonic quadruples, we have to assume that the diagonal points of a complete quadrangle are never collinear; therefore, in the rest of this section we assume that $\text{char}(F) \neq 2$.

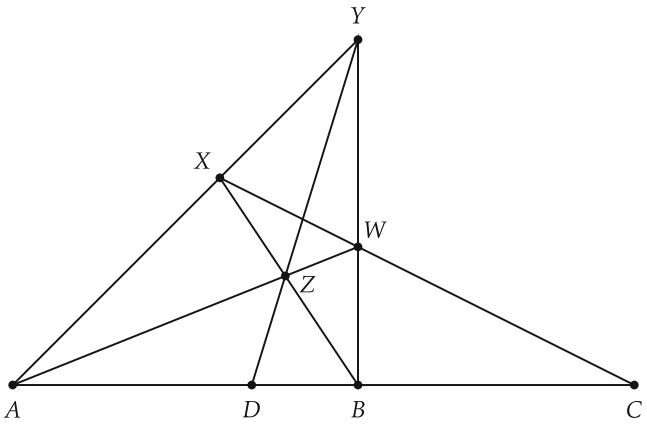


Fig. 1.17 H(AB, CD). $A = XY \cdot ZW$ and $B = XZ \cdot YW$ are two diagonal points of the complete quadrangle $XYZW$. C and D lie on the remaining two sides XW and YZ

Definition 1.8.5 A *harmonic quadruple of points* (or simply a *harmonic quadruple*) is an ordered quadruple of distinct collinear points A, B, C, D such that A and B are diagonal points of a complete quadrangle in $\mathbb{P}^2(F)$, thus lying on four sides of the complete quadrangle, and C and D lie on the remaining two sides of the complete quadrangle. If A, B, C, D form a harmonic quadruple, this is denoted by H(AB, CD). We call D the *harmonic conjugate* of C with respect to A and B.

Remark 1.8.6 In Definition 1.8.5, the ordering of A and B can be reversed, and the ordering of C and D can also be reversed. Thus, the statement H(AB, CD) is equivalent to any one of H(BA, CD), H(AB, DC), H(BA, DC).

For the most part, we shall use *harmonic quadruple* to refer to a *harmonic quadruple of points*.

Proposition 1.8.7 *Given three distinct collinear points $A, B, C \in \mathbb{P}^2(F)$, there exists a unique $D \in ABC$ such that H(AB, CD).*

Proof For the proof, we refer the reader to Fig. 1.17.

We first show the existence of D. Given three distinct collinear points A, B, C, we draw a triangle XYW whose sides XY, YW, XW pass through A, B, C, respectively. We then let $Z = AW \cdot BX$ and $D = YZ \cdot AB$.

We must now prove that point D is independent of the triangle XYW chosen. Imagine a second quadrilateral $X'Y'Z'W'$ (not shown in Fig. 1.17) used to construct the point D. We must show that $YZ \cdot Y'Z' = D$. We shall prove the most general case and leave the remaining cases to the reader; this most general case is when the triangles XZW and $X'Z'W'$ form a dual Desargues configuration and when the triangles XYW and $X'Y'W'$ form another dual Desargues configuration.

The triangles XZW and $X'Z'W'$ are perspective from $\ell = AC$. Hence, they are perspective from a point O by the dual of Desargues' Theorem; in other words, XX', ZZ', WW' pass through O. Similarly, XYW and $X'Y'W'$ are perspective from ℓ. Hence, XX', YY', WW' pass through a point, which must be O.

It follows that the triangles XYZ and $X'Y'Z'$ are perspective from O. Hence, by Desargues' Theorem, they are perspective from a line, which must be ℓ since $XY \cdot X'Y' = A$ and $XZ \cdot X'Z' = B$. This implies that YZ and $Y'Z'$ intersect at a point on ℓ; hence, $YZ \cdot Y'Z' = D$. □

Proposition 1.8.8 *Let $\phi \in Aut(\mathbb{P}^2(F))$ and $H(AB, CD)$ for distinct collinear points A, B, C, D. Then $H(\phi(A)\phi(B), \phi(C)\phi(D))$.*

Proof This is clear from the fact that harmonic quadruples are defined in terms of configurations of lines, and that automorphisms of $\mathbb{P}^2(F)$ preserve such configurations. □

The construction given in the proof of Proposition 1.8.7 is geometrically clear but difficult to do calculations with. Fortunately, there exists a criterion for four distinct collinear points to form a harmonic quadruple in terms of their cross-ratio.

Theorem 1.8.9 *Given four distinct collinear points A, B, C, D, $H(AB, CD)$ if and only if $(A, B; C, D) = -1$.*

Proof We refer the reader to Fig. 1.17 for the proof.

We first consider the special case $A = [1 : 0 : 0]$, $B = [0 : 1 : 0]$, $C = [1 : \lambda : 0]$ for some nonzero λ. Note that A, B, C lie on the line $\ell : z = 0$. If we take $t = \frac{y}{x}$ as a linear coordinate on ℓ, then A, B, C have coordinates $0, \infty, \lambda$, respectively. We will show that the harmonic conjugate D of C with respect to A and B has linear coordinate $-\lambda$ and therefore

$$(A, B; C, D) = (0, \infty; \lambda, -\lambda) = -1.$$

Draw the line $\ell_1 : y = z$ from A and choose the point $Y = [0 : 1 : 1] \in \ell_1$. Draw the line $\ell_2 : y = \lambda x$ from C and let $X = \ell_1 \cdot \ell_2 = [1 : \lambda : \lambda]$. Let $W = BY \cdot \ell_2 = [0 : 0 : 1]$. Then $AW : y = 0$ and $BX : \lambda x = z$. Hence, $Z = AW \cdot BX = [1 : 0 : \lambda]$, $YZ : \lambda x + y - z = 0$, and $D = YZ \cdot \ell = [1 : -\lambda : 0]$, which has linear coordinate $-\lambda$. For the other direction, if $(A, B; C, D) = -1$, the linear coordinate of D is $-\lambda$ and hence $D = [1 : -\lambda : 0]$. In this case, the construction above can be repeated until the equation of YZ has been obtained as $\lambda x + y - z = 0$ and it is clear that $D = YZ \cdot \ell$, hence H(AB, CD).

By Proposition 1.8.8, the general case can be reduced to the special case treated above, by using an automorphism of $\mathbb{P}^2(F)$ that maps A and B to $[1 : 0 : 0]$ and $[0 : 1 : 0]$, respectively. □

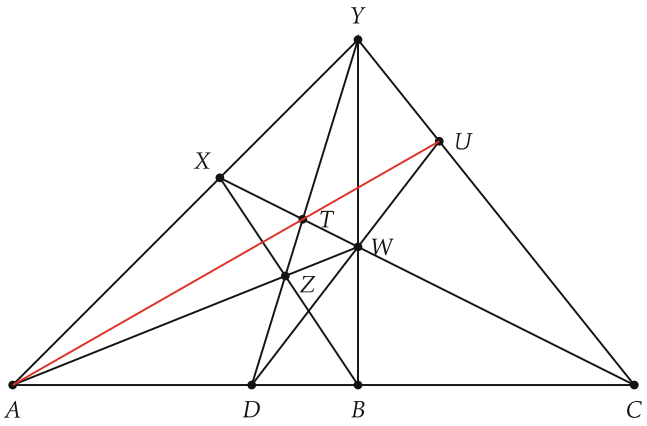


Fig. 1.18 H(AB, CD) if and only if H(CD, AB)

Corollary 1.8.10 *H*(AB, CD) *if and only if H*(CD, AB).

Proof This follows from Theorem 1.8.9 and Proposition 1.7.33. □

Proof We give another proof using Desargues' Theorem. We refer the reader to Fig. 1.18 for the proof.

Let $XYZW$ be a complete quadrangle as in Fig. 1.17. Let $U = CY \cdot DW$ and $T = YD \cdot XW$. Then $YTUW$ is a complete quadrangle with two diagonal points C and D, and YW intersects AC at B.

We claim that UT intersects AC at A. The triangles YWU and XZT are perspective from the line AC. Hence, by the dual of Desargues' Theorem, they are perspective from a point, which must be A since $XY \cdot WZ = A$. Hence, UT also passes through A. □

Corollary 1.8.11 *A projectivity takes harmonic quadruples to harmonic quadruples.*

Proof This follows from Theorem 1.8.9, Corollary 1.7.24, Theorem 1.7.28. □

Corollary 1.8.12 *H*(AB, CD) *is equivalent to any of H*(BA, CD), *H*(AB, DC), *H*(BA, DC), *H*(CD, AB), *H*(DC, AB), *H*(CD, BA), *and H*(DC, BA).

We finish this section with a brief discussion of the dual concept to that of a harmonic quadruple.

Definition 1.8.13 A *harmonic quadruple of lines* is an ordered quadruple of distinct concurrent lines a, b, c, d such that a and b are diagonal lines of a complete quadrilateral in $\mathbb{P}^2(F)$, thus passing through four vertices of the complete quadrilateral,

and c and d pass through the remaining two vertices of the complete quadrilateral. If a, b, c, d form a harmonic quadruple of lines, this is denoted by $\mathrm{H}(ab, cd)$. We call d the *harmonic conjugate* of c with respect to a and b.

It is clear that a harmonic quadruple of lines in $\mathbb{P}^2(F)$ corresponds to a harmonic quadruple of points in the dual projective plane $(\mathbb{P}^2(F))^*$. Therefore, checking whether four concurrent lines form a harmonic quadruple of lines is, in principle, the same as checking whether four collinear points form a harmonic quadruple of points. The major tool to do that is the cross-ratio, but first we must extend the definition to a concurrent quadruple of lines.

Definition 1.8.14 Let $P \in \mathbb{P}^2(F)$ be a point. Given four distinct lines a, b, c, d passing through P, their *cross-ratio* is defined as $(A, B; C, D)$, where A, B, C, D are the points in $(\mathbb{P}^2(F))^*$ that correspond to a, b, c, d via the correspondence outlined in Sect. 1.3.

With this definition, the proof of the following theorem is trivial:

Theorem 1.8.15 *Given four distinct concurrent lines a, b, c, d, $\mathrm{H}(ab, cd)$ if and only if $(a, b; c, d) = -1$.*

The following proposition establishes a relation between the cross-ratio of four concurrent lines and the cross-ratio of four collinear points obtained as the intersection of the four lines with another line.

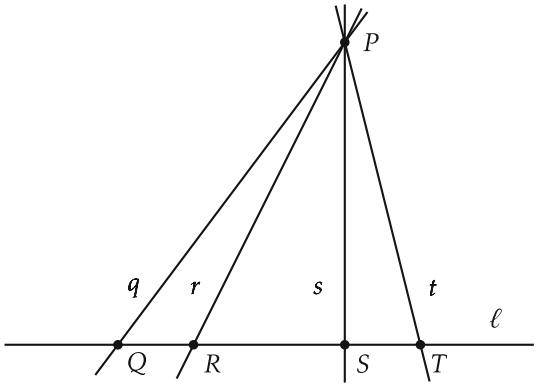
Proposition 1.8.16 *Let q, r, s, t be four distinct concurrent lines in $\mathbb{P}^2(F)$, and let P be the point through which they all pass. Let ℓ be a line that does not pass through P, and let $Q = q \cdot \ell$, $R = r \cdot \ell$, $S = s \cdot \ell$, $T = t \cdot \ell$. Then $(q, r; s, t) = (Q, R; S, T)$ (Fig. 1.19).*

Proof Let Q', R', S', T' be the points in $(\mathbb{P}^2(F))^*$ that correspond to q, r, s, t via the correspondence outlined in 1.3. Since the cross-ratio is invariant under $\mathrm{PGL}_3(F)$ by Proposition 1.7.31 we may assume, without loss of generality, that Q', R', S', T' lie on the line $c = 0$. Let

$$\begin{aligned} Q' &= [\alpha : 1 : 0], \\ R' &= [\beta : 1 : 0], \\ S' &= [\gamma : 1 : 0], \\ T' &= [\delta : 1 : 0], \end{aligned}$$

where $\alpha, \beta, \gamma, \delta \in F$. (We leave the case where one of Q', R', S', T' is $[1 : 0 : 0]$ to the reader.) Choosing $\frac{a}{b}$ as a linear coordinate on the line $c = 0$, we have

Fig. 1.19 Proposition 1.8.16



$$(q, r; s, t) = (Q', R'; S', T') = (\alpha, \beta; \gamma, \delta).$$

The equations of the lines q, r, s, t are then

$$\begin{aligned} q &: \alpha x + y = 0, \\ r &: \beta x + y = 0, \\ s &: \gamma x + y = 0, \\ t &: \delta x + y = 0. \end{aligned}$$

Note that if we choose two different lines for ℓ, the corresponding quadruples of four points are related by a perspectivity. Hence, by Theorems 1.7.22 and 1.7.28, we are free to choose ℓ as we wish; we choose $\ell : z = 0$ for simplicity. Then,

$$\begin{aligned} Q &= [1 : -\alpha : 0], \\ R &= [1 : -\beta : 0], \\ S &= [1 : -\gamma : 0], \\ T &= [1 : -\delta : 0]. \end{aligned}$$

If we choose $\frac{-y}{x}$ as a linear coordinate on ℓ, we have

$$(Q, R; S, T) = (\alpha, \beta; \gamma, \delta).$$

□

Corollary 1.8.17 *Under the hypotheses of Proposition 1.8.16, q, r, s, t form a harmonic quadruple of lines if and only if Q, R, S, T form a harmonic quadruple of points (Fig. 1.19).*

Proof This follows from Theorems 1.8.9 and 1.8.15, and Proposition 1.8.16. □

Exercises

1. Prove Proposition 1.8.4.
2. Complete the proof of Proposition 1.8.7 by examining the cases where XZW and $X'Z'W'$ fail to form a dual Desargues configuration or the triangles XYW and $X'Y'W'$ fail to form a dual Desargues configuration.
3. In Corollary 1.8.12, we gave a total of eight harmonic quadruples for eight permutations of the letters A, B, C, D. As there are a total of 24 such permutations, that leaves 16 that are unaccounted for. Investigate whether these 16 remaining permutations form harmonic quadruples.
4. Suppose that $\mathrm{H}(AB, CD)$. Show that there exists a projectivity on $\ell = AB$ such that $ABCD \,\overline{\wedge}\, BACD$. Show also that the projectivity that you found is a hyperbolic involution. (See Exercise 1.7.8 for the definitions of a hyperbolic projectivity.)

The Synthetic Approach: Axioms for Projective Planes

2

Abstract

Projective geometry is developed axiomatically. Axioms for affine and projective planes are given. Duality is investigated and Desargues' Theorem proved for projective planes that arise as planes in a projective 3-space. Pappus' Theorem and projective planes over division rings are introduced. The last section is dedicated to proving that every projective plane where Desargues' Theorem holds is a projective plane over a uniquely determined division ring and that this division ring is commutative if Pappus' Theorem holds as well.

In Chap. 1, we developed a substantial amount of projective geometry via the analytic approach. In this chapter, we start over, this time developing projective geometry out of certain axioms. In other words, we develop projective geometry from an axiomatic point of view.

We start by defining axioms for affine and projective planes in Sect. 2.1. Duality in a projective plane is investigated next in Sect. 2.2 via the four axioms **P1***–**P4*** that are duals to **P1**–**P4**. We then define the dual of a given projective plane and prove that the dual of any proposition that holds in the original projective plane holds in the dual of that projective plane (Definition 2.2.3 and Proposition 2.2.4).

We treat Desargues' Theorem in a projective plane in Sect. 2.3. We first define projective 3-spaces, and we prove that Desargues' Theorem holds in a projective plane that is obtained as a plane in projective 3-space (Theorem 2.3.12 and Corollary 2.3.13). We finish Sect. 2.3 with a discussion of the Moulton plane, as an example of a projective plane where Desargues' Theorem does not hold, another fundamental difference from the case of projective planes over fields. Pappus' Theorem is treated in Sect. 2.4. We also introduce here, for the first time, projective planes over division rings, as these shall be important in Sect. 2.5, where the main results on coordinatization of projective planes are proved. These are, first, that a projective plane satisfying Desargues' Theorem (**P5**) is isomorphic to $\mathbb{P}^2(D)$ for a

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E. Coşkun, *Projective Geometry*, Compact Textbooks in Mathematics,
https://doi.org/10.1007/978-3-032-02196-0_2

uniquely determined division ring D, and if Pappus' Theorem also holds (**P6**), then D is a field (Theorems 2.5.26 and 2.5.29). This brings us to a full circle, where we started in Chap. 1.

2.1 Axioms for Affine and Projective Planes

In Sect. 1.1, we defined affine planes over a field, and showed in Theorem 1.1.12 that they satisfy the three properties **A1**–**A3**. In fact, it is possible to take just those three properties as the basis for the definition of an affine plane. Hence, the affine plane $\mathbb{A}^2(F)$ over a field F provides our first examples of affine planes, defined via **A1**–**A3**.

Definition 2.1.1 An *affine plane* is a nonempty set $\mathbb{A}^2$, whose elements are called *points*, with some of its subsets called *lines*, satisfying the axioms **A1**–**A3** listed below.

A1. Given two distinct points, there exists a unique line that passes through both.
A2. Given a line ℓ and a point P, there is a unique line that passes through P and is parallel to ℓ.
A3. There exist three noncollinear points.

Remark 2.1.2 **A3** implies that $\mathbb{A}^2$ is not equal to any line in it.

Proposition 2.1.3 *Two lines in $\mathbb{A}^2$ that are not parallel intersect in a unique point.*

Proof Suppose that ℓ_1 and ℓ_2 are two lines that intersect in two distinct points P and Q. By **A1**, $\ell_1 = \ell_2$ and hence $\ell_1 \parallel \ell_2$, a contradiction. □

Proposition 2.1.4 *Parallelism is an equivalence relation on the set of lines in $\mathbb{A}^2$.*

Proof Reflexivity and symmetry follow from Definition 1.1.6.

For transitivity, take three lines ℓ_i, $i = 1, 2, 3$, such that $\ell_1 \parallel \ell_2$ and $\ell_2 \parallel \ell_3$. If $\ell_1 = \ell_3$, the proof is complete. Otherwise, suppose that $\ell_1 \nparallel \ell_3$ and let $\ell_1 \cdot \ell_3 = P$. But then ℓ_1 and ℓ_3 are two distinct lines, both passing through P and parallel to ℓ_2, contradicting **A2**. □

Is any of the axioms **A1**–**A3** redundant? That is, can one of the axioms be deduced from the others? It turns out this is not the case, but in order to prove it, one must exhibit, for any of the axioms, a set with some of its subsets designated as lines, that satisfy all axioms except the one chosen. Table 2.1 shows examples for all three axioms for the affine plane.

The following proposition collects some simple results about the number of points and lines in an affine plane. We leave the proof to the exercises.

Table 2.1 None of the axioms **A1**–**A3** is redundant

Set X	**A1**	**A2**	**A3**
$X = \{1\}$, the only line is $\{1\}$	+	+	−
$X = \{1, 2, 3\}$, lines are $\{1, 2\}$, $\{2, 3\}$, $\{1, 3\}$	+	−	+
$X = \mathbb{A}^2(\mathbb{R})$, lines are vertical Euclidean lines	−	+	+

Proposition 2.1.5 *Let $\mathbb{A}^2$ be an affine plane.*

(i) There exist at least four points in $\mathbb{A}^2$.
(ii) There exist at least three lines in $\mathbb{A}^2$, none of which is parallel to any other.
(iii) Given two lines $\ell, \ell' \subset \mathbb{A}^2$, there is a bijection between the points of ℓ and the points of ℓ'.
(iv) There exist at least two points on a given line in $\mathbb{A}^2$.
(v) $\mathbb{A}^2$ is finite if and only if the number of points on any line is finite. In this case, if the number of points on any line is N, the number of points in $\mathbb{A}^2$ is N^2.

In Sect. 1.2, we saw that there is a one-to-one correspondence between ideal points and points on $\ell_\infty \subset \mathbb{P}^2(F)$, for a field F. In the synthetic approach, this correspondence is taken as the basis for the construction of a projective plane from an affine plane. However, before we can do this, we must define a projective plane.

Definition 2.1.6 A *projective plane* is a nonempty set $\mathbb{P}^2$, whose elements are called *points*, with some of its subsets called *lines*, satisfying the properties **P1**–**P4** listed below.

P1. Given two distinct points, there exists a unique line that passes through both.
P2. Two distinct lines intersect in a unique point.
P3. There exist three noncollinear points.
P4. There exist at least three points on any line.

By Theorem 1.2.17, the projective plane $\mathbb{P}^2(F)$ over a field F is a projective plane.

Remark 2.1.7 **P3** implies that $\mathbb{P}^2$ is not equal to any line in it.

Projective planes can be obtained from affine planes, and vice versa, as Theorems 2.1.8 and 2.1.9 show.

Theorem 2.1.8 *Let $\mathbb{A}^2$ be an affine plane, and let ℓ_∞ be the set of its ideal points. Define $\mathbb{P}^2 = \mathbb{A}^2 \cup \ell_\infty$. Define a line in $\mathbb{P}^2$ to be an affine line together with the corresponding ideal point, or ℓ_∞. Then $\mathbb{P}^2$ is a projective plane, called the* completion *of the affine plane $\mathbb{A}^2$.*

Proof We must prove that **P1**–**P4** are satisfied.

P1. Let $P, Q \in \mathbb{P}^2$ be two distinct points. We distinguish three cases.
- (i) $P, Q \in \mathbb{A}^2$. Then the affine line PQ, together with its ideal point, is the unique projective line that passes through P and Q.
- (ii) Exactly one of P and Q lies on ℓ_∞. Assume without loss of generality that $P \in \mathbb{A}^2$ and $Q \in \ell_\infty$. By **A2**, there exists a unique affine line ℓ that passes through P and is parallel to the lines in the parallelism class Q. Then the affine line ℓ, together with its ideal point Q, is the unique projective line through P and Q.
- (iii) $P, Q \in \ell_\infty$. Then ℓ_∞ is the unique projective line that passes through P and Q.

P2. Let $\ell_1, \ell_2 \subset \mathbb{P}^2$ be two distinct lines. We distinguish three cases.
- (i) ℓ_1 and ℓ_2 are obtained from two lines in $\mathbb{A}^2$ and intersect at a point in $\mathbb{A}^2$. In this case, the proof of **P2** is complete since the ideal points of ℓ_1 and ℓ_2 are different; therefore, ℓ_1 and ℓ_2 cannot intersect at a point of ℓ_∞.
- (ii) ℓ_1 and ℓ_2 are obtained from two lines in $\mathbb{A}^2$, but they do not intersect at a point in $\mathbb{A}^2$. Then $\ell_1 \cdot \ell_2 = P$, where $P \in \ell_\infty$ is the ideal point corresponding to the parallelism class of ℓ_1 and ℓ_2.
- (iii) Exactly one of ℓ_1 and ℓ_2 is obtained from a line in $\mathbb{A}^2$. Assume without loss of generality that ℓ_1 is obtained from a line in $\mathbb{A}^2$ and $\ell_2 = \ell_\infty$. Then $\ell_1 \cdot \ell_2 = P$, where $P \in \ell_\infty$ is the ideal point corresponding to the parallelism class of ℓ_1.

P3. This follows from **A3**.

P4. Let $\ell \subset \mathbb{P}^2$. We distinguish two cases.
- (i) ℓ is obtained from a line in $\mathbb{A}^2$. By Proposition 2.1.5 (iv), there exist two points on the affine line. Together with the ideal point of this affine line, we obtain three points on ℓ.
- (ii) $\ell = \ell_\infty$. By Proposition 2.1.5 (ii), there exist three lines in $\mathbb{A}^2$, none of which are parallel to each other. The ideal points of these lines provide three points on ℓ.

□

Theorem 2.1.9 *Let $\mathbb{P}^2$ be a projective plane, and $\ell_0 \subset \mathbb{P}^2$ be a line. Then the nonempty set $\mathbb{P}^2 \setminus \ell_0$, with lines defined as the nonempty intersections of lines in $\mathbb{P}^2$ with this set, is an affine plane.*

Proof We must prove that **A1**–**A3** are satisfied.

A1. Let $P, Q \in \mathbb{P}^2 \setminus \ell_0$ be two distinct points. By **P1**, there exists a unique line $\ell \subset \mathbb{P}^2$ passing through both. Then $\ell \cap (\mathbb{P}^2 \setminus \ell_0)$ is the unique line in $\mathbb{P}^2 \setminus \ell_0$ passing through P and Q.

A2. Let $\ell \subset \mathbb{P}^2 \setminus \ell_0$ be a line obtained by restricting a line in $\mathbb{P}^2$, which we also denote by ℓ, and let $P \in \mathbb{P}^2 \setminus (\ell_0)$. Let $Q = \ell_0 \cdot \ell \in \mathbb{P}^2$. The line $PQ \cap (\mathbb{P}^2 \setminus \ell_0)$

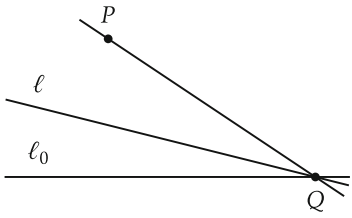


Fig. 2.1 The proof of **A2**, Theorem 2.1.9

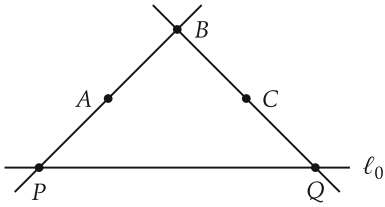


Fig. 2.2 The proof of **A3**, Theorem 2.1.9

passes through P and either does not intersect $\ell \subset \mathbb{P}^2 \setminus \ell_0$ or is equal to it; it is clear that it is the unique such line (Fig. 2.1).

A3. Note that **P3**, which guarantees the existence of three noncollinear points in $\mathbb{P}^2$, is not of help here since we do not know whether these points lie on ℓ_0. We follow a different approach. By **P4** and Remark 2.1.7, we choose points $P, Q \in \ell_0$ and $A \in \mathbb{P}^2 \setminus \ell_0$. By **P4**, we choose a point $B \in PA$ distinct from A or P. Note that $B \in \mathbb{P}^2 \setminus \ell_0$ by **P2**. Finally, we choose $C \in BQ$ distinct from B or Q, by **P4**. Note again that $C \in \mathbb{P}^2 \setminus \ell_0$ by **P2**. It can now be seen that $A, B, C \in \mathbb{P}^2 \setminus \ell_0$ are three noncollinear points (Fig. 2.2).

□

The following proposition collects some simple results about the number of points and lines in a projective plane. We leave the proof to the exercises.

Proposition 2.1.10 *Let $\mathbb{P}^2$ be a given projective plane.*

(i) There exist at least seven points in $\mathbb{P}^2$.
(ii) Given two lines $\ell, \ell' \subset \mathbb{P}^2$, there is a bijection between the points of ℓ and the points of ℓ'.
(iii) $\mathbb{P}^2$ is finite if and only if the number of points on any line is finite. In this case, if the number of points on any line is $N + 1$, the number of points in $\mathbb{P}^2$ is $N^2 + N + 1$.

Definition 2.1.11 Let $\mathbb{A}^2$ be a finite affine plane. By Proposition 2.1.5 (v), the number of points in $\mathbb{A}^2$ is N^2, where $N \geq 2$ is the number of points on any line. N is called the *order* of $\mathbb{A}^2$.

Let $\mathbb{P}^2$ be a finite projective plane. By Proposition 2.1.10 (iii), the number of points in $\mathbb{P}^2$ is $N^2 + N + 1$, where $N \geq 2$ is one less than the number of points on any line. N is called the *order* of $\mathbb{P}^2$.

Table 2.2 None of the axioms **P1**–**P4** is redundant

Set X	**P1**	**P2**	**P3**	**P4**
$X = \{1, 2, 3\}$, lines are $\{1, 2\}$, $\{2, 3\}$, $\{1, 3\}$	+	+	+	−
X is any finite set with 3 elements, the only line is X itself	+	+	−	+
$X = \mathbb{A}^2(F)$ for a field F, lines are affine lines	+	−	+	+
$X = \mathbb{P}^2(\mathbb{R})$, lines are subsets defined by equations of the form $ax + by + cz = 0$, where $a, b, c \in \mathbb{Q}$ (note the field!), not all zero	−	+	+	+

Table 2.2 exhibits examples, for all four axioms for the projective plane, that satisfy all axioms but the one chosen.

Exercises

1. Prove Propositions 2.1.5 and 2.1.10.
2. Prove that in an affine plane of order N, the number of lines that pass through any point is $N + 1$, and the total number of lines is $N^2 + N$.
3. Prove that in a projective plane of order N, the number of lines that pass through any point is $N + 1$, and the total number of lines is $N^2 + N + 1$.
4. Prove that the completion of an affine plane of order N is a projective plane of order N.
5. Given a projective plane $\mathbb{P}^2$ and a line $\ell \subset \mathbb{P}^2$, $\mathbb{P}^2 \setminus \ell$ is an affine plane by Theorem 2.1.9. Prove that the completion of this affine plane is isomorphic to $\mathbb{P}^2$.
6. Define the concept of an *isomorphism* of affine and projective planes. Prove that any affine plane of order 2 is isomorphic to $\mathbb{A}^2(\mathbb{F}_2)$ and hence any projective plane of order 2 is isomorphic to $\mathbb{P}^2(\mathbb{F}_2)$. Prove that any affine plane of order 3 is isomorphic to $\mathbb{A}^2(\mathbb{F}_3)$ and hence any projective plane of order 3 is isomorphic to $\mathbb{P}^2(\mathbb{F}_3)$.
7. Let X be a set, whose elements are called *points*, with some of its subsets called *lines*, satisfying the properties listed below:
   (i) The number of elements of X is N^2 for some $N \geq 2$.
   (ii) Given two distinct points $P, Q \in X$, there exists a unique line that passes through both.
   (iii) There exist N points on any line.
   We define parallel lines in the same way as in Definition 1.1.6. Prove that X is an affine plane.

2.2 The Dual Projective Plane

Recall the *principle of duality between points and lines* enunciated in Sect. 1.3: if certain words of a definition or proposition, stated for $\mathbb{P}^2(F)$, where F is a field, are changed according to the dictionary

$$\begin{array}{rcl} \text{point} & \longleftrightarrow & \text{line} \\ \text{lies on} & \longleftrightarrow & \text{passes through} \\ \text{collinear} & \longleftrightarrow & \text{concurrent} \\ \text{join} & \longleftrightarrow & \text{intersection,} \end{array}$$

then we obtain another definition or proposition.

Applying the dictionary to the projective plane axioms **P1**–**P4** yields the following[1] where we rephrase **P2** slightly to emphasize the resemblance to **P1**:

P1***.** Given two distinct lines, there exists a unique point that lies on both.
P2***.** Given two distinct points, there exists a unique line that passes through them both.
P3***.** There exist three nonconcurrent lines.
P4***.** There exist at least three lines through any point.

Of course, as already stated in Sect. 1.3, **P1*** is **P2** and **P2*** is **P1**. We cannot equate **P3*** or **P4*** to one of **P1**-**P4**. However, the following proposition implies that there is no difference between using **P1**–**P4** and **P1***–**P4***.

Proposition 2.2.1 (Cf. [6, Proposition 4.1]) *A projective plane satisfies **P1***–**P4***. Conversely, a nonempty set $\mathbb{P}^2$, whose elements are called* points*, with some of its subsets called* lines*, satisfying the properties **P1***–**P4***, is a projective plane if one leaves the definitions of points and lines unchanged.*

Proof For one direction, assume $\mathbb{P}^2$ is a projective plane. The discussion above shows that **P1*** and **P2*** hold in $\mathbb{P}^2$ since they are none other than **P2** and **P1**, respectively. For **P3***, we choose three noncollinear points A, B, C in $\mathbb{P}^2$ by **P3** and note that the lines AB, BC, AC are nonconcurrent. For **P4***, we choose a point P and a line ℓ that does not pass through P. We choose three distinct points $A, B, C \in \ell$ by **P4**. Then the three lines PA, PB, PC through P are distinct.

For the other direction, assume that $\mathbb{P}^2$ is a nonempty set, whose elements are called *points*, with some of its subsets called *lines*, satisfying the properties **P1***–**P4***. Then $\mathbb{P}^2$ obviously satisfies **P1** and **P2**. For **P3**, take three nonconcurrent lines, ℓ_1, ℓ_2, ℓ_3 by **P3***; note that these lines are distinct. The three points $\ell_1 \cdot \ell_2$, $\ell_1 \cdot \ell_3$, $\ell_2 \cdot \ell_3$ are then three noncollinear points. For **P4**, let $\ell \subset \mathbb{P}^2$ be a line and P a point not on ℓ. By **P4***, there exist three lines that pass through P; the intersections of these lines with ℓ, whose existence is guaranteed by **P1***, are then three distinct points on ℓ. □

Corollary 2.2.2 *If **P** is any proposition that can be proved using **P1–P4**, then **P*** can also be proved using **P1–P4**.*

[1] We adopt the convention of using an asterisk to indicate dual statements.

We can use Proposition 2.2.1 to define the *dual projective plane* to $\mathbb{P}^2$.

Definition 2.2.3 Let $\mathbb{P}^2$ be a given projective plane. Let $(\mathbb{P}^2)^*$ be the set of lines in $\mathbb{P}^2$, with a line defined as the pencil of all lines in $\mathbb{P}^2$ through a given point in $\mathbb{P}^2$. Then $(\mathbb{P}^2)^*$, which is a projective plane by Proposition 2.2.1, is called the *dual projective plane* to $\mathbb{P}^2$.

Proposition 2.2.4 *If* $\boldsymbol{P}$ *is any proposition that holds in a projective plane* $\mathbb{P}^2$*, then* $\boldsymbol{P}^*$ *holds in the dual projective plane* $(\mathbb{P}^2)^*$.

A projective plane is called *self-dual* if it is isomorphic as a projective plane to its dual.

Corollary 2.2.5 *If* $\boldsymbol{P}$ *is any proposition that holds in a self-dual projective plane* $\mathbb{P}^2$*, then* $\boldsymbol{P}^*$ *also holds in* $\mathbb{P}^2$.

The reader should be aware that there exist projective planes that are not self-dual. However, the projective plane $\mathbb{P}^2(F)$, where F is a field, is self-dual, as is clear from the discussion in Sect. 1.3. We shall examine the isomorphisms between $\mathbb{P}^2(F)$ and its dual projective plane further in Sect. 3.1.

Exercise

1. Given a projective plane $\mathbb{P}^2$, prove that there exists a natural isomorphism from $\mathbb{P}^2$ to its *double dual* $(\mathbb{P}^2)^{**}$. Therefore, if $\mathbb{P}^2$ is self-dual, so is $(\mathbb{P}^2)^*$.

2.3 Projective 3-Spaces and Desargues' Theorem

Corollary 2.2.2 concerns propositions that can be proved using **P1**–**P4**. This immediately raises the question: are there propositions that cannot be proved using **P1**–**P4**? The answer is "yes," and an example is given by Desargues' Theorem itself; we shall see at the end of this section that there exist projective planes where Desargues' Theorem does not hold.

Are there conditions beside **P1**–**P4** we can impose on a given projective plane $\mathbb{P}^2$ to ensure that Desargues' Theorem holds on it? In fact, if $\mathbb{P}^2$ can be obtained as a plane in projective 3-space, then Desargues' Theorem holds on $\mathbb{P}^2$, as Corollary 2.3.13 shows.

Definition 2.3.1 A *projective 3-space* is a nonempty set $\mathbb{P}^3$, whose elements are called *points*, with some of its subsets called *lines*, and with some others of its subsets called *planes*, satisfying the properties **S1**–**S6** listed below.

S1. Given two distinct points, there exists a unique line that passes through both.

S2. Given three noncollinear points, there exists a unique plane that passes through all three.
S3. A plane and a line that does not lie on the plane intersect in a unique point.
S4. Two distinct planes intersect in a unique line.
S5. There exist four noncoplanar points, no three of which are collinear.
S6. There exist at least three points on any line.

Remark 2.3.2 The axioms listed above imply the following:

(i) $\mathbb{P}^3$ is not equal to any line or plane in it,
(ii) there exist at least 6 different lines in $\mathbb{P}^3$,
(iii) there exist at least 4 different planes in $\mathbb{P}^3$,
(iv) there exist at least three points on any plane in $\mathbb{P}^3$.

Proposition 2.3.3 *We have the following in* $\mathbb{P}^3$*:*

(i) Two distinct lines can intersect in at most one point.
(ii) If two distinct points lie on a plane, then the line passing through these points lies on that plane.
(iii) A line and a point not on the line lie in a unique plane.
(iv) A plane is not a line.
(v) A plane, together with the lines that lie on it, is a projective plane.

Proof *Case (i).* Suppose that two distinct lines ℓ_1 and ℓ_2 intersect in two distinct points P and Q. By **S1**, there exists a unique line that passes through P and Q, which must then be equal to both ℓ_1 and ℓ_2. Hence $\ell_1 = \ell_2$, a contradiction.

Case (ii). Suppose that we are given two distinct points that lie on the plane $\mathcal{P}$, and that the line ℓ passing through these two points does not lie on $\mathcal{P}$. By **S3**, ℓ and $\mathcal{P}$ intersect in a unique point, a contradiction.

Case (iii). Let ℓ be a line and P a point not on ℓ. By **S6**, we can choose two distinct points Q and R on ℓ. **S1** implies that these three points are noncollinear. By **S2**, there exists a unique plane $\mathcal{P}$ that passes through P, Q, R. By (ii), ℓ lies on $\mathcal{P}$. It is clear that $\mathcal{P}$ is the unique plane passing through P and ℓ.

Case (iv). Suppose that a plane $\mathcal{P}$ is also a line. We note that it is sufficient to find another plane $\mathcal{Q}$, such that $\mathcal{P}$ does not lie on $\mathcal{Q}$. For in that case $\mathcal{P} \cap \mathcal{Q}$ is both a unique point by **S3** and a unique line by **S4**, which is a contradiction since a line contains at least three points by **S6**.

Since $\mathcal{P}$ is a line, there exist at least two points A and B that do not lie on $\mathcal{P}$ by **S2** and (i) of Remark 2.3.2. By **S1**, **S6**, and (i), we can choose a point $P \in \mathcal{P}$ such that the three points A, B, P are not collinear. Let $\mathcal{Q}$ be the unique plane through A, B, P, whose existence is guaranteed by **S2**. If $\mathcal{P}$ does not lie on $\mathcal{Q}$, the proof is finished.

On the other hand, if $\mathcal{P}$ lies on $\mathcal{Q}$, by (i) of Remark 2.3.2 again, we can choose a point C outside of $\mathcal{Q}$. Since the line AB lies on $\mathcal{Q}$ by (ii), C is not on this line. Let $\mathcal{R}$ be the unique plane through A, B, C. Then $\mathcal{Q}$ and $\mathcal{R}$ are distinct planes, hence

intersect in a unique line by **S4**; it is clear that this line is simply AB. Since $P \notin AB$, $P \notin \mathcal{R}$ and hence $\mathcal{P}$ does not lie on $\mathcal{R}$.

Case (v). Let the plane be $\mathcal{P}$. By (iv) of Remark 2.3.2, $\mathcal{P}$ contains at least three points, hence is nonempty.

We must now prove that **P1**–**P4** are satisfied.

P1. Let $P, Q \in \mathcal{P}$ be two distinct points. By **S1**, there exists a unique line that passes through P and Q; by (ii), this line lies on $\mathcal{P}$.

P2. Let there be given two distinct lines $\ell_1, \ell_2 \subset \mathcal{P}$. By (i) and **S6**, we can choose two distinct points $P, Q \in \ell_1$ and two distinct points $R, S \in \ell_2$ such that none of the points lies on both ℓ_1 and ℓ_2. Choose a fifth point $T \in \mathbb{P}^3 \setminus \mathcal{P}$, possible by (i) of Remark 2.3.2. Let $\mathcal{Q}$ be the unique plane that passes through the three noncollinear points P, Q, T; and let $\mathcal{R}$ be the unique plane that passes through the three noncollinear points R, S, T.

Note that $\mathcal{P}$ is different from both $\mathcal{Q}$ and $\mathcal{R}$. Hence, $\mathcal{P} \cap \mathcal{Q}$ is a line by **S4**; it is then equal to ℓ_1. Similarly, $\mathcal{P} \cap \mathcal{R} = \ell_2$. It is then clear that $\mathcal{Q}$ and $\mathcal{R}$ are also distinct; hence they intersect in a unique line m by **S4**, and $T \in m$. This implies that m does not lie on $\mathcal{P}$ and, by **S3**, $m \cap \mathcal{P}$ consists of a unique point X. Then $X \in \mathcal{P} \cap \mathcal{Q} = \ell_1$ and $X \in \mathcal{P} \cap \mathcal{R} = \ell_2$ (Fig. 2.3).

P3. Suppose that all groups of three points on $\mathcal{P}$ are collinear. Then $\mathcal{P}$ is a line, contradicting (iv).

P4. This follows from **S6**.

□

Remark 2.3.2 and Proposition 2.3.3 (iv) together imply that points, lines, planes, and $\mathbb{P}^3$ itself are distinct objects.

We leave the proof of the following proposition to the exercises.

Proposition 2.3.4 *Two lines in a projective 3-space intersect if and only if they lie on a plane.*

If two lines ℓ_1 and ℓ_2 in $\mathbb{P}^3$ intersect in a unique point P, we write $\ell_1 \cdot \ell_2 = P$. If a line ℓ and a plane $\mathcal{P}$ in $\mathbb{P}^3$ intersect in a unique point P, we write $\ell \cdot \mathcal{P} = P$. If two planes $\mathcal{P}$ and $\mathcal{Q}$ in $\mathbb{P}^3$ intersect in a unique line ℓ, we write $\mathcal{P} \cdot \mathcal{Q} = \ell$.

The projective 3-space over a field F can be defined in a way similar to that followed in Sect. 1.2.

Definition 2.3.5 The *projective 3-space* $\mathbb{P}^3(F)$ *over* F is defined as the set of 1-dimensional subspaces of the vector space F^4.

Alternatively, given two points $P = (x_1, y_1, z_1, w_1)$, $Q = (x_2, y_2, z_2, w_2) \in F^4 \setminus \{0\}$, we say that P *is equivalent to* Q, denoted by $P \sim Q$, if there exists $\lambda \neq 0$ such that $x_2 = \lambda x_1$, $y_2 = \lambda y_1$, $z_2 = \lambda z_1$, $w_2 = \lambda w_1$.

Proposition 2.3.6 *$P \sim Q$ as defined above is an equivalence relation.*

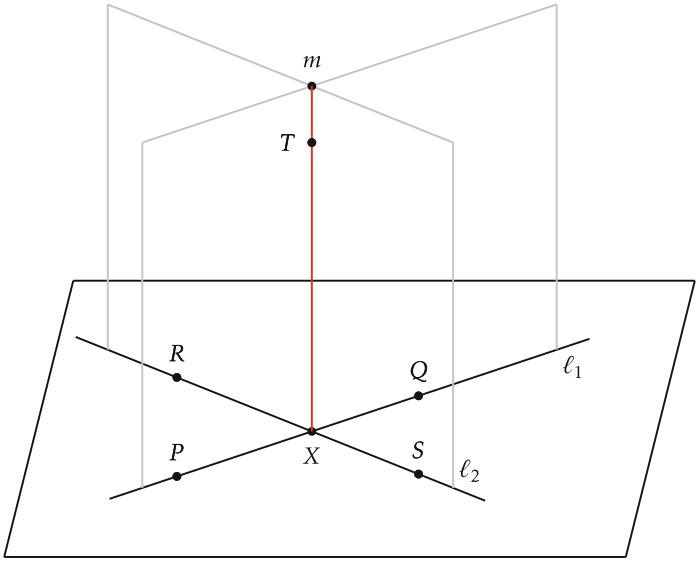


Fig. 2.3 The proof of **P2**, (v), Proposition 2.3.3

Definition 2.3.7 The *projective 3-space* $\mathbb{P}^3(F)$ *over* F is defined as the set of equivalence classes of points in $F^4 \setminus \{0\}$ under the equivalence relation given above. Elements of $\mathbb{P}^3(F)$ are called *points*. A point P in $\mathbb{P}^3(F)$ that is the equivalence class of $(x_0, y_0, z_0, w_0) \in F^4 \setminus \{0\}$ will be denoted by $[x_0 : y_0 : z_0 : w_0]$. The elements $x_0, y_0, z_0, w_0 \in F$ are called the *projective coordinates* of P.

As in the case of $\mathbb{P}^2(F)$, subsets of $\mathbb{P}^3(F)$ can be defined by the vanishing of homogeneous polynomials in the variables x, y, z, w.

Definition 2.3.8 A *plane* in $\mathbb{P}^3(F)$ is a subset defined by an equation of the form $ax + by + cz + dw = 0$, where a, b, c, d are not all zero. A *line* in $\mathbb{P}^3(F)$ is a subset defined by two such equations, $ax + by + cz + dw = 0$ and $ex + fy + gz + hw = 0$, where the matrix

$$\begin{bmatrix} a & b & c & d \\ e & f & g & h \end{bmatrix}$$

has rank 2.

Remark 2.3.9 A plane $\mathcal{P}$ with equation $ax + by + cz + dw = 0$ will be simply denoted as $\mathcal{P} : ax + by + cz + dw = 0$.

Let $\mathcal{P} : ax + by + cz + dw = 0$ be a plane in $\mathbb{P}^3(F)$. The union of all the 1-dimensional subspaces of F^4 corresponding to the points in $\mathcal{P}$ under Definition 2.3.5

forms a 3-dimensional subspace in F^4 and this subspace is also defined by the equation $ax + by + cz + dw = 0$.

Similarly, let ℓ be a line in $\mathbb{P}^3(F)$, defined by $ax+by+cz+dw = 0$ and $ex+fy+gz+hw = 0$. The union of all the 1-dimensional subspaces of F^4 corresponding to the points in ℓ under Definition 2.3.5 forms a 2-dimensional subspace in F^4. This subspace is also defined by $ax + by + cz + dw = 0$ and $ex + fy + gz + hw = 0$.

Proposition 2.3.10 *We have the following criteria to determine when points are collinear or coplanar in* $\mathbb{P}^3(F)$*:*

(i) Three points $P_i = [x_i : y_i : z_i : w_i] \in \mathbb{P}^3(F)$, $i = 1, 2, 3$, *are collinear if and only if the matrix*

$$\begin{bmatrix} x_1 & y_1 & z_1 & w_1 \\ x_2 & y_2 & z_2 & w_2 \\ x_3 & y_3 & z_3 & w_3 \end{bmatrix}$$

has rank 2 or less.

(ii) Four points $P_i = [x_i : y_i : z_i : w_i] \in \mathbb{P}^3(F)$, $i = 1, 2, 3, 4$, *are coplanar if and only if*

$$\begin{vmatrix} x_1 & y_1 & z_1 & w_1 \\ x_2 & y_2 & z_2 & w_2 \\ x_3 & y_3 & z_3 & w_3 \\ x_4 & y_4 & z_4 & w_4 \end{vmatrix} = 0.$$

Proof *Case (i).* The projective coordinates of the points P_1, P_2, P_3 give three vectors $\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3} \in F^4$. The statement that P_1, P_2, P_3 are collinear is equivalent to the statement that $\mathbf{v_1}$, $\mathbf{v_2}$, $\mathbf{v_3}$ are contained in a 2-dimensional subspace of F^4. The result now follows by linear algebra.

Case (ii). The projective coordinates of the points P_1, P_2, P_3, P_4 give four vectors $\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3}, \mathbf{v_4} \in F^4$. The statement that P_1, P_2, P_3, P_4 are coplanar is equivalent to the statement that $\mathbf{v_1}$, $\mathbf{v_2}$, $\mathbf{v_3}$, $\mathbf{v_4}$ are contained in a 3-dimensional subspace of F^4; in other words, that they are linearly dependent. The result now follows by linear algebra. □

Theorem 2.3.11 *Let F be a given field. Then $\mathbb{P}^3(F)$, with planes and lines as defined in Definition 2.3.8, is a projective 3-space.*

Proof We must prove that **S1**–**S6** are satisfied.

S1. Let P and Q be two distinct points. Their projective coordinates give vectors $\mathbf{v}$ and $\mathbf{w}$ in F^4 that are linearly independent. Hence, there exists a unique 2-

dimensional vector subspace of F^4 that contains both, which implies that there exists a unique line in $\mathbb{P}^3(F)$ that passes through P and Q.

S2. Let P, Q, R be three noncollinear points. By Proposition 2.3.10, their projective coordinates give vectors $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ in F^4 that span a 3-dimensional vector subspace, implying that there exists a unique plane in $\mathbb{P}^3(F)$ that passes through P, Q, R.

S3. Let a line ℓ and a plane $\mathcal{P}$ be given such that ℓ does not lie on $\mathcal{P}$; by Remark 2.3.9, they correspond to a 2-dimensional subspace V and a 3-dimensional subspace W of F^4, respectively, such that V is not contained in W. These together span all of F^4 and they intersect in a subspace of dimension 1, which gives a point equal to $\ell \cap \mathcal{P}$.

S4. Let two distinct planes $\mathcal{P}$ and $\mathcal{Q}$ be given; by Remark 2.3.9, they correspond to two distinct 3-dimensional subspaces of F^4. Hence, together they span all of F^4 and intersect in a subspace of dimension 2, which gives a line equal to $\mathcal{P} \cap \mathcal{Q}$.

S5. It suffices to consider the points $[1 : 0 : 0 : 0]$, $[0 : 1 : 0 : 0]$, $[0 : 0 : 1 : 0]$, $[0 : 0 : 0 : 1]$. By Proposition 2.3.10, these points are not coplanar and no three of them are collinear.

S6. A line ℓ corresponds to a 2-dimensional subspace V of F^4. Since it is possible to choose three vectors in V such that no two are scalar multiples of each other, there exist three distinct points on ℓ.

□

Theorem 2.3.12 (Desargues' Theorem in Projective 3-Space, Cf. [6, Theorem 2.1]) *Two triangles in* $\mathbb{P}^3$ *in a Desargues configuration are perspective from a line.*

Proof Let the two triangles be ABC and $A'B'C'$. We refer the reader to Fig. 1.10 in Sect. 1.6. Since each of the triangles lies in a unique plane by **S2**, these two planes could be different or the same. For the proof, we distinguish the two cases.

Case 1. Assume that the triangles ABC and $A'B'C'$ lie in different planes $\mathcal{P}$ and $\mathcal{Q}$, respectively. By **S4**, $\mathcal{P} \cap \mathcal{Q}$ is a line. Using (ii) of Proposition 2.3.3, it can be seen that the two distinct lines AB and $A'B'$ lie in the plane that passes through the three noncollinear points O, A, B. Hence, they intersect in a unique point by (v) of Proposition 2.3.3. Let $P = AB \cdot A'B'$. Similarly, let $Q = AC \cdot A'C'$ and $R = BC \cdot B'C'$. By (ii) of Proposition 2.3.3, P, Q, R lie on both $\mathcal{P}$ and $\mathcal{Q}$; hence they are collinear.

Case 2. Assume that the triangles ABC and $A'B'C'$ lie on the same plane $\mathcal{P}$. We can choose a point $X \in \mathbb{P}^3 \setminus \mathcal{P}$ by (i) of Remark 2.3.2. By **S6**, we can choose a point $D \in XA$ such that $D \neq A$ and $D \neq X$. There exists a unique plane $\mathcal{Q}$ through the three noncollinear points O, X, A by **S2**. Hence, the lines OA and XA lie on $\mathcal{Q}$ by (ii) of Proposition 2.3.3, and D and A' lie on $\mathcal{Q}$. Therefore, the lines OD and XA' lie on $\mathcal{Q}$ again by (ii) of Proposition 2.3.3 and since $\mathcal{Q}$ is a projective plane by (v) of Proposition 2.3.3, OD and XA' intersect in a unique point. Let $D' = OD \cdot XA'$. Then the triangles DBC and $D'B'C'$ form a Desargues

configuration: the line ODD' is not equal to OBB' or OCC' because it does not lie on $\mathcal{P}$, and O, D, D' are three distinct points. It is also clear that the triangles DBC and $D'B'C'$ do not lie on the same plane.

We can therefore use the previous case to conclude that the three points $S = DB \cdot D'B'$, $T = DC \cdot D'C'$, $R = BC \cdot B'C'$ are collinear. Since X and S are distinct (why?), and XS does not lie on $\mathcal{P}$, $XS \cdot \mathcal{P}$ is a unique point and it can be seen that it is $P = AB \cdot A'B'$. By a similar argument, $XT \cdot \mathcal{P}$ is $Q = AC \cdot A'C'$. Since S, T, R are collinear and are projected from X onto $\mathcal{P}$ at the points P, Q, R, it follows that these are also collinear. □

Corollary 2.3.13 *Desargues' Theorem is valid in any projective plane obtained as a plane in* $\mathbb{P}^3$.

Corollary 2.3.13 implies that Desargues' Theorem (Theorem 1.6.5) holds in $\mathbb{P}^2(F)$, where F is a field, since $\mathbb{P}^2(F)$ is embedded in $\mathbb{P}^3(F)$ as a plane. This provides another proof of Theorem 1.6.5.

The example of the *Moulton plane*, described below, shows that Desargues' Theorem cannot be proved using only **P1**–**P4**. For this reason, we will consider Desargues' Theorem as an axiom that may or may not be satisfied for projective planes, and we will denote it by **P5**.

Definition 2.3.14 The *affine Moulton plane* is the set $\mathbb{R} \times \mathbb{R}$, with lines (sometimes called "Moulton lines" to distinguish them from Euclidean lines) defined as one of the following types of sets:

- $\{(x, y) \in \mathbb{R} \times \mathbb{R} \mid x = k\}$, for some $k \in \mathbb{R}$,
- $\{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y = mx + b\}$, for some $m, b \in \mathbb{R}$, where $m \geq 0$,
- $\{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y = mx + b,\ x \leq 0\} \cup \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y = 2mx + b,\ x \geq 0\}$, for some $m, b \in \mathbb{R}$, where $m < 0$.

Informally, lines in the affine Moulton plane are the same as lines in $\mathbb{A}^2(\mathbb{R})$, with the exception that if the slope of the line is negative, it doubles as one crosses the y-axis from left to right (Fig. 2.4).

That the name "affine Moulton plane" is not a misnomer is shown by the following proposition.

Proposition 2.3.15 *The* affine Moulton plane *satisfies **A1**–**A3**.*

Proof

A1. Let $P, Q \in \mathbb{R} \times \mathbb{R}$ be two distinct points. If the ordinary Euclidean line through P and Q is vertical or has nonnegative slope, then it is also the unique Moulton line through P and Q. If P and Q lie on the same side of the y-axis, and the

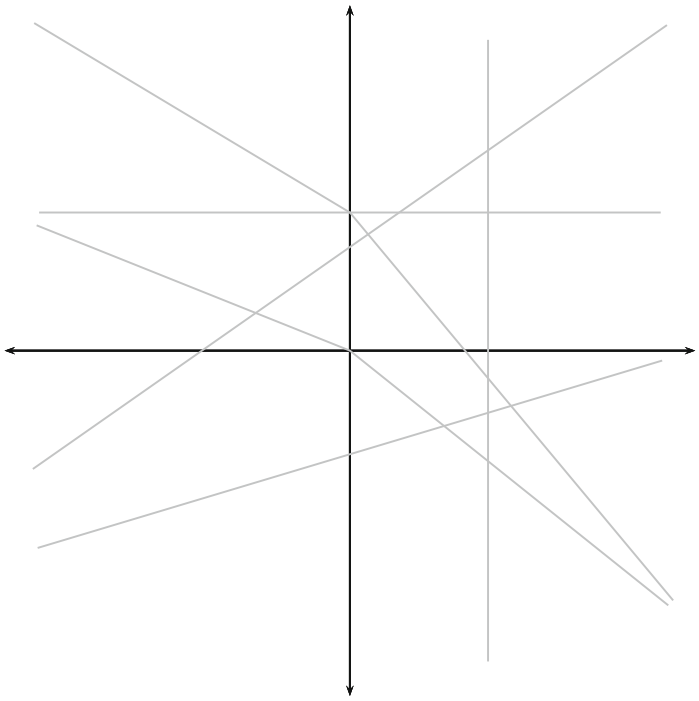


Fig. 2.4 Some lines in the affine Moulton plane

slope of the ordinary Euclidean line through P and Q is negative, one can take that line and extend it to the other side of the y-axis by doubling or halving the slope appropriately.

For the last remaining possibility, suppose, without loss of generality, that $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ where $x_1 < 0$, $x_2 > 0$, and $y_1 > y_2$. We have to find $m, b \in \mathbb{R}$, $m < 0$, such that

$$m = \frac{b - y_1}{-x_1},$$
$$2m = \frac{y_2 - b}{x_2}.$$

Solving this system, we get

$$b = \frac{2x_2y_1 - x_1y_2}{2x_2 - x_1},$$
$$m = \frac{y_2 - y_1}{2x_2 - x_1}.$$

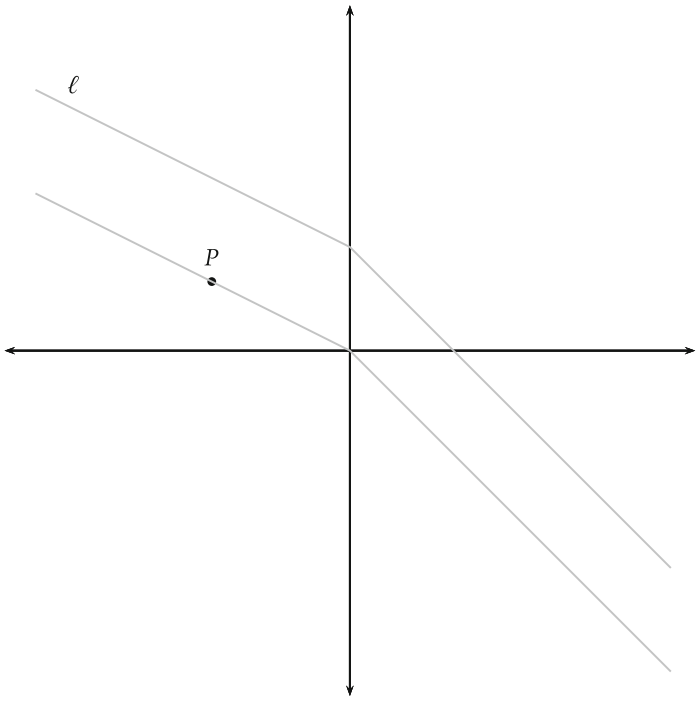


Fig. 2.5 Parallel line to a given line ℓ passing through a given point P in the Moulton plane

Note that $m < 0$. The Moulton line of the third type in Definition 2.3.14 with parameters equal to m and b as given above is then the unique line through P and Q.

A2. Let ℓ be a Moulton line and $P = (x_1, y_1)$ in $\mathbb{R} \times \mathbb{R}$. It is clear that there exists a unique Moulton line through P that is parallel to ℓ: this line has the same type of equation as ℓ itself, but the parameters k, m, b are adjusted so that it passes through P (Fig. 2.5).

A3. The points $(0, 0)$, $(1, 0)$, $(0, 1)$ are easily seen to be noncollinear.

□

Definition 2.3.16 The *Moulton plane* is the completion of the affine Moulton plane in the sense of Theorem 2.1.8.

We can now show that **P5** does not hold in the Moulton plane. Let O be the point at infinity for vertical lines in the affine Moulton plane, and let $A = (-1, 2)$, $B = (-2, 1)$, $C = (0, 1)$, $A' = (-1, -2)$, $B' = (-2, 0)$, $C' = (0, -1)$. It is clear that the triangles ABC and $A'B'C'$ are perspective from O, as can be seen in Fig. 2.6. We then calculate

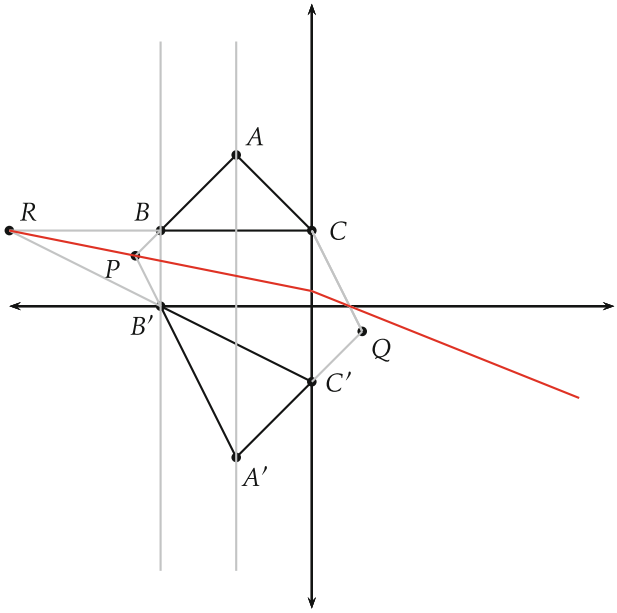


Fig. 2.6 **P5** does not hold in the Moulton plane

$$P = AB \cdot A'B' = (-7/3, 2/3),$$
$$Q = AC \cdot A'C' = (2/3, -1/3),$$
$$R = BC \cdot B'C' = (-4, 1).$$

(We remind the reader that since Q lies to the right of the y-axis, one must double the slope of the line segment AC in calculations.)

Hence, the line PR has equations

$$y = -\frac{x}{5} + \frac{1}{5}, \quad x \leq 0,$$
$$y = -\frac{2x}{5} + \frac{1}{5}, \quad x \geq 0,$$

and it is clear that Q does not lie on PR.

Finally, we want to prove that in a projective plane, **P5** implies its dual. However, before we can do that, we must dualize the concept of a Desargues configuration (see also Exercise 1.6.1).

Definition 2.3.17 A *dual Desargues configuration* in $\mathbb{P}^2$ consists of two triangles abc and $a'b'c'$ that are perspective from a line o, such that the points $o \cdot a \cdot a'$, $o \cdot b \cdot b'$,

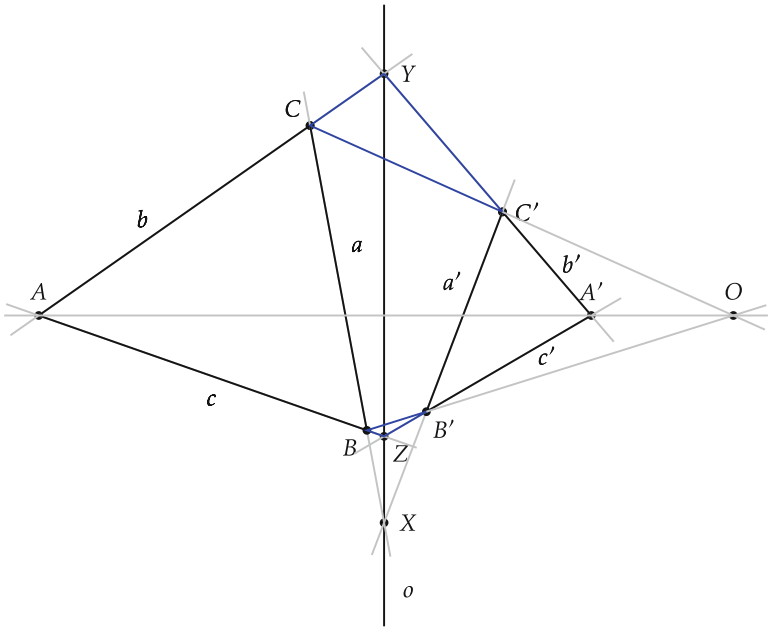


Fig. 2.7 **P5** implies **P5***

$o \cdot c \cdot c'$ are distinct; the lines o, a, a' are distinct; the lines o, b, b' are distinct; and the lines o, c, c' are distinct.

Theorem 2.3.18 *In a projective plane,* ***P5*** *holds if and only if* ***P5**** *holds.*

Proof We shall only prove the "only if" direction, leaving the "if" direction to the reader.

We must prove that two triangles in a dual Desargues configuration are perspective from a point. Let two triangles abc and $a'b'c'$ be perspective from line o, as in Fig. 2.7. Let $X = a \cdot a'$, $Y = b \cdot b'$, $Z = c \cdot c'$. Moreover, let $A = b \cdot c$, $B = a \cdot c$, $C = a \cdot b$, and similarly let $A' = b' \cdot c'$, $B' = a' \cdot c'$, $C' = a' \cdot b'$. We must prove that the lines AA', BB', CC' are concurrent.

Let $O = BB' \cdot CC'$. It can be seen from Fig. 2.7 that the two blue triangles ZBB' and YCC' are perspective from the point X and form a Desargues configuration. Hence, by **P5**, $O = BB' \cdot CC'$, $A' = C'Y \cdot ZB'$, $A = CY \cdot ZB$ are collinear. This shows that the triangles abc and $a'b'c'$ are perspective from the point O. □

Exercises

1. Prove the assertions in Remark 2.3.2.

2. Prove Propositions 2.3.4 and 2.3.6.
3. Check carefully the unproved claims in Case 2 of the proof of Theorem 2.3.12.
4. In how many ways can $\mathbb{P}^2(F)$ be embedded in $\mathbb{P}^3(F)$ as a plane?
5. Calculate the number of points, lines, and planes in $\mathbb{P}^3(F)$ for a finite field F.
6. Define the concept of an *isomorphism* of projective 3-spaces. Prove that any projective 3-space contains at least 15 points and that any projective 3-space that contains at least 15 points is isomorphic to $\mathbb{P}^3(\mathbb{F}_2)$.
7. In the proof of Theorem 2.3.18, show that $B \neq B'$, $C \neq C'$, BB' and CC' intersect in a unique point, ZBB' and YCC' are triangles perspective from the point X and they form a Desargues configuration.
8. Prove that in a projective plane, if **P5*** holds, then **P5** holds.

2.4 Projective Planes over Division Rings and Pappus' Theorem

In Chap. 1, we have proved Desargues' and Pappus' Theorems—Theorems 1.6.5 and 1.6.6—for projective planes over fields. However, as the example of the Moulton plane examined in Sect. 2.3 shows, these theorems do not necessarily hold in an arbitrary projective plane: we have already seen that Desargues' Theorem (**P5**) fails to hold in the Moulton plane, and Theorem 2.4.10 below implies that Pappus' Theorem fails to hold in it as well.

In this section, we study some properties of projective planes over division rings. Recall that a *division ring* is a ring with identity such that every nonzero element has a two-sided multiplicative inverse. Division rings have all the properties of fields except for commutativity of multiplication. A noncommutative division ring the reader is likely to be familiar with is the ring of quaternions, denoted $\mathbb{H}$. We shall see below that Desargues' Theorem holds, but Pappus' Theorem does not hold, in the projective plane over $\mathbb{H}$. However, the interest of projective planes over division rings is far more than providing examples. In Sect. 2.5, we shall see that any projective plane where Desargues' Theorem holds is a projective plane over a division ring, and if Pappus' Theorem also holds, that division ring is commutative; that is, it is a field. Thus, projective planes where Desargues' Theorem holds are characterized completely as projective planes over division rings.

In Chap. 1, we have used vector spaces over fields extensively in defining projective spaces and studying their properties. Over a division ring D, it is necessary to use left or right D-modules since D is not necessarily commutative. In what follows, we shall work with right D-modules, that is, modules where scalar multiplication is effected on the right, and we shall call right D-modules simply *vector spaces* over D.[2] Vector spaces over D satisfy many properties of vector spaces over a field, most importantly those concerning bases and dimensions: every

[2] This is a matter of convention, and some authors may use left D-modules instead. The two points of view are equivalent.

vector space has a basis, and any two bases for the same vector space have the same cardinality. This number is called the dimension of the vector space. If we start with a finite-dimensional vector space, isomorphic to D^n for some positive integer n, then every subspace of D^n has dimension at most n. On the other hand, the concept of determinant cannot be defined over a division ring, and hence, criteria for collinearity or coplanarity that use the determinant cannot be used without change. For the remainder of this section, we shall omit proofs that proceed identically to those given in Chap. 1, mentioning only the changes that are necessary. For general facts about division rings and vector spaces over them, the reader is referred to [4] and [11].

Definition 2.4.1 Let D be a division ring. The *projective plane* $\mathbb{P}^2(D)$ *over* D is the set of 1-dimensional subspaces of the vector space D^3.

Equivalently,

Definition 2.4.2 Let D be a division ring. Given two points $P = (x_1, y_1, z_1)$, $Q = (x_2, y_2, z_2) \in D^3 \setminus \{0\}$, we say that P *is equivalent to* Q, denoted by $P \sim Q$, if there exists a nonzero $\lambda \in D$ such that $x_2 = x_1\lambda$, $y_2 = y_1\lambda$, $z_2 = z_1\lambda$. $P \sim Q$ is an equivalence relation. The *projective plane* $\mathbb{P}^2(D)$ *over* D is defined as the set of equivalence classes of points in $D^3 \setminus \{0\}$ under this equivalence relation. Elements of $\mathbb{P}^2(D)$ are called *points*. A point P in $\mathbb{P}^2(D)$ that is the equivalence class of (x_0, y_0, z_0) will be denoted by $[x_0 : y_0 : z_0]$. x_0, y_0, z_0 are called *projective coordinates* of P.

It is obvious that $[x_0 : y_0 : z_0] = [x_0\lambda : y_0\lambda : z_0\lambda]$ for any nonzero $\lambda \in D$.

Definition 2.4.3 Let D be a division ring. A *line* in $\mathbb{P}^2(D)$ is the subset of points obtained as the equivalence classes of vectors in a 2-dimensional subspace of D^3. Equivalently, a *line* in $\mathbb{P}^2(D)$ is the subset of points defined by an equation of the form $ax + by + cz = 0$, where $a, b, c \in D$ are not all zero.

Remark 2.4.4 Note that in the equation of a line above, the coefficients precede the variables, as before, but in Definition 2.4.2, equivalence is defined by letting the nonzero constant λ *follow* the projective coordinates. This ensures that lines are well-defined for a projective plane over a noncommutative division ring D.

With these definitions, $\mathbb{P}^2(D)$ satisfies **P1**–**P4**, hence, it is a projective plane by Definition 2.1.6.

The definition of projective 3-spaces over division rings is similar to the case of fields:

Definition 2.4.5 Let D be a division ring. The *projective 3-space* $\mathbb{P}^3(D)$ *over* D is the set of 1-dimensional subspaces of the vector space D^4. $\mathbb{P}^3(D)$ can also be defined as the set of equivalence classes of points $(x_0, y_0, z_0, w_0) \in D^4 \setminus \{0\}$,

where two points $P = (x_1, y_1, z_1, w_1)$, $Q = (x_2, y_2, z_2, w_2) \in D^4 \setminus \{0\}$ are called *equivalent*, denoted $P \sim Q$, if there exists a nonzero $\lambda \in D$ such that $x_2 = x_1\lambda$, $y_2 = y_1\lambda$, $z_2 = z_1\lambda$, $w_2 = w_1\lambda$. Elements of $\mathbb{P}^3(D)$ are called *points*. A point P in $\mathbb{P}^3(D)$ that is the equivalence class of (x_0, y_0, z_0, w_0) will be denoted by $[x_0 : y_0 : z_0 : w_0]$. x_0, y_0, z_0, w_0 are called *projective coordinates* of P.

Again, it is obvious that $[x_0 : y_0 : z_0 : w_0] = [x_0\lambda : y_0\lambda : z_0\lambda : w_0\lambda]$ for any nonzero $\lambda \in D$.

Definition 2.4.6 Let D be a division ring. A *plane* in $\mathbb{P}^3(D)$ is the subset of points obtained as the equivalence classes of vectors in a 3-dimensional subspace of D^4. Equivalently, a *plane* in $\mathbb{P}^3(D)$ is the subset of points defined by an equation of the form $ax + by + cz + dw = 0$, where $a, b, c, d \in D$ are not all zero. A *line* in $\mathbb{P}^3(D)$ is the subset of points obtained as the equivalence classes of vectors in a 2-dimensional subspace of D^4.

With these definitions, $\mathbb{P}^3(D)$ satisfies **S1**–**S6**, hence, it is a projective 3-space by Definition 2.3.1.

Theorem 2.4.7 ***P5*** *holds in a projective plane $\mathbb{P}^2(D)$ over a division ring D.*

Proof It is clear that $\mathbb{P}^2(D)$ can be obtained as a plane in $\mathbb{P}^3(D)$ (for example, with equation $w = 0$). The result now follows from Corollary 2.3.13. □

Theorem 2.4.8 *Let D be a division ring. Then $(\mathbb{P}^2(D))^* \cong \mathbb{P}^2(D^{op})$. In particular, if $D \cong D^{op}$, then $\mathbb{P}^2(D)$ is self-dual.*

Proof We need only to prove the first statement. Recall that, by Definition 2.2.3, the dual projective plane $(\mathbb{P}^2(D))^*$ is the set of lines in $\mathbb{P}^2(D)$. A line in $\mathbb{P}^2(D)$ has equation $ax + by + cz = 0$, where $a, b, c \in D$ are not all zero; two such equations,

$$a_1x + b_1y + c_1z = 0,$$
$$a_2x + b_2y + c_2z = 0,$$

define the same line if and only if there exists a nonzero $\lambda \in D$ such that $a_2 = \lambda a_1$, $b_2 = \lambda b_1$, $c_2 = \lambda c_1$. We define a map $\phi : (\mathbb{P}^2(D))^* \to \mathbb{P}^2(D^{op})$ by mapping a line $\ell : ax + by + cz = 0$ to $[a : b : c] \in \mathbb{P}^2(D^{op})$; ϕ is well-defined due to the fact that multiplication in D^{op} is defined to be multiplication in D in reverse order. It is clear that ϕ is also bijective.

It remains to show that ϕ maps lines to lines. Let a line in $(\mathbb{P}^2(D))^*$ be given. Recall that, again by Definition 2.2.3, a line in $(\mathbb{P}^2(D))^*$ is the pencil of all lines in $\mathbb{P}^2(D)$ through a given point in $\mathbb{P}^2(D)$. Let this point be $P = [x_0 : y_0 : z_0]$. Then any line $ax + by + cz = 0$ in $\mathbb{P}^2(D)$ that passes through P satisfies

$$ax_0 + by_0 + cz_0 = 0.$$

Using the notation $d \cdot e = ed$ for the product of d and e in D^{op}, the above equation can be written as

$$x_0 \cdot a + y_0 \cdot b + z_0 \cdot c = 0.$$

But this is precisely the equation of a line in $\mathbb{P}^2(D^{op})$. □

Corollary 2.4.9 *The projective plane $\mathbb{P}^2(F)$ over a field F is self-dual.*

Proof Observe that $F \cong F^{op}$ since F is commutative. □

We now give an example of a projective plane where Pappus' Theorem does not hold. Consider $\mathbb{P}^2(\mathbb{H})$, the projective plane over the quaternions. Let $\ell_1 : jy + iz = 0$ and $\ell_2 : ix + jy = 0$ be two lines in $\mathbb{P}^2(\mathbb{H})$. Then the points $A = [0 : 1 : k]$, $B = [1 : 1 : k]$, $C = [1 : -j : i]$ lie on ℓ_1, and $D = [1 : -k : 0]$, $E = [1 : -k : 2]$, $F = [-i : j : 1]$ lie on ℓ_2. We note that none of these points is equal to $\ell_1 \cdot \ell_2 = [1 : -k : 1]$.

We now compute

$$\begin{aligned}
AE &: x + ky - z = 0,\\
BD &: kx + y - (1-k)z = 0,\\
AF &: (1-i)x - ky + z = 0,\\
CD &: x - ky + (1+i)z = 0,\\
BF &: (1 - i + j - k)x - 2y + (1 + i + j - k)z = 0,\\
CE &: -(1-i)x + (j - 3k)y + 2z = 0.
\end{aligned}$$

Hence,

$$\begin{aligned}
X &= AE \cdot BD = [-k : 1 - 2k : 2],\\
Y &= AF \cdot CD = [-1 : -j : 1],\\
Z &= BF \cdot CE = [3 : 3 + i - 2j + k : -1 + i + 5k].
\end{aligned}$$

Now, the line XY has equation

$$(-4 - 2i - 4j + 3k)x + 5y + (-4 - 2i + j + 3k)z = 0,$$

and it can be seen, after a tedious but straightforward calculation, that $Z \notin XY$.

Since Pappus' Theorem cannot be proved using **P1**–**P4**, we will consider it as an axiom that may or may not be satisfied for projective planes, and we will denote it

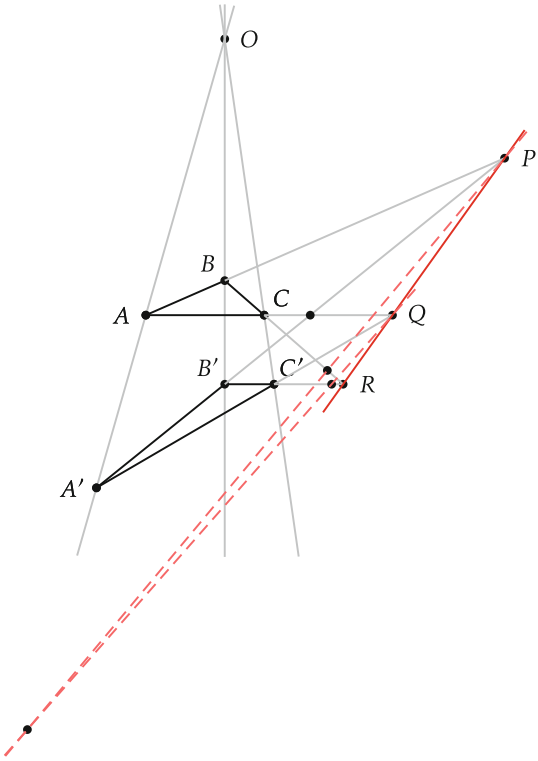


Fig. 2.8 **P6** implies **P5**. In order to avoid clutter, the points S, T, U, V are left unlabeled

by **P6**. In Sect. 2.3, we pointed out that **P5** cannot be proved using **P1**–**P4**. However, as the following theorem shows, **P5** and **P6** are not independent.

Theorem 2.4.10 (Cf. [6, Proposition 5.8]) *In a projective plane, if **P6** holds, then **P5** holds.*

Proof Let $S = AC \cdot A'B'$. Applying **P6** to the two lines OBB' and CSA, we get that $T = OS \cdot BC$, $U = OA \cdot B'C$, $P = AB \cdot B'S$ are collinear. Applying **P6** again to the two lines OCC' and $B'A'S$, we get that U, $V = OS \cdot B'C'$, $Q = CS \cdot A'C'$ are collinear. Applying **P6** one last time to the two lines $B'CU$ and TVS, we get that R, P, Q are collinear (Fig. 2.8). □

Therefore, there is no projective plane where **P6** holds but **P5** does not hold. However, we have seen examples showing that all other combinations of **P5** and **P6** are possible: $\mathbb{P}^2(F)$ for a field F satisfies both **P5** and **P6**; $\mathbb{P}^2(\mathbb{H})$ satisfies **P5** by Theorem 2.4.7 but does not satisfy **P6**; the Moulton plane does not satisfy **P5** and hence does not satisfy **P6** either, by Theorem 2.4.10 (Table 2.3).

Theorem 2.4.11 *In a projective plane, **P6** holds if and only if **P6*** holds.*

Table 2.3 By Theorem 2.4.10, there is no projective plane where **P6** holds but **P5** does not hold. All other combinations are possible, as shown by these examples

Projective plane	**P5**	**P6**
$\mathbb{P}^2(\mathbb{R})$	Holds	Holds
$\mathbb{P}^2(\mathbb{H})$	Holds	Does not hold
Moulton plane	Does not hold	Does not hold

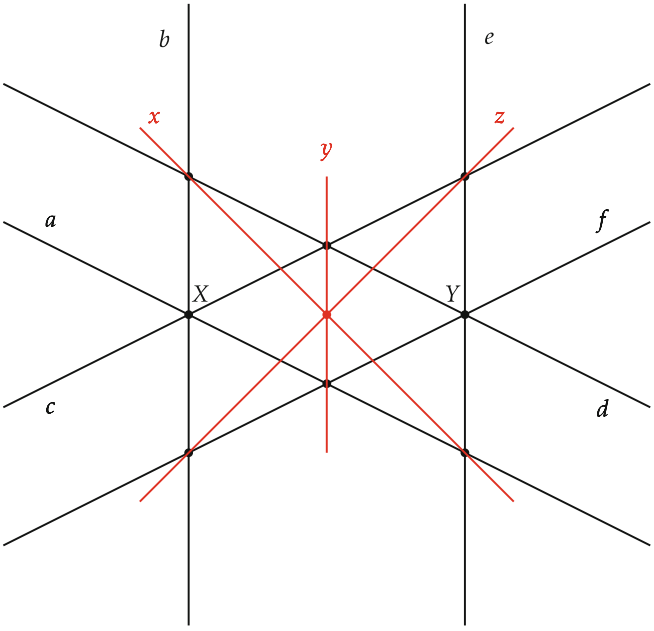


Fig. 2.9 **P6** implies **P6***. In order to avoid clutter, the points $a \cdot e$, $a \cdot f$, $b \cdot d$, $b \cdot f$, $c \cdot d$, $c \cdot e$ are left unlabeled

Proof We shall only prove the "only if" direction, leaving the "if" direction to the reader. We refer the reader to Fig. 2.9 throughout the proof.

Dualizing the statement of **P6**, we must prove the following: Let six distinct lines a, b, c, d, e, f be given such that a, b, c pass through the point X, d, e, f pass through a second, different, point Y, and suppose that none of the six lines is the join of the two points. Then the joins $x = (a \cdot e)(b \cdot d)$, $y = (a \cdot f)(c \cdot d)$, $z = (b \cdot f)(c \cdot e)$ are concurrent.

In order to apply **P6**, we note that the lines a and d are two distinct lines; points X, $a \cdot e$, $a \cdot f$ are three distinct points on a; points Y, $c \cdot d$, $b \cdot d$ are three distinct points on d; and none of the six points is $a \cdot d$. Then the intersection points $(a \cdot e)(b \cdot d) \cdot (a \cdot f)(c \cdot d)$, $X(c \cdot d) \cdot Y(a \cdot e)$, $X(b \cdot d) \cdot Y(a \cdot f)$ are collinear. But $X(c \cdot d) = c$, $Y(a \cdot e) = e$, $X(b \cdot d) = b$, $Y(a \cdot f) = f$. Hence, the three points $(a \cdot e)(b \cdot d) \cdot (a \cdot f)(c \cdot d)$, $c \cdot e$, $b \cdot f$ are collinear. That implies that $x = (a \cdot e)(b \cdot d)$, $y = (a \cdot f)(c \cdot d)$, $z = (b \cdot f)(c \cdot e)$ are concurrent. □

Exercises

1. Let D be a division ring. Prove that $\mathbb{P}^2(D)$ satisfies **P1**–**P4**. (Hint: Only the proof of **P3** needs slight modification.)
2. Let D be a division ring. Prove that $\mathbb{P}^3(D)$ satisfies **S1**–**S6**. (Hint: Again, only the proofs of a small number of axioms need slight modification.)
3. Using the quaternions $\mathbb{H}$, construct an example to show that lines would not be well-defined if we placed the coefficients after the variables in the equations of lines.
4. Prove that $\mathbb{H}$ is isomorphic to $\mathbb{H}^{op}$. Hence, by Theorem 2.4.8, $\mathbb{P}^2(\mathbb{H})$ is self-dual.
5. Let $\mathbb{P}^2(D)$ be a projective plane over a division ring D. Suppose, moreover, that $\mathbb{P}^2(D)$ is finite. Prove that $\mathbb{P}^2(D)$ is, in fact, isomorphic to $\mathbb{P}^2(F)$ for some field F, and therefore **P6** holds in it. (Hint: Use Wedderburn's Theorem.)
6. In the proof of Theorem 2.4.10, verify that the hypotheses of Pappus' Theorem are satisfied in all places where it is used.
7. Prove that in a projective plane, if **P6*** holds, then **P6** holds.

2.5 Coordinatization of Projective Planes

We finally come to the end of the chapter, where we introduce coordinates on a projective plane satisfying **P5**. The main results are Theorems 2.5.26 and 2.5.29. These state that any projective plane satisfying **P5** is isomorphic to $\mathbb{P}^2(D)$, where D is a division ring, uniquely determined up to isomorphism, and also that D is a field if and only if **P6** holds in the projective plane.

Throughout this section, we work with a projective plane $\mathbb{P}^2$ that satisfies **P5**. We choose an arbitrary line $\ell_\infty \subset \mathbb{P}^2$ as the line at infinity; it follows that $\mathbb{A}^2 = \mathbb{P}^2 \setminus \ell_\infty$ is an affine plane by Theorem 2.1.9. By Exercise 2.1.5, the completion of $\mathbb{A}^2$ is again $\mathbb{P}^2$. Our strategy in this section will be to coordinatize $\mathbb{A}^2$ first by investigating a large class of its automorphisms; then the coordinatization of $\mathbb{P}^2$ will follow. Many of the ideas in this section are due to [6, Chapter 7].

We start with the following two properties of $\mathbb{A}^2$, which will prove crucial later:

Proposition 2.5.1 *$\mathbb{A}^2$ satisfies the following properties:*

A5a (Small Desargues' Axiom). *Suppose that six distinct points A, A', B, B', C, C' are given such that the lines AA', BB', CC' are distinct. Suppose further that $AB \parallel A'B'$ and $AC \parallel A'C'$. Then $BC \parallel B'C'$.*

A5b (Big Desargues' Axiom). *Suppose that two triangles ABC and $A'B'C'$ are perspective from the point O, that is, AA', BB', CC' are concurrent at the point O. Suppose further that OA, OB, OC are distinct lines; A and A' are different from O; B and B' are different from O; and C and C' are different from O. Suppose finally that $AB \parallel A'B'$ and $AC \parallel A'C'$. Then $BC \parallel B'C'$.*

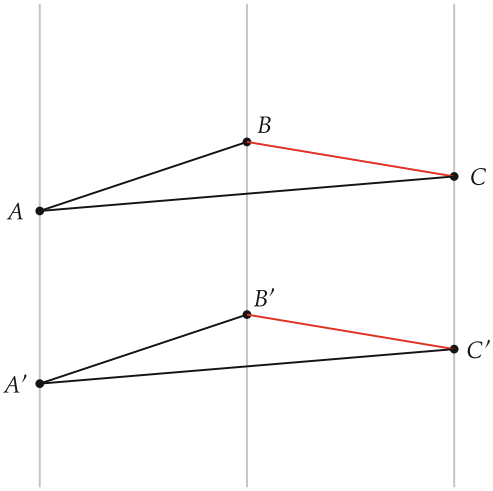


Fig. 2.10 **A5a (Small Desargues' Axiom)** in $\mathbb{A}^2$, Proposition 2.5.1

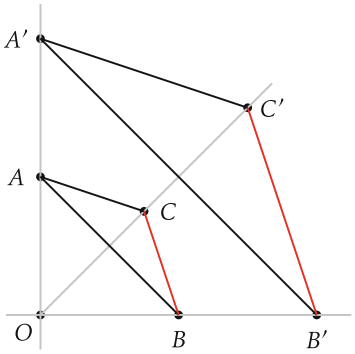


Fig. 2.11 **A5b (Big Desargues' Axiom)** in $\mathbb{A}^2$, Proposition 2.5.1. (The figure is identical to Fig. 1.3.)

Proof The proofs follow from **P5** in $\mathbb{P}^2$ (Figs. 2.10 and 2.11). □

Dilations are the class of automorphisms we shall use to coordinatize $\mathbb{P}^2$.

Definition 2.5.2 An element $\phi \in \mathrm{Aut}(\mathbb{A}^2)$ is called a *dilation* if $\phi(\ell) \parallel \ell$ for every line $\ell \subset \mathbb{A}^2$.

Remark 2.5.3 The reader who has not already done so may wish to work through Exercise 1.5.1 in order to get working knowledge of dilations in an affine plane over a field.

Proposition 2.5.4 *The dilations form a normal subgroup, denoted* $Dil(\mathbb{A}^2)$*, of* $Aut(\mathbb{A}^2)$.

Proposition 2.5.5 (Cf. [6, Proposition 7.2]) *A dilation that is not the identity can have at most one fixed point. Given two distinct points $X, Y \in \mathbb{A}^2$, if two dilations ϕ and ψ satisfy $\phi(X) = \psi(X)$ and $\phi(Y) = \psi(Y)$, then $\phi = \psi$.*

Proof We shall prove the first claim only and leave the proof of the second claim to the reader. Suppose that a dilation has two distinct fixed points P and Q. Let R be any point not on the line PQ. Then $\phi(PR)$ is a line through P parallel to PR; therefore, it is PR itself. Similarly, $\phi(QR) = QR$. Since $R = PR \cdot QR$, $\phi(R) = R$.

For points on PQ, the above argument, used either with the pair P and R or the pair Q and R, shows that ϕ fixes all the points of PQ, finishing the proof. □

We continue by examining dilations without fixed points.

Definition 2.5.6 A *translation* is a dilation that is either the identity or one that does not have fixed points.

Proposition 2.5.7 (Cf. [6, Proposition 7.4]) *Let ϕ be a translation that is not the identity, and let $P, Q \in \mathbb{A}^2$ be two distinct points. Then $P\phi(P) \parallel Q\phi(Q)$.*

Proof Write $\ell = P\phi(P)$ and $m = Q\phi(Q)$. By Definition 2.5.2, $\phi(\ell) = \ell$ and $\phi(m) = m$. Suppose now that ℓ and m are not parallel. Then these two lines intersect at a unique point $O \in \mathbb{A}^2$. But then $\phi(O) = O$, contradicting the assumption that ϕ is a translation that is not the identity. □

Definition 2.5.8 A *parallelogram* $ABCD$ in $\mathbb{A}^2$ is a collection of four points A, B, C, D, no three of which are collinear, such that $AB \parallel CD$ and $AD \parallel BC$. The points A, B, C, D are called the *vertices* of the parallelogram.

Remark 2.5.9 Note that the vertices of a parallelogram $ABCD$ can be ordered in a total of 8 ways: $ABCD$, $BCDA$, $CDAB$, $DABC$, $ADCB$, $DCBA$, $CBAD$, $BADC$.

Given 3 noncollinear points $A, B, C \in \mathbb{A}^2$, it is possible to find a fourth point $D \in \mathbb{A}^2$ such that $ABCD$ is a parallelogram. We simply draw the line ℓ through C parallel to AB and the line m through A parallel to BC. Since AB is not parallel to BC, ℓ is not parallel to m and hence these two lines intersect at a unique point $D \in \mathbb{A}^2$. Then $ABCD$ is a parallelogram by construction. Note that the point D is not unique as it depends on the order of the other three points being used (Fig. 2.12).

Corollary 2.5.10 *Let ϕ be a translation that is not the identity, and let $P, Q \in \mathbb{A}^2$ be two distinct points such that $Q \notin P\phi(P)$. Then there is a parallelogram $\phi(P)PQ\phi(Q)$. Conversely, given a parallelogram $ABCD$ and a translation ϕ that is not the identity, if $\phi(A) = B$, then $\phi(D) = C$.*

Proof The first claim follows from Definition 2.5.2 and Proposition 2.5.7.

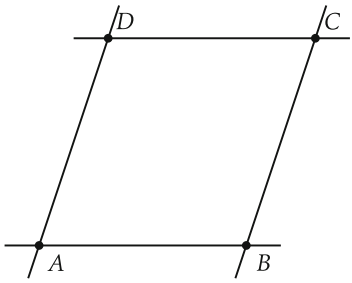


Fig. 2.12 Parallelogram $ABCD$

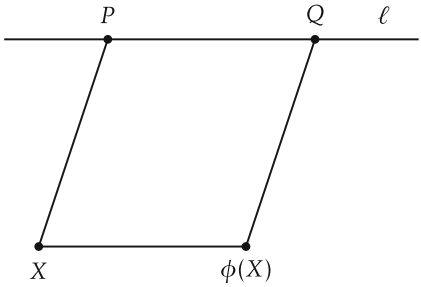


Fig. 2.13 Proposition 2.5.12, $X \notin \ell$

For the second claim, note that $D\phi(D) \parallel AB$ by Proposition 2.5.7 and $AD \parallel B\phi(D)$ by Definition 2.5.2. This forces $\phi(D) = C$ by Definition 2.5.8. □

Definition 2.5.11 Given a translation ϕ that is not the identity, the parallelism class of $P\phi(P)$ is independent of the point $P \in \mathbb{A}^2$ by Proposition 2.5.7. We call this parallelism class the *direction* of ϕ.

Proposition 2.5.12 (Cf. [6, Theorem 7.6]) *Let $P, Q \in \mathbb{A}^2$ be two points. Then there exists a unique translation ϕ such that $\phi(P) = Q$.*

Proof It suffices to prove the case where $P \neq Q$, as otherwise it is clear that the identity is the unique translation mapping P to $Q = P$. We also assume that any line in $\mathbb{A}^2$ contains at least 3 points; otherwise, Exercise 2.1.6 implies that $\mathbb{A}^2 \cong \mathbb{A}^2(\mathbb{F}_2)$ and hence it is already coordinatized and outside of the scope of this section.

We first prove existence. Let $X \in \mathbb{A}^2$ be a point that does not lie on $\ell = PQ$. We define $\phi(X)$ to be the unique point such that $QPX\phi(X)$ is a parallelogram. (See Fig. 2.13.) It can be easily checked that this defines a bijective function $\alpha : \mathbb{A}^2 \setminus \ell \to \mathbb{A}^2 \setminus \ell$. We now choose a line $m \parallel \ell, m \neq \ell$, a point $R \in m$, and we let S be the point such that $PQSR$ is a parallelogram. (See Fig. 2.14.) Using the line m and the points R and S, going through the same procedure as above, we obtain another bijective function $\beta : \mathbb{A}^2 \setminus m \to \mathbb{A}^2 \setminus m$. Now, by Corollary 2.5.10, a translation ϕ satisfies $\phi(P) = Q$ if and only if $\phi(R) = S$; moreover, ϕ agrees with α on $\mathbb{A}^2 \setminus \ell$ and with β on $\mathbb{A}^2 \setminus m$.

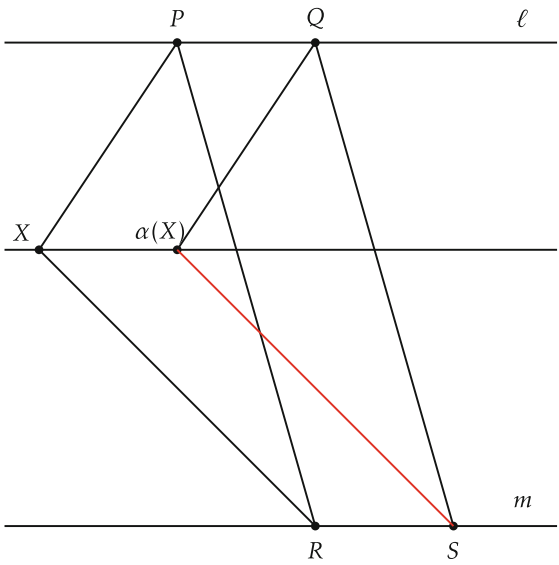


Fig. 2.14 Proposition 2.5.12

The idea is therefore to ensure that α and β agree on $\mathbb{A}^2 \setminus (\ell \cup m)$ and then "glue" them together to obtain the desired translation ϕ. Let $X \in \mathbb{A}^2 \setminus (\ell \cup m)$, and construct $\alpha(X)$ as before. (See Fig. 2.14.) To prove $\alpha(X) = \beta(X)$, the only thing that is needed is to show that $XR \parallel \alpha(X)S$. (Why is $X\alpha(X) \parallel m$?) This follows immediately from **A5a**, applied to the triangles PXR and $Q\alpha(X)S$.

We leave the proof of uniqueness to the reader. □

Proposition 2.5.13 (Cf. [6, Propositions 7.5 and 7.7]) *The translations form an abelian normal subgroup, denoted Tran*$(\mathbb{A}^2)$*, of Dil*$(\mathbb{A}^2)$.

Proof Since the identity is always a translation by Definition 2.5.6, Tran$(\mathbb{A}^2)$ is nonempty. Tran$(\mathbb{A}^2)$ is closed under inverses since the inverse of an automorphism of $\mathbb{A}^2$ with no fixed points has no fixed points. In order to prove that Tran$(\mathbb{A}^2)$ is closed under compositions, it suffices to prove that for $\phi, \psi \in$ Tran$(\mathbb{A}^2)$ that are not the identity, then $\phi\psi$ is a translation; for that, by Definition 2.5.6 and Proposition 2.5.5, we must prove that if $\psi\phi$ has a fixed point, then it has two distinct fixed points.

Suppose then that $\psi\phi$ has a fixed point P; write $Q = \phi(P)$. Note that $Q \neq P$. Choose a point $R \notin PQ$ and let $S = \phi(R)$. Then by Corollary 2.5.10, there exists a parallelogram $PQSR$. But by the same corollary, using the same parallelogram $PQSR$, we have $\psi(S) = R$ since $\psi(Q) = P$. Hence, $\psi\phi(R) = R$, and R is a second fixed point of $\psi\phi$.

We must next prove that, given $\phi, \psi \in$ Tran$(\mathbb{A}^2)$ that are not the identity, then $\phi\psi = \psi\phi$. We first consider the case where ϕ and ψ have different directions. Choose a point P. Let S be the unique point such that $PQSR$ is a parallelogram,

where $Q = \phi(P)$ and $R = \psi(P)$. Then $S = \psi(Q) = \phi(R)$ by Corollary 2.5.10. But then $\phi\psi(P) = \psi\phi(P)$; by Proposition 2.5.12, it follows that $\phi\psi = \psi\phi$.

At this point in the proof, we introduce some notation. Given two points $P, Q \in \mathbb{A}^2$, we denote the translation that maps P to Q, unique by Proposition 2.5.12, by τ_{PQ}; it is clear that the same translation can be so denoted in many ways. We have the following properties:

- Given $P \in \mathbb{A}^2$, $\tau_{PP} = \mathrm{id}$.
- Given $P, Q \in \mathbb{A}^2$, $\tau_{PQ}^{-1} = \tau_{QP}$. (We now know that the inverse of a translation is a translation.)
- Given $P, Q, R \in \mathbb{A}^2$, $\tau_{QR}\tau_{PQ} = \tau_{PR}$. (We now know that the composition of two translations is a translation.)

Suppose now that ϕ and ψ have the same direction. Choose a point P; let $\ell = P\phi(P)$; and choose a point $T \notin \ell$. Then we can write $\phi = \tau_{T\phi(P)}\tau_{PT}$ and $\psi = \tau_{T\psi(\phi(P))}\tau_{\phi(P)T}$. Note that all the τs appearing in these formulas have different directions except $\tau_{\phi(P)T}$ and $\tau_{T\phi(P)}$, and these are inverses of each other. Hence, all the τs that appear in ϕ and ψ commute with each other; this implies that $\phi\psi = \psi\phi$.

It remains to prove normality. Let $\tau \in \mathrm{Tran}(\mathbb{A}^2)$ and $\sigma \in \mathrm{Dil}(\mathbb{A}^2)$ be given. Since $\sigma\tau\sigma^{-1}$ is a dilation, it suffices to prove that if it has a fixed point, then it is the identity. Suppose then that there exists a point P such that $\sigma\tau\sigma^{-1}(P) = P$. This implies that τ has the fixed point $\sigma^{-1}(P)$. Hence τ is the identity and $\sigma\tau\sigma^{-1}$ is the identity. □

Proposition 2.5.14 *Let $\ell \subset \mathbb{A}^2$ be any line. The translations mapping ℓ to itself form a subgroup of $\mathrm{Tran}(\mathbb{A}^2)$, denoted by $\mathrm{Tran}_\ell(\mathbb{A}^2)$. $\mathrm{Tran}_\ell(\mathbb{A}^2)$ is abelian.*

Proof It is clear that $\mathrm{Tran}_\ell(\mathbb{A}^2)$ is nonempty and is closed under compositions and inverses. Hence, it is a subgroup of $\mathrm{Tran}(\mathbb{A}^2)$. The last claim follows from the fact that a subgroup of an abelian group is abelian. □

We can now define an abelian group, denoted D, which will form part of the division ring over which the projective plane $\mathbb{P}^2$ will be defined. Let $\ell \subset \mathbb{A}^2$ be any line and let $D = \ell$ as a set. We also fix, until the end of the section, a point $O \in D$ that will play the role of the identity element of D for addition.

Theorem 2.5.15 *Define an operation, called* addition*, on D, as follows: given $P, Q \in D$, $P + Q = \tau_{OP}(Q)$. D, together with $+$, is an abelian group isomorphic to $\mathrm{Tran}_\ell(\mathbb{A}^2)$; its identity element is O.*

Proof Define a map $\Phi : \mathrm{Tran}_\ell(\mathbb{A}^2) \to D$ by $\Phi(\phi) = \phi(O)$. It is now easy to verify that Φ has inverse $\Phi^{-1} : D \to \mathrm{Tran}_\ell(\mathbb{A}^2)$ defined by $\Phi^{-1}(P) = \tau_{OP}$ and that Φ is a group homomorphism. □

We have now completed half of the work on the construction of the division ring D. For the other half, namely multiplication, we consider another kind of dilation. Recall that, by Proposition 2.5.5, a dilation that is not the identity can have at most one fixed point. As translations (except for the identity) have no fixed point, we are now going to investigate dilations with exactly one fixed point (plus the identity).

Definition 2.5.16 A *scaling* is a dilation that is either the identity or that has exactly one fixed point. Given a scaling that is not the identity, its fixed point is called the *center*. We consider the identity to be a scaling with center equal to any given point.

Remark 2.5.17 A scaling could also have been defined as a dilation with at least one fixed point; it is trivial to see that the two definitions are equivalent.

Proposition 2.5.18 *Let ϕ be a scaling that is not the identity, let P be its center, and let $Q \neq P$ be any point. Then $P, Q, \phi(Q)$ are three distinct collinear points.*

Proof That $P, Q, \phi(Q)$ are three distinct points follows directly from Definition 2.5.16. For collinearity, note that $\phi(PQ) = P\phi(Q)$ is a line that passes through P and is parallel to PQ by Definition 2.5.2; it follows that $\phi(PQ) = P\phi(Q) = PQ$ and hence $P, Q, \phi(Q)$ are collinear. □

Proposition 2.5.19 *Let ϕ be a scaling that is not the identity, with center $P \in \mathbb{A}^2$, let ℓ and m be two distinct lines in $\mathbb{A}^2$ that pass through P, let $A \in \ell$, and let $B, C \in m$, with all three points different from P. Then $C = \phi(B)$ if and only if $AB \parallel \phi(A)C$.*

Proof The "only if" part follows from Definition 2.5.2. For the "if" part, suppose that $AB \parallel \phi(A)C$ and $C \neq \phi(B)$. Then, by Definition 2.5.2, $\phi(A)\phi(B)$ and $\phi(A)C$ would be two distinct lines, both parallel to AB, and both passing through the same point, namely $\phi(A)$. This contradicts **A2**. □

Proposition 2.5.20 (Cf. [6, Theorem 7.9]) *Let P, Q, R be three collinear points with $Q \neq P$ and $R \neq P$. Then there exists a unique scaling ϕ with center P such that $\phi(Q) = R$.*

Proof The proof is similar in spirit to that of Proposition 2.5.12. It suffices to prove the case where $Q \neq R$, as otherwise it follows from Proposition 2.5.18 that the identity is the unique scaling with center P mapping Q to $R = Q$. Again, as in the proof of Proposition 2.5.12, we assume that any line in $\mathbb{A}^2$ passes through at least 3 points.

We first prove existence. Let $X \in \mathbb{A}^2$ be a point that does not lie on $\ell = PQ$. We define $\phi(X)$ as follows: we draw the parallel line to QX passing through R and let $\phi(X)$ be the intersection of that line with PX. (See Fig. 2.15.) It can be easily checked that this defines a bijective function $\alpha : \mathbb{A}^2 \setminus \ell \to \mathbb{A}^2 \setminus \ell$. We now choose a line m passing through P, $m \neq \ell$, a point $S \in m$, $S \neq P$, and we let $T \in m$ be the

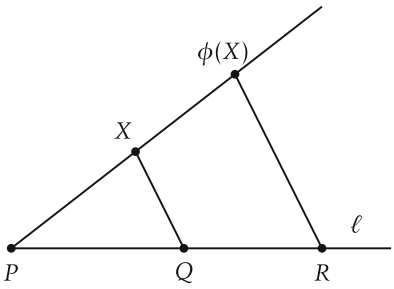


Fig. 2.15 Proposition 2.5.20, $X \notin \ell$

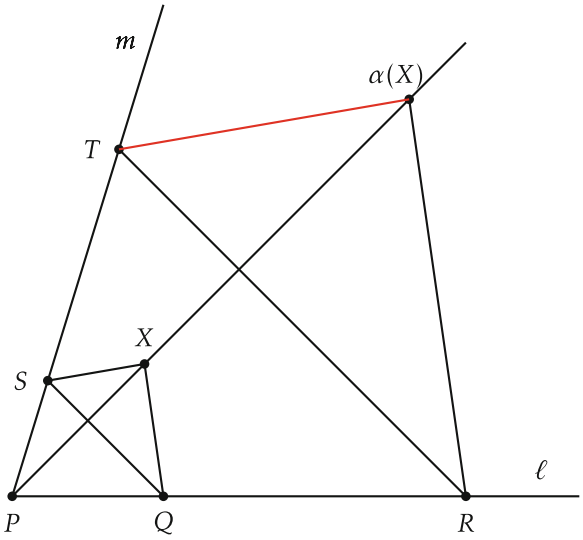


Fig. 2.16 Proposition 2.5.20

unique point such that $QS \parallel RT$. (See Fig. 2.16.) Using the line m and the points S and T, going through the same procedure as above, we obtain another bijective function $\beta : \mathbb{A}^2 \setminus m \to \mathbb{A}^2 \setminus m$. By Proposition 2.5.19, a scaling ϕ with center P satisfies $\phi(Q) = R$ if and only if $\phi(S) = T$; moreover, ϕ agrees with α on $\mathbb{A}^2 \setminus \ell$ and with β on $\mathbb{A}^2 \setminus m$.

The idea is to ensure that α and β agree on $\mathbb{A}^2 \setminus (\ell \cup m)$ and then "glue" them together to get the desired scaling ϕ. Let $X \in \mathbb{A}^2 \setminus (\ell \cup m)$, and construct $\alpha(X)$ as before. (See Fig. 2.16.) To prove $\alpha(X) = \beta(X)$, the only thing that is needed is to show that $XS \parallel \alpha(X)T$. This follows immediately from **A5b**, applied to the triangles SXQ and $T\alpha(X)R$.

We leave the proof of uniqueness to the reader. □

Proposition 2.5.21 *Let $P \in \mathbb{A}^2$. The scalings with center P form a subgroup, denoted $Scal_P(\mathbb{A}^2)$, of $Dil(\mathbb{A}^2)$.*

Proof By Definition 2.5.16, a dilation ϕ is a scaling with center P if and only if $\phi(P) = P$. It is now easy to check that $\mathrm{Scal}_P(\mathbb{A}^2)$ is nonempty and is closed under compositions and inverses. □

Let P, Q, R be three collinear points with $Q \neq P$ and $R \neq P$. We denote the scaling with center P that maps Q to R, unique by Proposition 2.5.20, by σ^P_{QR}; it is clear that the same scaling can be so denoted in many ways. We have the following properties:

- Given $Q \in \mathbb{A}^2$, $\sigma^P_{QQ} = \mathrm{id}$.
- Given $Q, R \in \mathbb{A}^2$ such that P, Q, R are collinear, $(\sigma^P_{QR})^{-1} = \sigma^P_{RQ}$. (We now know that the inverse of a scaling with center P is a scaling with center P.)
- Given $Q, R, S \in \mathbb{A}^2$ such that P, Q, R, S are collinear, $\sigma^P_{RS}\sigma^P_{QR} = \sigma^P_{QS}$. (We now know that the composition of two scalings with center P is a scaling with center P.)

We shall suppress the center P from the notation whenever it is clear from the context.

We are now in a position to define the multiplication on D. We choose a point $U \in D$, $U \neq O$, to play the role of the multiplicative identity. To simplify notation, all scalings below have center O unless otherwise specified.

Theorem 2.5.22 *Define an operation, called* multiplication*, on D, as follows: for any point $X \in D$, $O \cdot X = X \cdot O = O$; for $X, Y \in D$ different from O, define $X \cdot Y = \sigma_{UY}(X) = \sigma_{UY}\sigma_{UX}(U)$. Then $D^* = D \setminus \{O\}$, together with the operation $\cdot$, is a group.*

Proof In order to prove the theorem, we must check the following:

(i) D^* is closed under multiplication. This follows from the definition of $X \cdot Y$ as $\sigma_{UY}(X)$ for $X, Y \in D^*$ since σ_{UY} has a fixed point, namely O, and cannot map any other point to O.
(ii) Multiplication is associative. This follows since by Proposition 2.5.21 we already know that scalings with center O form a group under composition.
(iii) The element U is an identity for multiplication. This follows since, by definition, $X \cdot U = \sigma_{UU}(X) = X$ and $U \cdot X = \sigma_{UX}(U) = X$.
(iv) Every element of D^* has a multiplicative inverse. Let $X \in D^*$ and let $Y = \sigma^{-1}_{UX}(U)$. (Why is $Y \in D^*$?) Note that $\sigma_{UY} = \sigma^{-1}_{UX}$. Then it can be easily seen that $X \cdot Y = \sigma_{UY}\sigma_{UX}(U) = U$ and $Y \cdot X = \sigma_{UX}\sigma_{UY}(U) = U$.

□

Remark 2.5.23 Note that being closed under multiplication, the associativity of multiplication, and the element U being an identity element for multiplication extend to D.

Lemma 2.5.24 *We have the following identities in Dil($\mathbb{A}^2$):*

(i) $\tau_{O(X\cdot Y)} = \sigma_{UY}\tau_{OX}\sigma_{UY}^{-1}$,
(ii) $\sigma_{U(X\cdot Y)} = \sigma_{UY}\sigma_{UX}$.

Proof The second identity follows from the definition of multiplication in Theorem 2.5.22. To prove the first identity, note that $\sigma_{UY}\tau_{OX}\sigma_{UY}^{-1}$ is a translation by Proposition 2.5.13. To fully determine it, it suffices to compute its action on O; it can be easily seen that $\sigma_{UY}\tau_{OX}\sigma_{UY}^{-1}$ maps O to $X \cdot Y$. □

Theorem 2.5.25 *In D, multiplication distributes over addition. Hence, D is a division ring.*

Proof Let $X, Y, Z \in D$. We first consider $(X + Y) \cdot Z$. It is clear that $(X + Y) \cdot Z = X \cdot Z + Y \cdot Z$ if one of X, Y, Z is O. Otherwise, we compute

$$\begin{aligned}
\tau_{O((X+Y)\cdot Z)} &= \sigma_{UZ}\tau_{O(X+Y)}\sigma_{UZ}^{-1} \\
&= \sigma_{UZ}\tau_{OX}\tau_{OY}\sigma_{UZ}^{-1} \\
&= \sigma_{UZ}\tau_{OX}\sigma_{UZ}^{-1}\sigma_{UZ}\tau_{OY}\sigma_{UZ}^{-1} \\
&= \tau_{O(X\cdot Z)}\tau_{O(Y\cdot Z)} \\
&= \tau_{O(X\cdot Z+Y\cdot Z)}.
\end{aligned}$$

Calculating the actions of the first and last terms on O, we see that $(X + Y) \cdot Z = X \cdot Z + Y \cdot Z$.

We now consider $X\cdot(Y+Z)$. Consider the map $\alpha : \mathrm{Tran}_\ell(\mathbb{A}^2) \to \mathrm{Tran}_\ell(\mathbb{A}^2)$ that is defined by $\alpha(\tau_{OP}) = \tau_{O(X\cdot P)}$. By [6, Lemma 7.10], α is a group homomorphism. Hence, we have

$$\alpha(\tau_{O(Y+Z)}) = \alpha(\tau_{OY}\tau_{OZ}) = \alpha(\tau_{OY})\alpha(\tau_{OZ}),$$

which implies

$$\tau_{O(X\cdot(Y+Z))} = \tau_{O(X\cdot Y)}\tau_{O(X\cdot Z)} = \tau_{O(X\cdot Y+X\cdot Z)},$$

finishing the proof. □

The following theorem is the first main result of this section.

Theorem 2.5.26 (Cf. [6, Theorems 7.14 and 7.15]) *The projective plane $\mathbb{P}^2$ is isomorphic to $\mathbb{P}^2(D)$. D is unique up to isomorphism.*

Proof We first introduce coordinates on $\mathbb{A}^2$ using the division ring D defined above, as this implies that $\mathbb{P}^2 \cong \mathbb{P}^2(D)$. Let us recall, from Chap. 1, that the coordinates on $\mathbb{A}^2(D)$ are defined using two intersecting axes, namely the x-axis and the y-axis. Assuming that we have one-to-one mappings between the points of the x-axis, the points of the y-axis, and the elements of the division ring D, a point $P \in \mathbb{A}^2(D)$ is determined by the x and the y coordinates, and these can be determined as follows: we draw two lines from P, one parallel to the x-axis and the other parallel to the y-axis, and we consider the intersection points of these lines with the axes. These intersection points then give us the coordinates of P. For example, if P is on the x-axis, the procedure gives the coordinates of P as $(a, 0)$, where $a \in D$ is the element that corresponds to P. Similarly, if P is on the y-axis, we obtain $P = (0, a)$, where $a \in D$ is the element that corresponds to P.

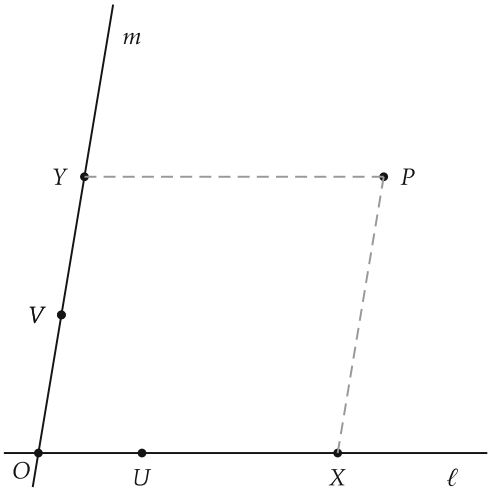
We now imitate this procedure. (See Fig. 2.17.) We consider two lines, ℓ and m, which intersect at the point O, and we take points $U \in \ell$ and $V \in m$ that will play the role of the identity element for multiplication. Using the line ℓ, we determine the division ring D as before. Let $P \in \mathbb{A}^2$ be an arbitrary point. If P lies on the line ℓ, we coordinatize P as $(a, 0)$, where $a \in D$ is the element that corresponds to P. If P lies on m instead, we consider $b = \sigma_{VP}(U) \in D$, and we coordinatize P as $(0, b)$. If P does not lie on either line, we draw two lines from P, one parallel to ℓ and the other parallel to m, we consider the intersection points of these lines with ℓ and m, we find the elements $a, b \in D$ that correspond to these two intersection points and we coordinatize P as (a, b). The lines ℓ and m then play the roles of the x-axis and the y-axis, respectively. It is also clear that this construction establishes a one-to-one correspondence between the points of $\mathbb{A}^2$ and $D \times D$.

It remains to prove that the lines in $\mathbb{A}^2$ are given by equations of the form $ax + by + c = 0$, where $a, b, c \in D$ and at least one of a and b is nonzero. For that, we make use of Lemmas 2.5.27 and 2.5.28 below. First, we note that a line parallel to ℓ has an equation of the form $y = b$ for some $b \in D$ by construction. (Again, the reader is referred to Fig. 2.17.) Similarly, a line parallel to m has an equation of the form $x = a$ for some $a \in D$. Next, suppose that a line n, different from ℓ and m, passes through O. Let $P = (1, a)$ be the intersection point of n with the line $x = 1$. If $Q = (x, y)$ is any point on n, different from O, then there is a unique scaling σ_{PQ} that maps P to Q. By Lemma 2.5.28 there exists a unique $k \in D^*$ such that $x = k$ and $y = ak$. It is clear that this is equivalent to $y = ax$. Finally, suppose that a line r, not parallel to ℓ or m, is given, and that r does not pass through O. Let s be the line that is parallel to r and that passes through O. Then r and m intersect in a point $(0, b)$, where $b \in D^*$, and r is obtained from s by the translation $\tau_{O(0,b)}$. Applying this translation to $y = ax$, and using Lemma 2.5.27, we obtain the equation of r as $y = ax + b$.

We leave the proof of the uniqueness of D to the reader. □

Lemma 2.5.27 *Let $\tau \in Tran(\mathbb{A}^2)$ and suppose that $\tau(O) = (a, b)$ for some $a, b \in D$. Then for every $P = (x, y) \in \mathbb{A}^2$, we have $\tau(P) = (x + a, y + b)$.*

Fig. 2.17 Theorem 2.5.26



Proof The reader is advised to refer to Fig. 2.17 again for the proof. Suppose $X = (x, 0)$ and $Y = (0, y)$. Since $OXPY$ is a parallelogram, we have $\tau_{OP} = \tau_{OX}\tau_{OY}$ by Corollary 2.5.10. Similarly, $\tau = \tau_{OA}\tau_{OB}$, where $A = (a, 0)$ and $B = (0, b)$. Now consider the map $\alpha : \mathrm{Tran}_\ell(\mathbb{A}^2) \to \mathrm{Tran}_m(\mathbb{A}^2)$ that is defined by $\alpha(\tau_{O(a,0)}) = \tau_{O(0,a)}$. By [6, Lemma 7.10], α is a group homomorphism. This implies that

$$\tau_{O(0,s+t)} = \tau_{O(0,s)}\tau_{O(0,t)}$$

for all $s, t \in D$. Using this fact, and the commutativity of the group $\mathrm{Tran}(\mathbb{A}^2)$, we compute

$$\begin{aligned}
\tau(P) &= \tau\tau_{OP}(O) \\
&= \tau_{OA}\tau_{OB}\tau_{OX}\tau_{OY}(O) \\
&= \tau_{OA}\tau_{OX}\tau_{OB}\tau_{OY}(O) \\
&= \tau_{O(X+A)}\tau_{O(Y+B)}(O) \\
&= (x + a, y + b).
\end{aligned}$$

□

Lemma 2.5.28 *Let $\sigma \in Scal_O(\mathbb{A}^2)$. Then there exists a unique $a \in D^*$ such that for every $P = (x, y) \in \mathbb{A}^2$, we have $\sigma(P) = (xa, ya)$.*

Proof Let a be defined by $\sigma(U) = (a, 0)$. Uniqueness follows immediately. Suppose $X = (x, 0)$ and $Y = (0, y)$. We again consider the map $\alpha : \mathrm{Tran}_\ell(\mathbb{A}^2) \to$

$\mathrm{Tran}_m(\mathbb{A}^2)$ that is defined by $\alpha(\tau_{O(a,0)}) = \tau_{O(0,a)}$. As in Lemma 2.5.27, α is a group homomorphism. Hence

$$\tau_{O(0,st)} = \sigma_{U(t,0)}\tau_{O(0,s)}\sigma_{U(t,0)}^{-1}$$

for all $s, t \in D$. We again have $\tau_{OP} = \tau_{OX}\tau_{OY}$ as in the proof of Lemma 2.5.27. We compute

$$\begin{aligned}
\sigma(P) &= \sigma_{U(a,0)}\tau_{OX}\tau_{OY}(O) \\
&= \sigma_{U(a,0)}\tau_{OX}\tau_{OY}\sigma_{U(a,0)}^{-1}(O) \\
&= \sigma_{U(a,0)}\tau_{OX}\sigma_{U(a,0)}^{-1}\sigma_{U(a,0)}\tau_{OY}\sigma_{U(a,0)}^{-1}(O) \\
&= \tau_{O(xa,0)}\tau_{O(0,ya)}(O) \\
&= (xa, ya).
\end{aligned}$$

□

We now come to the second main result.

Theorem 2.5.29 (Cf. [6, Theorem 6.9]) *$\mathbb{P}^2(D)$ satisfies **P6** if and only if D is a field.*

Proof The "if" direction is Pappus' Theorem (Theorem 1.6.6). For the "only if" direction, we use the geometric procedure, described in Sect. 1.4, to multiply two elements $a, b \in D$. We ignore the cases where one of a and b is 0 or 1 or $a = b$ as we trivially have $ab = ba$ in all these cases. Let us recall the procedure: since the lines $\ell_1 : x = a$ and $\ell_2 : y = x$ are given, their intersection $Y = \ell_1 \cdot \ell_2 = (a, a)$ is given. Since the points $X = (1, 1)$ and $Y' = (b, 0)$ are given, their join $\ell_3 : (b-1)y + x = b$ is given. Then the line ℓ_4 through (a, a) parallel to ℓ_3 is given; its equation is $\ell_4 : (b-1)y + x = ba$. Finally, the intersection $X' = \ell_4 \cdot \ell_x = (ba, 0)$ is given. This allows us to find $ba \in D$. To find $ab \in D$, we only need to reverse the roles of a and b and go through the same procedure. This gives us $m_1 : x = b$, $Z = m_1 \cdot \ell_2 = (b, b)$, $m_2 : (a-1)y + x = a$, and finally $m_3 : (a-1)y + x = ab$, whose intersection with ℓ_x gives us $ab \in D$. Let $Z' = (a, 0)$. We now apply the affine version of Pappus' Theorem (Theorem 1.1.17) to the lines ℓ_x and ℓ_2, together with the triples of points $X, Y, Z \in \ell_2$ and $X', Y', Z' \in \ell_x$: since $YZ' \parallel Y'Z$ and $XY' \parallel X'Y$, we get $XZ' \parallel X'Z$. (See Fig. 2.18.) This implies $ab = ba$. □

Exercises

1. Prove Proposition 2.5.4 and the second claim in Proposition 2.5.5.

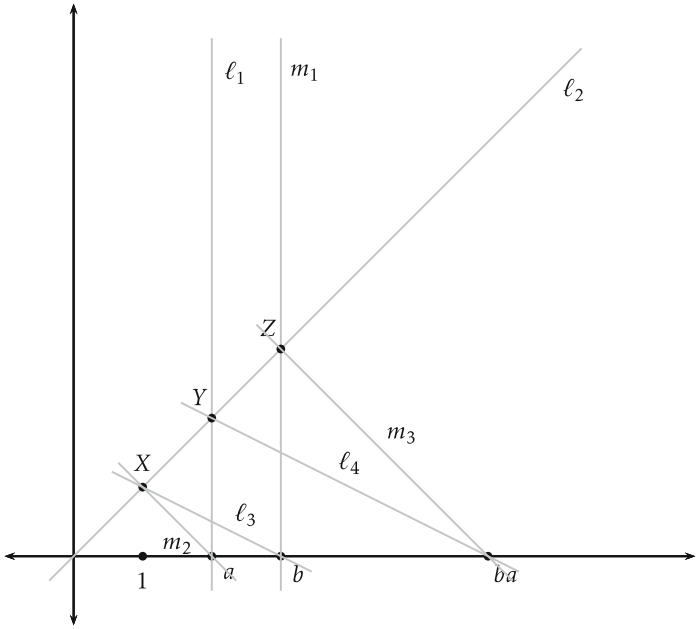


Fig. 2.18 Theorem 2.5.29

2. Complete the details in the proofs of Propositions 2.5.12 and 2.5.20.
3. Check the details in the proof of Theorem 2.5.15.
4. Prove the uniqueness of D in Theorem 2.5.26.

Conics

3

Abstract

Conics are studied on projective planes over fields. Polarities are introduced and used to define conics. After proving Pascal's and Bézout's Theorems, we investigate the pencils of conics passing through four given points, as well as the conics passing through five given points. The last section examines a classical topic, that is, the nondegenerate conics over the field of real numbers.

In this chapter, we use the theory developed in Chap. 1 to study conic sections (or conics, for short) in a projective plane. This starts with a study of polarities in Sect. 3.1, setting the stage for the definition of conics later. In Sect. 3.2, we define and study the basic properties of conics. Special attention is given here to investigating the intersections of conics and lines, and the sets of points of conics over arbitrary fields. This is necessitated by the fact that the set of points of a conic can change as one passes from the base field to an extension field. Pascal's Theorem, a classical result in the theory of conics in a projective plane, is proved in Sect. 3.3, as well as a weak form of Bézout's Theorem for curves in projective plane. In Sect. 3.4 we show that, given four points in projective plane such that no three are collinear, the conics that pass through all four form a pencil of conics. We also show that through five given points in the plane such that no three are collinear, a unique conic passes and this conic is nondegenerate (Theorem 3.4.14). In the final section, Sect. 3.5, we examine conic sections over $\mathbb{R}$ and we determine the three-way classification of the affine part (ellipse, hyperbola, parabola) in terms of the intersection points of the conic with the line at infinity.

Throughout this chapter, F denotes an arbitrary but fixed field with $\text{char}(F) \neq 2$. As in Chap. 1, we avoid mentioning that coordinates of points and coefficients of equations belong to the base field F where these are clear from the context. Vector spaces are also over F and finite-dimensional unless otherwise specified.

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E. Coşkun, *Projective Geometry*, Compact Textbooks in Mathematics,
https://doi.org/10.1007/978-3-032-02196-0_3

3.1 Polarities

Let us briefly recall the explanation, offered in Sect. 1.3, of why the principle of duality holds for $\mathbb{P}^2(F)$. Lines in $\mathbb{P}^2(F)$ are defined by equations of the form $ax + by + cz = 0$, where a, b, c are not all zero, and two such equations $a_i x + b_i y + c_i z = 0$, $i = 1, 2$, define the same line if and only if there is a nonzero $\lambda \in F$ such that $\lambda a_1 = a_2$, $\lambda b_1 = b_2$, $\lambda c_1 = c_2$. Hence, to a line in $\mathbb{P}^2(F)$ with equation $a_0 x + b_0 y + c_0 z = 0$ corresponds the point $[a_0 : b_0 : c_0] \in \mathbb{P}^2(F)$. This led us to consider another copy of $\mathbb{P}^2(F)$, denoted $(\mathbb{P}^2(F))^*$, whose points correspond in a one-to-one manner to lines in $\mathbb{P}^2(F)$. We have also seen that points of $\mathbb{P}^2(F)$ are in one-to-one correspondence with lines in $(\mathbb{P}^2(F))^*$. The reason for the validity of the principle of duality in $\mathbb{P}^2(F)$ is simply that the one-to-one correspondences just described are incidence-preserving.

In order to discuss polarities, we examine $(\mathbb{P}^2(F))^*$ in greater detail. Recall that, given a vector space V, its *dual vector space*, or simply *dual*, is defined as

$$V^\vee = \mathrm{Hom}_F(V, F).$$

We can restate the observations made above in terms of the elements of the dual vector space $V^\vee$. Given a line $\ell \subset \mathbb{P}^2(F)$ with equation $ax + by + cz = 0$, the kernel of the nonzero element $\phi \in (F^3)^\vee$ defined by $\phi(x, y, z) = ax + by + cz$ is precisely the 2-dimensional subspace of F^3 corresponding to ℓ. Conversely, if we start by choosing a nonzero element $\phi \in (F^3)^\vee$, its kernel is a 2-dimensional subspace that then defines a line in $\mathbb{P}^2(F)$. Finally, it can be easily verified that two nonzero elements of $(F^3)^\vee$ have the same kernel if and only if they are scalar multiples of each other.

These remarks show that there is a one-to-one correspondence between lines in $\mathbb{P}^2(F)$ and elements of $\mathbb{P}((F^3)^\vee)$ (see Remark 1.2.5), which maps a line ℓ in $\mathbb{P}^2(F)$ with equation $ax + by + cz = 0$ to the point $[\phi] \in \mathbb{P}((F^3)^\vee)$, where the nonzero element $\phi \in (F^3)^\vee$ is defined by $\phi(x, y, z) = ax + by + cz$. It is therefore clear that the projective plane $(\mathbb{P}^2(F))^*$ mentioned in Sect. 1.3 is none other than $\mathbb{P}((F^3)^\vee)$. We call it the *dual projective plane* to $\mathbb{P}^2(F)$. In general, for any 3-dimensional vector space V, one can construct $\mathbb{P}(V^\vee)$; it is again called the *dual projective plane* to $\mathbb{P}(V)$.

Since any (finite-dimensional) vector space V is isomorphic to its dual $V^\vee$, it is perhaps not surprising that $\mathbb{P}(V)$ is isomorphic to $\mathbb{P}(V^\vee)$, under a suitable definition of isomorphism (Exercise 2.1.6).

Definition 3.1.1 Let V and W be two 3-dimensional vector spaces. An *isomorphism* from $\mathbb{P}(V)$ to $\mathbb{P}(W)$ is a bijective map $\phi : \mathbb{P}(V) \to \mathbb{P}(W)$ that maps lines to lines.

This raises the question of whether there exist isomorphisms from $\mathbb{P}(V)$ to $\mathbb{P}(W)$ for 3-dimensional vector spaces V and W. As in Sect. 1.4, we consider a vector

space isomorphism $T : V \to W$. T induces an isomorphism $\phi_T : \mathbb{P}(V) \to \mathbb{P}(W)$ as follows: given a point $[\mathbf{v}] \in \mathbb{P}(V)$, we define $\phi_T([\mathbf{v}]) = [T(\mathbf{v})]$. Many of the results of Sect. 1.4 can be generalized to isomorphisms. We give two of these below and leave the rest to the interested reader.

Proposition 3.1.2 *Given a vector space isomorphism $T : V \to W$, ϕ_T is an isomorphism from $\mathbb{P}(V)$ to $\mathbb{P}(W)$.*

Proof The proof is similar to that of Proposition 1.4.4. We omit the details. □

Proposition 3.1.3 *Given $A, B, C, D \in \mathbb{P}(V)$ and $A', B', C', D' \in \mathbb{P}(W)$, such that no three points in either collection of four points are collinear, there exists a unique vector space isomorphism $T : V \to W$ such that $\phi_T(A) = A'$, $\phi_T(B) = B'$, $\phi_T(C) = C'$, $\phi_T(D) = D'$.*

Proof Again, the proof is similar to that of Proposition 1.4.6. We omit the details. □

Example 3.1.4 Let $V = F^3$ and $W = V^\vee$. V and $V^\vee$ are isomorphic vector spaces: after choosing a basis for V, one can write an isomorphism $V \cong V^\vee$ by sending the basis elements to three linearly independent vectors in $V^\vee$, but there is no way to do this canonically.

Let $\mathbf{e_1}, \mathbf{e_2}, \mathbf{e_3} \in V$ be the standard basis, and let $\mathbf{f_1}, \mathbf{f_2}, \mathbf{f_3} \in V^\vee$ be the *dual basis*, which is the unique basis that satisfies

$$\mathbf{f_i}(\mathbf{e_j}) = \delta_{ij}$$

for $i, j = 1, 2, 3$. A vector space isomorphism $T : V \to V^\vee$ is defined by $T(\mathbf{e_i}) = \mathbf{f_i}$ for $i = 1, 2, 3$. Now, given a point

$$P = [x_0 : y_0 : z_0] = [x_0\mathbf{e_1} + y_0\mathbf{e_2} + z_0\mathbf{e_3}] \in \mathbb{P}(V),$$

we compute

$$\phi_T(P) = [T(x_0\mathbf{e_1} + y_0\mathbf{e_2} + z_0\mathbf{e_3})] = x_0\mathbf{f_1} + y_0\mathbf{f_2} + z_0\mathbf{f_3} = [x_0 : y_0 : z_0],$$

where both sets of projective coordinates are computed with respect to the bases given above. The point $\phi_T(P) \in \mathbb{P}(V^\vee)$ defines a line $\ell \subset \mathbb{P}(V)$ whose equation is

$$(x_0\mathbf{f_1} + y_0\mathbf{f_2} + z_0\mathbf{f_3})(x\mathbf{e_1} + y\mathbf{e_2} + z\mathbf{e_3}) = x_0x + y_0y + z_0z = 0.$$

Example 3.1.5 Let V and W, as well as their bases, be as in Example 3.1.4. This time, we define a vector space isomorphism $T : V \to V^\vee$ by $T(\mathbf{e_1}) = \mathbf{f_1}$, $T(\mathbf{e_2}) = \mathbf{f_1} + \mathbf{f_2}$, $T(\mathbf{e_3}) = \mathbf{f_1} + \mathbf{f_2} + \mathbf{f_3}$. Given a point

$$P = [x_0 : y_0 : z_0] = [x_0\mathbf{e_1} + y_0\mathbf{e_2} + z_0\mathbf{e_3}] \in \mathbb{P}(V),$$

we compute

$$\begin{aligned}\phi_T(P) &= [T(x_0\mathbf{e_1} + y_0\mathbf{e_2} + z_0\mathbf{e_3})] \\ &= (x_0 + y_0 + z_0)\mathbf{f_1} + (y_0 + z_0)\mathbf{f_2} + z_0\mathbf{f_3} \\ &= [x_0 + y_0 + z_0 : y_0 + z_0 : z_0].\end{aligned}$$

The point $\phi_T(P) \in \mathbb{P}(V^\vee)$ defines a line $\ell \subset \mathbb{P}(V)$ whose equation is

$$\begin{aligned}0 &= ((x_0 + y_0 + z_0)\mathbf{f_1} + (y_0 + z_0)\mathbf{f_2} + z_0\mathbf{f_3})(x\mathbf{e_1} + y\mathbf{e_2} + z\mathbf{e_3}) \\ &= (x_0 + y_0 + z_0)x + (y_0 + z_0)y + z_0 z.\end{aligned}$$

Giving a vector space isomorphism $T : V \to V^\vee$ is equivalent to giving a nondegenerate bilinear form on V, also denoted by T, by $(\mathbf{v}, \mathbf{v}') \mapsto T(\mathbf{v}, \mathbf{v}') = T(\mathbf{v})(\mathbf{v}')$.

Definition 3.1.6 A *polarity* on $\mathbb{P}^2(F)$ is an isomorphism

$$\phi_T : \mathbb{P}^2(F) = \mathbb{P}(F^3) \to \mathbb{P}((F^3)^\vee),$$

where $T : F^3 \to (F^3)^\vee$ is a vector space isomorphism whose associated bilinear form on V is symmetric.

Remark 3.1.7 The isomorphism ϕ_T given in Example 3.1.4 is a polarity, as can easily be checked.

A polarity assigns to a point $P \in \mathbb{P}^2(F)$ a line $\ell_P = \phi_T(P) \in \mathbb{P}((F^3)^\vee)$, called the *polar* of P. Conversely, it assigns to a line $\ell \subset \mathbb{P}^2(F)$ a point $P_\ell = \phi_T^{-1}(\ell)$, called the *pole* of ℓ. It is clear that $\ell_{P_\ell} = \ell$ and $P_{\ell_P} = P$ for every $P \in \mathbb{P}^2(F)$ and every line $\ell \subset \mathbb{P}^2(F)$. It is also clear that these assignments are bijective.

Until the end of the section, we assume that a polarity on $\mathbb{P}^2(F)$ has been given.

The requirement of symmetry in Definition 3.1.6 is crucial for the following result:

Proposition 3.1.8 (The Pole and Polar Property) *Let $P, Q \in \mathbb{P}^2(F)$. Then $P \in \ell_Q$ if and only if $Q \in \ell_P$.*

Proof Let $\mathbf{v}, \mathbf{w} \in F^3$ such that $P = [\mathbf{v}]$ and $Q = [\mathbf{w}]$. $P \in \ell_Q$ is equivalent to $T(\mathbf{w}, \mathbf{v}) = 0$ which, by symmetry, is equivalent to $T(\mathbf{v}, \mathbf{w}) = 0$ and hence $Q \in \ell_P$. □

Definition 3.1.9 Two points $P, Q \in \mathbb{P}^2(F)$ are *conjugate* if $P \in \ell_Q$ or, equivalently, $Q \in \ell_P$. Two lines $\ell, m \subset \mathbb{P}^2(F)$ are *conjugate* if $P_\ell \in m$ or, equivalently, $P_m \in \ell$. A point P is *self-conjugate* if it is conjugate to itself. A line ℓ is *self-conjugate* if it is conjugate to itself.

Remark 3.1.10 Conjugacy is a symmetric relation by the Pole and Polar Property.

Proposition 3.1.11 *If $P, Q \in \mathbb{P}^2(F)$ are two distinct self-conjugate points, then PQ is not a self-conjugate line.*

Proof Suppose that $a = PQ$ is self-conjugate, with pole $A \in PQ$. Since P and Q are distinct, at least one of them is different from A. Suppose, without loss of generality, that $P \neq A$. Since $P \in a = \ell_A$, $A \in \ell_P$ by the Pole and Polar Property. As $P \in \ell_P$ also, $\ell_P = PA = a$ is the polar of P. This is a contradiction, since the same line cannot be the polar of two distinct points. □

Corollary 3.1.12 *A self-conjugate line in $\mathbb{P}^2(F)$ contains exactly one self-conjugate point, its pole.*

Exercises

1. A *self-polar triangle* is a triangle where the polar of every vertex is the opposite side and the pole of every side is the opposite vertex. Find self-polar triangles for Example 3.1.4. Can you find a criterion to determine when three points $P_i = [x_i : y_i : z_i] \in \mathbb{P}^2(F)$ form a self-polar triangle with respect to the polarity of Example 3.1.4?
2. Show that the bilinear form associated to the isomorphism $T : V \to V^\vee$ in Example 3.1.5 is not symmetric. Find examples showing that the Pole and Polar Property does not hold in that case. Therefore, the symmetry assumption is necessary in the proof of the Pole and Polar Property.
3. Let $V = F^3$. After choosing a basis for V, and the associated dual basis for $V^\vee$, a vector space isomorphism T can be represented by a 3×3 matrix A. Prove that A is also the matrix of the bilinear form T with respect to the chosen basis of V. Prove also that the bilinear form T is symmetric if and only if A is symmetric.
4. Let V, $W = V^\vee$, as well as their bases, be as in Examples 3.1.4 and 3.1.5. Go through the calculations in those examples for the following vector space isomorphisms $T : V \to V^\vee$:
   (i) $T(\mathbf{e_1}) = \mathbf{f_1}, T(\mathbf{e_2}) = \mathbf{f_3}, T(\mathbf{e_3}) = \mathbf{f_2}$,
   (ii) $T(\mathbf{e_1}) = \mathbf{f_1} + 2\mathbf{f_2}, T(\mathbf{e_2}) = 2\mathbf{f_1} + 2\mathbf{f_2} - \mathbf{f_3}, T(\mathbf{e_3}) = -\mathbf{f_2} + \mathbf{f_3}$,
   (iii) $T(\mathbf{e_1}) = \mathbf{f_1} + \mathbf{f_2} + \mathbf{f_3}, T(\mathbf{e_2}) = \mathbf{f_2} + 2\mathbf{f_3}, T(\mathbf{e_3}) = -4\mathbf{f_3}$,
   (iv) $T(\mathbf{e_1}) = -\mathbf{f_2} + \mathbf{f_3}, T(\mathbf{e_2}) = -\mathbf{f_1} - \mathbf{f_3}, T(\mathbf{e_3}) = \mathbf{f_1} - \mathbf{f_2}$.
   Which cases above define polarities on $\mathbb{P}^2(F)$? For those that do, write down criteria for a point $P = [x_0 : y_0 : z_0] \in \mathbb{P}^2(F)$ to be self-conjugate.

3.2 Conics

In Example 3.1.4, where we had a polarity on $\mathbb{P}^2(F)$, it is possible to easily deduce a criterion for a given point $P = [x_0 : y_0 : z_0]$ to be self-conjugate: P is self-conjugate—that is, $P \in \ell_P$—if and only if $x_0^2 + y_0^2 + z_0^2 = 0$. Exercise 3.1.4 contained other such examples. In general, given a polarity on $\mathbb{P}^2(F)$, the set of points that are self-conjugate can be described as the set of solutions of a homogeneous polynomial of degree 2.

Definition 3.2.1 A *conic* is a subset $C \subset \mathbb{P}^2(F)$ defined by a homogeneous equation of degree 2, which we write as

$$C : ax^2 + by^2 + cz^2 + 2dxy + 2exz + 2fyz = 0,$$

where $a, b, c, d, e, f \in F$, not all zero. If

$$\begin{vmatrix} a & d & e \\ d & b & f \\ e & f & c \end{vmatrix} \neq 0,$$

then the conic is *nondegenerate*; otherwise it is *degenerate*. Unless otherwise stated, by a "conic" we shall mean a nondegenerate conic.

Remark 3.2.2 Both nondegenerate and degenerate conics always exist over F: $C : x^2 - yz = 0$ is nondegenerate, while $D : xy = 0$ is degenerate.

Remark 3.2.3 Giving a nondegenerate conic $C \subset \mathbb{P}^2(F)$ is equivalent to giving a polarity on $\mathbb{P}^2(F)$. Indeed, let $\mathbf{e_1}, \mathbf{e_2}, \mathbf{e_3} \in F^3$ be the standard basis and $\mathbf{f_1}, \mathbf{f_2}, \mathbf{f_3} \in (F^3)^\vee$ be the dual basis, as in Example 3.1.4. There is a unique vector space isomorphism $T : F^3 \to (F^3)^\vee$ whose matrix with respect to the chosen bases of F^3 and $(F^3)^\vee$ is the nonsingular matrix

$$\begin{bmatrix} a & d & e \\ d & b & f \\ e & f & c \end{bmatrix},$$

and T defines a polarity on $\mathbb{P}^2(F)$. It is clear that the set of self-conjugate points of this polarity is precisely the conic C, and that this construction can be reversed. In other words, a point $P = [\mathbf{v}] \in \mathbb{P}^2(F)$ lies on C if and only if $T(\mathbf{v}, \mathbf{v}) = 0$.

Remark 3.2.4 Write

$$\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$

and

$$A = \begin{bmatrix} a & d & e \\ d & b & f \\ e & f & c \end{bmatrix}.$$

Then the equation of the conic in Definition 3.2.1 can also be written in the form

$$\mathbf{v}^t A \mathbf{v} = 0.$$

Just as an automorphism of $\mathbb{P}^2(F)$ maps lines to lines, it maps conics to conics.

Proposition 3.2.5 *An automorphism $\phi \in Aut(\mathbb{P}^2(F))$ transforms a conic in $\mathbb{P}^2(F)$ into another conic in $\mathbb{P}^2(F)$. Nondegenerate conics are transformed into nondegenerate conics, and degenerate conics are transformed into degenerate conics.*

Proof By Theorem 1.4.13, $\mathrm{Aut}(\mathbb{P}^2(F))$ is generated by two types of automorphisms: ϕ_T for $T \in \mathrm{PGL}_3(F)$ and χ_σ for $\sigma \in \mathrm{Aut}(F)$. It therefore suffices to prove the claims for these two kinds of automorphisms.

Given a conic $C : ax^2 + by^2 + cz^2 + 2dxy + 2exz + 2fyz = 0$ and $\sigma \in \mathrm{Aut}(F)$ it is easy to see that χ_σ transforms C to

$$\chi_\sigma(C) : \sigma(a)x^2 + \sigma(b)y^2 + \sigma(c)z^2 + 2\sigma(d)xy + 2\sigma(e)xz + 2\sigma(f)yz = 0,$$

which is nondegenerate if and only if C is, in view of the identity

$$\begin{vmatrix} \sigma(a) & \sigma(d) & \sigma(e) \\ \sigma(d) & \sigma(b) & \sigma(f) \\ \sigma(e) & \sigma(f) & \sigma(c) \end{vmatrix} = \sigma\left(\begin{vmatrix} a & d & e \\ d & b & f \\ e & f & c \end{vmatrix} \right).$$

Now let $T \in \mathrm{PGL}_3(F)$. ϕ_T transforms C into the conic whose equation is, using the notation of Remark 3.2.4,

$$\phi_T(C) : (T^{-1}\mathbf{v})^t A (T^{-1}\mathbf{v}) = \mathbf{v}^t ((T^{-1})^t A T^{-1})\mathbf{v} = 0.$$

It is clear that $\phi_T(C)$ is nondegenerate if and only if C is nondegenerate, since the symmetric matrix $(T^{-1})^t A T^{-1}$ is nonsingular if and only if A is nonsingular. □

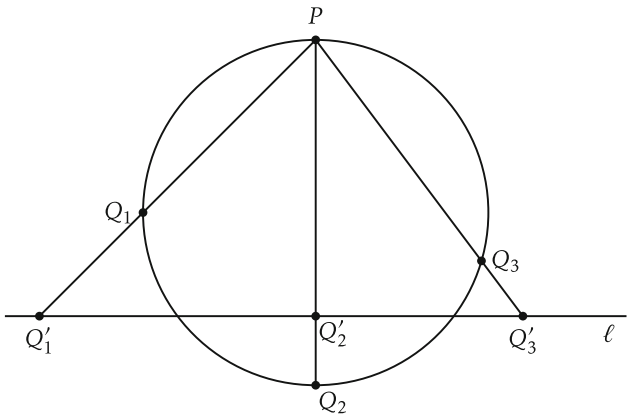


Fig. 3.1 A one-to-one correspondence between $\mathbb{C}$-valued points of the conic $C : x^2+y^2+z^2 = 0$ and the line $\ell : x = 0$ in $\mathbb{P}^2(\mathbb{C})$

Recall that a line in $\mathbb{P}^2(F)$ is defined by a homogeneous equation of degree 1 and the points on a line are in one-to-one correspondence with $F \cup \{\infty\}$ (Definition 1.2.7 and Proposition 1.7.19). There are therefore always points on a line. This is not necessarily the case with conics, as Example 3.2.7 below shows. Given a conic $C \subset \mathbb{P}^2(F)$, it is therefore sometimes necessary to consider the equation of C as defining a conic in $\mathbb{P}^2(K)$ for a field extension K of F. This is known as *extending the base field*. In order to distinguish points of the conic C in $\mathbb{P}^2(F)$ or $\mathbb{P}^2(K)$, we shall use the notations $C(F)$ and $C(K)$ to denote the *F-valued* and *K-valued* points of C, respectively. Exercise 1 asks you to prove that any conic C over a field F necessarily has points over at most a quadratic extension of F.

Example 3.2.6 (Projection from a Point) Let $C : x^2+y^2+z^2 = 0$ in $\mathbb{P}^2(\mathbb{C})$. C is a nondegenerate conic and $C(\mathbb{C}) \neq \emptyset$. In fact, there is a one-to-one correspondence between the points of $C(\mathbb{C})$ and the points of any line in $\mathbb{P}^2(\mathbb{C})$. In order to establish this one-to-one correspondence, let $P = [1 : 0 : i] \in C(\mathbb{C})$ and $\ell : x = 0$. Given any point $Q \in C(\mathbb{C})$ with $Q \neq P$, we let $Q' = PQ \cdot \ell$. We now claim that the mapping $Q \mapsto Q'$ is a bijection between $C(\mathbb{C})$ and $\ell(\mathbb{C})$ (Fig. 3.1).

We note that any line through P in $\mathbb{P}^2(\mathbb{C})$ intersects ℓ at a unique point and this establishes a bijection—indeed, an elementary correspondence—between lines through P and points on ℓ. Let $[0 : \lambda : \mu] \in \ell$ be an arbitrary point, where λ and μ are not both zero. The line passing through P and $[0 : \lambda : \mu]$ has equation $-i\lambda x - \mu y + \lambda z = 0$ and to find its intersections with C, we must solve the system

$$\begin{aligned} x^2 + y^2 + z^2 &= 0, \\ -i\lambda x - \mu y + \lambda z &= 0. \end{aligned} \tag{3.1}$$

If $\lambda \neq 0$, we eliminate z from system (3.1) and obtain

$$y[(\lambda^2 + \mu^2)y + 2i\lambda\mu x] = 0. \tag{3.2}$$

The solution $y = 0$ yields P. The other solution yields

$$Q = [\lambda^2 + \mu^2 : -2i\lambda\mu : i(\lambda^2 - \mu^2)]. \tag{3.3}$$

If $\lambda = 0$, the system (3.1) has the solutions P and $[1 : 0 : -i]$ the latter of which, we note, is also given by (3.3). Therefore, Eq. (3.3) can be used to establish a bijection from $\ell(\mathbb{C})$ to $C(\mathbb{C})$.

Note that when $\lambda = 1$ and $\mu = 0$, Eq. (3.2) becomes $y^2 = 0$, having a unique solution with multiplicity 2, namely $y = 0$, which yields P. We therefore imagine the line $-ix + z = 0$ as intersecting the conic C twice at P. In other words, the line $-ix + z = 0$ is *tangent* to C at P. Informally, the tangent line can be thought of as the "limit" of the line PQ as Q approaches P.

Example 3.2.7 Let $C : x^2 + y^2 + z^2 = 0$ in $\mathbb{P}^2(\mathbb{R})$. C is nondegenerate but $C(\mathbb{R}) = \emptyset$. However, after extension of the base field to $\mathbb{C}$, $C(\mathbb{C})$ is the same as in Example 3.2.6.

Example 3.2.8 Let $C : x^2 - y^2 = 0$ in $\mathbb{P}^2(\mathbb{R})$. C is a degenerate conic, and $C(\mathbb{R})$ is the union of the two lines $\ell_1 : x - y = 0$ and $\ell_2 : x + y = 0$ in $\mathbb{P}^2(\mathbb{R})$ since $x^2 - y^2$ can be factorized as $(x-y)(x+y)$ over $\mathbb{R}$. A special case of this is when the two factors are in fact identical, such as the conic $D : x^2 = 0$ in $\mathbb{P}^2(\mathbb{R})$. In this case, $D(\mathbb{R})$ is simply the set of points on the line $\ell : x = 0$. Informally, D can be thought to consist of two copies of ℓ since x has the exponent 2 in the defining equation of D (Fig. 3.2).

Example 3.2.9 $C : x^2 + y^2 = 0$ in $\mathbb{P}^2(\mathbb{R})$ is a degenerate conic. After extension of the base field to $\mathbb{C}$, $C(\mathbb{C})$ can be decomposed as the union of the two lines $\ell_1 : x - iy = 0$ and $\ell_2 : x + iy = 0$ in $\mathbb{P}^2(\mathbb{C})$ since $x^2 + y^2$ can be factorized as $(x - iy)(x + iy)$ over $\mathbb{C}$. Over $\mathbb{R}$, the only point of C is $[0 : 0 : 1]$, which is simply $\ell_1 \cdot \ell_2$ (Fig. 3.3).

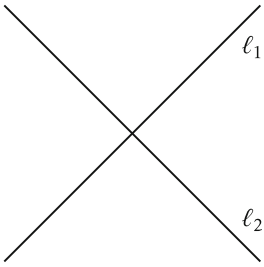


Fig. 3.2 A degenerate conic in $\mathbb{P}^2(\mathbb{R})$ that is the union of the two lines $\ell_1 : x - y = 0$ and $\ell_2 : x + y = 0$ in $\mathbb{P}^2(\mathbb{R})$ (Example 3.2.8)

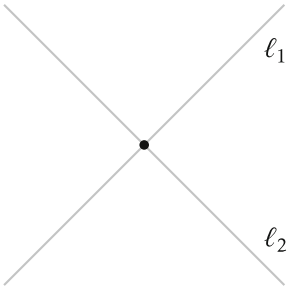


Fig. 3.3 A degenerate conic in $\mathbb{P}^2(\mathbb{R})$ whose set of points over $\mathbb{C}$ is the union of the two lines $\ell_1 : x - iy = 0$ and $\ell_2 : x + iy = 0$ in $\mathbb{P}^2(\mathbb{C})$ and whose only point over $\mathbb{R}$ is the point of intersection $\ell_1 \cdot \ell_2$ (Example 3.2.9). The parts of the lines drawn in gray indicate that points on them are $\mathbb{C}$-valued and not $\mathbb{R}$-valued points

One of the axioms for projective planes, namely **P2**, concerns intersections between lines: "Two distinct lines intersect in a unique point." Lines in $\mathbb{P}^2(F)$ are defined by the vanishing of homogeneous linear polynomials; therefore **P2** tells us completely what happens when we intersect two "degree-1 curves" in $\mathbb{P}^2(F)$ (that is, subsets of $\mathbb{P}^2(F)$ defined by homogeneous linear polynomials). A natural follow-up question would be to ask about intersections of a degree-1 curve and a degree-2 curve in $\mathbb{P}^2(F)$; that is, a line and a conic. The following proposition answers that question for nondegenerate conics; the case of degenerate conics is left to the reader.

Proposition 3.2.10 *Let $C \subset \mathbb{P}^2(F)$ be a nondegenerate conic and $\ell \subset \mathbb{P}^2(F)$ be a line. Then C and ℓ intersect in at most 2 points. If an intersection point is F-valued, then all intersection points are F-valued. If there are no intersection points over F, then there exists a quadratic extension K of F such that there are intersection points over K.*

Proof Choose two distinct points $[\mathbf{v_1}], [\mathbf{v_2}] \in \ell$. Then any point in ℓ can be written as $[\lambda\mathbf{v_1} + \mu\mathbf{v_2}]$ for $\lambda, \mu \in F$, not both zero. Since C is a nondegenerate conic, it is defined by a nondegenerate, symmetric bilinear form $T : F^3 \times F^3 \to F$. Given an arbitrary point $[\lambda\mathbf{v_1} + \mu\mathbf{v_2}] \in \ell$, we calculate

$$T(\lambda\mathbf{v_1} + \mu\mathbf{v_2}, \lambda\mathbf{v_1} + \mu\mathbf{v_2}) = \lambda^2 T(\mathbf{v_1}, \mathbf{v_1}) + 2\lambda\mu T(\mathbf{v_1}, \mathbf{v_2}) + \mu^2 T(\mathbf{v_2}, \mathbf{v_2}).$$

We note that $T(\mathbf{v_1}, \mathbf{v_1}), T(\mathbf{v_1}, \mathbf{v_2}), T(\mathbf{v_2}, \mathbf{v_2})$ cannot all be zero: if that were the case, by adding a suitably chosen third vector $\mathbf{v_3}$ to the linearly independent vectors $\mathbf{v_1}$ and $\mathbf{v_2}$, we would have a basis of V, and the matrix of the bilinear form T with respect to this basis would be of the form

$$\begin{bmatrix} 0 & 0 & * \\ 0 & 0 & * \\ * & * & * \end{bmatrix},$$

contradicting the nondegenerateness of T. Therefore, the points of intersection of ℓ with C are given by the quadratic equation

$$\lambda^2 T(\mathbf{v_1}, \mathbf{v_1}) + 2\lambda\mu T(\mathbf{v_1}, \mathbf{v_2}) + \mu^2 T(\mathbf{v_2}, \mathbf{v_2}) = 0,$$

where not all the coefficients are zero. The claims now follow from general properties of solutions of quadratic equations. □

Proposition 3.2.10 raises the question of when a line and a conic intersect in a unique point. In fact, given a nondegenerate conic $C \subset \mathbb{P}^2(F)$ and a point $P = [x_0 : y_0 : z_0] \in C(F)$, there is always a unique line passing through P, called the *tangent line to C at P* and denoted t_P^C or t_P when C is clear from the context, that intersects C uniquely at P. Let

$$\mathbf{v_0} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} \in F^3$$

be a vector corresponding to P, and let

$$\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in F^3$$

be any vector such that the line passing through $[\mathbf{v_0}]$ and $[\mathbf{v}]$ is tangent to C. From the proof of Proposition 3.2.10, we get the equation

$$\lambda^2 T(\mathbf{v_0}, \mathbf{v_0}) + 2\lambda\mu T(\mathbf{v_0}, \mathbf{v}) + \mu^2 T(\mathbf{v}, \mathbf{v}) = 0,$$

where $\lambda, \mu \in F$, not both zero. Now, since $P \in C(F)$, $T(\mathbf{v_0}, \mathbf{v_0}) = 0$. In order for the equation

$$2\lambda\mu T(\mathbf{v_0}, \mathbf{v}) + \mu^2 T(\mathbf{v}, \mathbf{v}) = 0$$

to have a unique solution, a necessary and sufficient condition is $T(\mathbf{v_0}, \mathbf{v}) = 0$. To summarize, the tangent line t_P^C consists of the points $[\mathbf{v}] \in \mathbb{P}^2(F)$ such that $T(\mathbf{v_0}, \mathbf{v}) = 0$. An equation for t_P^C can be found as

$$0 = T(\mathbf{v_0}, \mathbf{v})$$

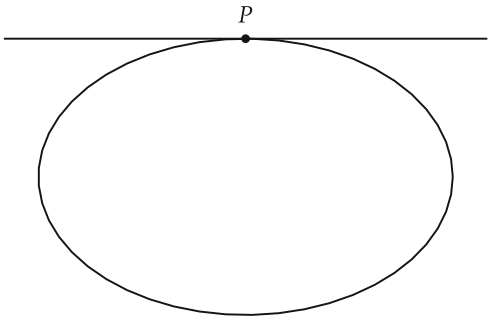


Fig. 3.4 The tangent line to C at P

$$= \mathbf{v}_0^t A \mathbf{v}$$

$$= \begin{bmatrix} x_0 & y_0 & z_0 \end{bmatrix} \begin{bmatrix} a & d & e \\ d & b & f \\ e & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= (ax_0 + dy_0 + ez_0)x + (dx_0 + by_0 + fz_0)y + (ex_0 + fy_0 + cz_0)z.$$

Note that the coefficients of x, y, z in the above equation do not vanish simultaneously since C is nondegenerate; the tangent line t_P^C is therefore uniquely determined by C and P (Fig. 3.4). At this point, we introduce some notation. Given a polynomial $f(x, y, z)$ with coefficients in F, let f_x, f_y, f_z denote the (formal) partial derivatives of f with respect to x, y, z. The equation of t_P^C can now be written as

$$t_P^C : f_x(P)x + f_y(P)y + f_z(P)z = 0.$$

Proposition 3.2.11 *Let $C \subset \mathbb{P}^2(F)$ be a nondegenerate conic. Then there are two possibilities:*

(i) $C(F) = \emptyset$ but there exists a quadratic extension K of F such that $C(K) \neq \emptyset$,
(ii) $C(F) \neq \emptyset$. In this case, choosing a point $P \in C(F)$ and a line $\ell \subset \mathbb{P}^2(F)$ that does not pass through P, projection of C onto ℓ from P as in Example 3.2.6 provides a bijection between $C(F)$ and the points of ℓ. Under this projection, P itself corresponds to $t_P^C \cdot \ell$.

In particular, if F is quadratically closed, then $C(F) \neq \emptyset$.

Proof *Case (i).* Consider any line $\ell \subset \mathbb{P}^2(F)$. By Proposition 3.2.10, there exists a quadratic extension K of F such that C and ℓ have intersection points over K; hence $C(K) \neq \emptyset$.

Case (ii). The proof is very similar to Example 3.2.6. The key observation is that any line through P other than the tangent line t_P^C intersects C in an F-valued point that is different from P. We leave the details to the reader. □

Proposition 3.2.12 *Let $C \subset \mathbb{P}^2(F)$ be a degenerate conic. Then there are two possibilities:*

(i) The equation defining C can be factorized as a product of two linear homogeneous polynomials (not necessarily distinct) over F. In this case, $C(F)$ is the union of the F-valued points of two lines (not necessarily distinct) in $\mathbb{P}^2(F)$.
(ii) The equation defining C is irreducible over F but can be factored as a product of two distinct linear homogeneous polynomials over a quadratic extension K of F. In this case, $C(K)$ is the union of the K-valued points of two distinct lines in $\mathbb{P}^2(K)$. $C(F)$ consists of one point, namely the intersection of the two lines given by the linear homogeneous factors of C over K.

Proof Let A be the matrix of coefficients of C as in Remark 3.2.4. Since A is singular, we can find a nonzero $\mathbf{v_0} \in F^3$ such that $\mathbf{v}_\mathbf{0}^t A = 0$; such a vector satisfies $P_0 = [\mathbf{v_0}] \in C(F)$. By Proposition 3.2.5 we can assume, without loss of generality, that $P_0 = [1 : 0 : 0]$. This implies that $a = d = e = 0$, and the equation of C is therefore $C : by^2 + cz^2 + 2fyz = 0$. The statement now follows from well-known facts about the factorization of quadratic polynomials (see Exercise 3.2.8). □

We finish this section with a brief discussion of *diagonalization of conics*. In many cases, calculations and proofs of theorems would be simplified if the equations of our conics could be written in *diagonal* form

$$C : \alpha x^2 + \beta y^2 + \gamma z^2 = 0,$$

as was the case in Examples 3.2.6, 3.2.7, 3.2.8, and 3.2.9.

We can try to use Proposition 3.2.5 to transform the equation of a given conic into diagonal form. For that purpose, we must find a matrix $Q \in \mathrm{GL}_3(F)$ such that the conic $C : ax^2 + by^2 + cz^2 + 2dxy + 2exz + 2fyz = 0$ in $\mathbb{P}^2(F)$ is transformed into

$$\phi_{Q^{-1}}(C) : \mathbf{v}^t (Q^t A Q)\mathbf{v} = 0,$$

where

$$A = \begin{bmatrix} a & d & e \\ d & b & f \\ e & f & c \end{bmatrix},$$

$$\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$

and $Q^t A Q$ is a diagonal matrix

$$Q^t A Q = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & \gamma \end{bmatrix}$$

with $\alpha, \beta, \gamma \in F$. It turns out that it is always possible to find such a Q.

Theorem 3.2.13 *Given a conic $C \subset \mathbb{P}^2(F)$ defined by a symmetric matrix A as in the above discussion, there is a $Q \in GL_3(F)$ such that $Q^t A Q$ is diagonal.*

Proof See [10, Corollary 2.4]. □

Exercises

1. Suppose that we are given a conic $C : ax^2 + by^2 + cz^2 + 2dxy + 2exz + 2fyz = 0$ in $\mathbb{P}^2(F)$. Prove that C always has points over a quadratic extension K of F; in other words, prove that there exists a point $[x_0 : y_0 : z_0]$ with $x_0, y_0, z_0 \in K$, not all zero, satisfying the equation of C. (Hint: Examine the proof of Proposition 3.2.10 carefully.)
2. Classify the following conics as nondegenerate or degenerate:
   (i) $C : 2x^2 + xy + y^2 - 2z^2 = 0$ in $\mathbb{P}^2(\mathbb{R})$,
   (ii) $C : x^2 + 4xy + 5y^2 + 6yz + 9z^2 = 0$ in $\mathbb{P}^2(\mathbb{R})$,
   (iii) $C : x^2 + 4xy + 5y^2 + 6yz + 9z^2 = 0$ in $\mathbb{P}^2(\mathbb{C})$,
   (iv) $C : x^2 + 2xz - y^2 + z^2 = 0$ in $\mathbb{P}^2(\mathbb{C})$.
3. For the conics listed in the previous exercise, investigate their sets of points in view of Propositions 3.2.11 and 3.2.12.
4. Finish the proof of part (ii) of Proposition 3.2.11.
5. Verify that the following conics are nondegenerate, and find equations for their tangent lines at the given points:
   (i) $C : x^2 + 4y^2 + 5z^2 = 0$ in $\mathbb{P}^2(\mathbb{C})$, $P = [3 : i : -i]$,
   (ii) $C : x^2 - y^2 - z^2 = 0$ in $\mathbb{P}^2(\mathbb{R})$, $P = [\sqrt{2} : 1 : 1]$,
   (iii) $C : 4x^2 + y^2 - z^2 = 0$ in $\mathbb{P}^2(\mathbb{R})$, $P = [-1 : -2 : 1]$,
   (iv) $C : x^2 + y^2 - 25z^2 = 0$ in $\mathbb{P}^2(\mathbb{Q})$, $P = [4 : -3 : 1]$.
6. Investigate the points of intersection between a line and a degenerate conic in $\mathbb{P}^2(F)$, similarly to Proposition 3.2.10.
7. Let C be a nondegenerate conic over F and $T \in \mathrm{PGL}_3(F)$. Let $C' = \phi_T(C)$, which is a nondegenerate conic by Proposition 3.2.5. Prove that $C'(F) \neq \emptyset$ if and only if $C(F) \neq \emptyset$.

8. Let $f(x, y)$ be a homogeneous quadratic polynomial over F. Prove that either f can be factorized as the product of two linear homogeneous polynomials over F, or f can be so factorized over a quadratic extension of F but not F itself.
9. Let $C \subset \mathbb{P}^2(F)$ be a nondegenerate conic such that $C(F) \neq \emptyset$.
   (i) Given a point $P \in C(F)$, prove that the tangent line t_P cannot intersect C at a point other than P.
   (ii) Given a point $P \in C(F)$, prove that the tangent line t_P is the polar of P.
   (iii) Given two distinct points $P, Q \in C(F)$, prove that the secant line PQ is the polar of the point $R = t_P \cdot t_Q$.
   (iv) Given a point $P \notin C(F)$, prove that there are at most two tangent lines to C that pass through P.
10. Let $C \subset \mathbb{P}^2(F)$ be a conic over a quadratically closed field F. Prove that C is nondegenerate if and only if there do not exist three distinct collinear points in $C(F)$.
11. Let C and D be degenerate conics over a quadratically closed field F such that $C(F) = D(F)$. Prove that the equations defining C and D are scalar multiples of each other.
12. The process of diagonalizing the equation of a conic is different from the diagonalization of the matrix A, as this exercise shows. Let $C : x^2 - y^2 + z^2 + 4xy = 0$ be a conic in $\mathbb{P}^2(\mathbb{Q})$. Prove that the corresponding symmetric matrix A is not diagonalizable over $\mathbb{Q}$. Find a matrix $Q \in \mathrm{GL}_3(\mathbb{Q})$ such that $Q^t AQ$ is diagonal. (Hint: For the second part, read the proofs of [10, Representation Criterion 2.3 and Corollary 2.4] carefully.)
13. Let $C \subset \mathbb{P}^2(F)$ be a nondegenerate conic, and $P, Q, R, S \in C(F)$ four distinct points. Note that $PQRS$ is a complete quadrangle. Let X, Y, Z be the diagonal points of $PQRS$. Prove that XYZ is a self-polar triangle. (See Exercise 3.1.1 for the definition.)

3.3 The Theorems of Pascal and Bézout

In this section, we use the tools developed in Sect. 3.2 to prove two important theorems. The first of these, namely Pascal's Theorem, is a generalization of Pappus' Theorem (Theorem 1.6.6). Recall that in Pappus' Theorem we had two distinct lines, three distinct points on one line, and three distinct points on the other line, such that none of the six points was the point of intersection of the two lines. We have seen in Proposition 3.2.12, as well as in Examples 3.2.8 and 3.2.9, that the union of two lines can be considered to be a degenerate conic. Pascal's Theorem generalizes Pappus' Theorem to nondegenerate conics.

Theorem 3.3.1 (Pascal's Theorem) *Let $C \subset \mathbb{P}^2(F)$ be a nondegenerate conic, and let six distinct points $A, B, C, D, E, F \in C(F)$ be given. Then the intersection points $X = AE \cdot BD$, $Y = AF \cdot CD$, $Z = BF \cdot CE$ are collinear.*

Proof By Proposition 3.2.10, no 3 of the points A, B, C, D, E, F are collinear. Therefore, using Proposition 1.4.6 we can assume, without loss of generality, that $A = [1 : 0 : 0]$, $B = [0 : 1 : 0]$, $C = [0 : 0 : 1]$, $D = [1 : 1 : 1]$. Let the equation of C be

$$ax^2 + by^2 + cz^2 + 2dxy + 2exz + 2fyz = 0.$$

Since $A, B, C, D \in C(F)$, we get $a = b = c = 0$ and $d + e + f = 0$. We can then write the equation of C as

$$dxy + exz - (d + e)yz = 0.$$

Note that $d \neq 0$ since if $d = 0$ then the defining equation of C would be reducible and hence C would be degenerate. Similarly, $e \neq 0$. Note also that all coordinates of E and F are nonzero, since otherwise they would lie on one of $AB : z = 0$, $BC : x = 0$, $AC : y = 0$. Write $E = [1 : \alpha : \beta]$ and $F = [1 : \gamma : \delta]$, with $\alpha, \beta, \gamma, \delta \in F$. Since $E, F \in C(F)$, we can calculate β and δ in terms of α and γ as follows:

$$\beta = \frac{-d\alpha}{e - (d + e)\alpha},$$
$$\delta = \frac{-d\gamma}{e - (d + e)\gamma}.$$

We now calculate the coordinates of X, Y, Z:

$$AE : d\alpha y + \alpha(e - (d + e)\alpha)z = 0,$$

$$BD : x = z,$$

$$X = [d\alpha : \alpha((d + e)\alpha - e) : d\alpha],$$

$$AF : d\gamma y + \gamma(e - (d + e)\gamma)z = 0,$$

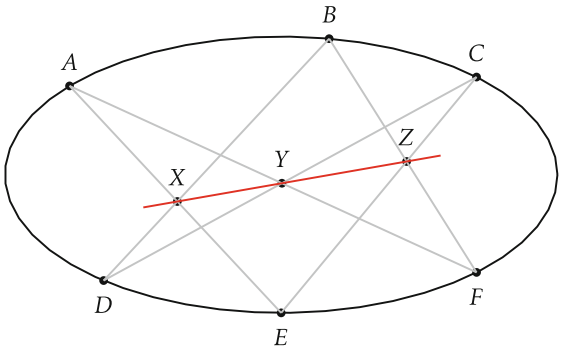
$$CD : x = y,$$

$$Y = [\gamma((d + e)\gamma - e) : \gamma((d + e)\gamma - e) : d\gamma],$$

$$BF : d\gamma x + (e - (d + e)\gamma)z = 0,$$

$$CE : -\alpha x + y = 0,$$

Fig. 3.5 Pascal's Theorem



$$Z = [((d+e)\gamma - e) : \alpha((d+e)\gamma - e) : d\gamma].$$

Proposition 1.2.12 now implies that X, Y, Z are collinear (Fig. 3.5). □

The second main result of this section, Bézout's Theorem (Theorem 3.3.6), concerns the number of intersection points of two conics. In its full generality, as proved in courses in algebraic geometry, Bézout's Theorem states that two distinct plane algebraic curves of degrees m and n over an algebraically closed field intersect in a total of mn points, counted with a suitable definition of multiplicity. We shall prove a weaker form of Bézout's Theorem that states that two distinct nondegenerate conics in $\mathbb{P}^2(F)$, both of which have points over F, intersect in at most 4 points. Our main tool to prove this will be Theorem 3.3.3, which allows us to put equations of nondegenerate conics with points over F into *standard form*; this allows us to express the intersection points of two conics in terms of solutions of a polynomial equation.

Example 3.3.2 Consider the nonsingular conic $C : f(x, y, z) = 0$, where $f(x, y, z) = yz - x^2$, and the points $O = [0 : 0 : 1]$, $A = [2 : 4 : 1]$, $B = [-2 : 4 : 1]$ on it. Of course, if $F = \mathbb{R}$, then the part of this conic that lies in $\mathbb{A}^2(F) = \mathbb{P}^2(F) \setminus \ell_\infty$, where $\ell_\infty : z = 0$, is the parabola $y = x^2$. (See Fig. 3.6.) It is clear that C contains a unique point at infinity: $C \cap \ell_\infty = \{[0 : 1 : 0]\}$.

Let us find equations for the tangent lines to C at O and A. We calculate

$$f_x(A) = (-2x)(A) = -4,$$
$$f_y(A) = (z)(A) = 1,$$
$$f_z(A) = (y)(A) = 4.$$

Hence, the tangent line to C at A has equation $t_A : -4x + y + 4z = 0$.

Similarly, $f_x(O) = 0$, $f_y(O) = 1$, $f_z(O) = 0$, and the tangent line to C at O has equation $t_O : y = 0$. Note that the two tangent lines intersect at $D = [1 : 0 : 1]$ and that no three of the points O, A, B, D are collinear.

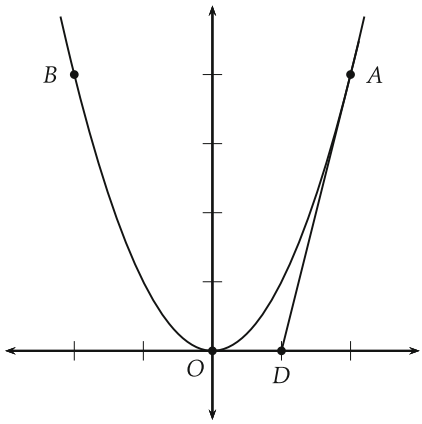


Fig. 3.6 The parabola $y = x^2$ in Example 3.3.2

Theorem 3.3.3 *Let $C \subset \mathbb{P}^2(F)$ be a nondegenerate conic such that $C(F) \neq \emptyset$. Then there exists an automorphism ϕ of $\mathbb{P}^2(F)$ such that $\phi(C)$ is the conic $yz - x^2 = 0$. We call this equation the* standard form *of C.*

Proof By Proposition 3.2.11, it is possible to choose three distinct points $O, A, B \in C(F)$; by Proposition 3.2.10 these three points are not collinear. Let $D = t_O \cdot t_A$. By Exercise 3.2.9 (i), no three of the points O, A, B, D are collinear. Hence, by Proposition 1.4.6, we can find a $T \in \mathrm{PGL}_3(F)$ such that $\phi_T(O) = [0 : 0 : 1]$, $\phi_T(A) = [2 : 4 : 1]$, $\phi_T(B) = [-2 : 4 : 1]$, $\phi_T(D) = [1 : 0 : 1]$. Then $\phi_T(C)$ is also a nondegenerate conic with points over F by Proposition 3.2.5 and Exercise 3.2.7. We claim that $\phi_T(C)$ has equation $yz - x^2 = 0$. To simplify the notation in what follows, we shall put a prime ($'$) next to all objects transformed by $\phi = \phi_T$, such as $C' = \phi(C)$.

Let the equation of C' be

$$C' : ax^2 + by^2 + cz^2 + 2dxy + 2exz + 2fyz = 0.$$

Since $O' \in C'(F)$, $c = 0$. Since $A' \in C'(F)$, $a + 4b + d + e + 2f = 0$. Since $B' \in C'(F)$, $a + 4b - d - e + 2f = 0$. These last two equations give $d = -e$ and $a + 4b + 2f = 0$.

We now calculate the tangent lines to C' at O' and A':

$$\begin{aligned} f_x(O') &= (2ax + 2dy - 2dz)(O') = -2d, \\ f_y(O') &= (2by + 2dx + 2fz)(O') = 2f, \\ f_z(O') &= (-2dx + 2fy)(O') = 0. \end{aligned}$$

Hence, the tangent line to C' at O' has equation $t_{O'} : -dx + fy = 0$. Similarly,

$$f_x(A') = (2ax + 2dy - 2dz)(A') = 4a + 6d,$$
$$f_y(A') = (2by + 2dx + 2fz)(A') = 8b + 4d + 2f,$$
$$f_z(A') = (-2dx + 2fy)(A') = -4d + 8f.$$

The tangent line to C' at A' has equation $t_{A'} : (2a+3d)x+(4b+2d+f)y+(-2d+4f)z = 0$. As D' must lie on both $t_{O'}$ and $t_{A'}$, we have, after a brief calculation, $d = 0$ and $a = -2f$. Hence, C' has equation $-x^2 + yz = 0$. □

Remark 3.3.4 Note that in Theorem 3.3.3, any point on the conic C can be made to correspond to $[0 : 0 : 1]$.

Corollary 3.3.5 *Let $C, D \subset \mathbb{P}^2(F)$ be nondegenerate conics such that $C(F) \neq \emptyset$ and $D(F) \neq \emptyset$. Then there exists an automorphism ϕ of $\mathbb{P}^2(F)$ such that $\phi(C) = D$.*

Theorem 3.3.6 (Bézout's Theorem for Conics (Weak Form)) *Let C and D be two distinct nondegenerate conics over F such that $C(F) \neq \emptyset$ and $D(F) \neq \emptyset$. Then C and D intersect in at most 4 points (Fig. 3.7).*

Proof By Theorem 3.3.3 we can assume, without loss of generality, that C is the conic $yz - x^2 = 0$. Let $\ell_\infty : z = 0$ be the line at infinity. Then C meets ℓ_∞ at $[0 : 1 : 0]$. We now distinguish two cases.

First, assume that $[0 : 1 : 0] \notin D(F)$. Let the equation of D be

$$ax^2 + by^2 + cz^2 + 2dxy + 2exz + 2fyz = 0.$$

Since $[0 : 1 : 0] \notin D(F)$, $b \neq 0$. Working on $\mathbb{A}^2(F) = \mathbb{P}^2(F) \setminus \ell_\infty$, whose points can be taken to satisfy $z = 1$, C and D intersect at the points where

$$bx^4 + 2dx^3 + (a + 2f)x^2 + 2ex + c = 0,$$

which has at most 4 solutions, and for each of these solutions, $y = x^2$ is uniquely determined.

Next, assume that $[0 : 1 : 0] \in D(F)$, which forces $b = 0$. In this case, we already have one point of intersection of C and D, namely $[0 : 1 : 0]$ itself. The remaining intersection points can be found as above, by setting $z = 1$ and plugging $y = x^2$ into the equation of D. This gives

$$2dx^3 + (a + 2f)x^2 + 2ex + c = 0,$$

which has at most 3 solutions. □

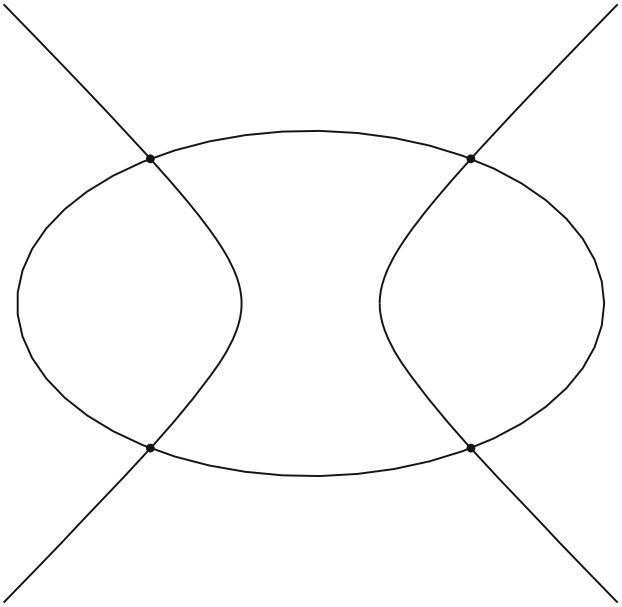


Fig. 3.7 Bézout's Theorem for Conics: two conics intersecting in 4 points

Corollary 3.3.7 *Let F be algebraically closed. Then any two nondegenerate conics over F intersect in at least one point.*

Proof This follows from the fact that any polynomial equation with coefficients in F has at least one solution. □

Exercises

1. Give examples of pairs of nondegenerate conics over $\mathbb{C}$ that intersect in 1, 2, 3, 4 points.
2. This exercise and the next outline Bézout's Theorem in its strong form for two conics. Let F be an algebraically closed field, so that any conic over F has points over F. Let C and D be two distinct nondegenerate conics in $\mathbb{P}^2(F)$, and suppose that $P \in C(F) \cap D(F)$. The *intersection multiplicity* of C and D at P, denoted $m_P(C, D)$, is defined as follows: using Theorem 3.3.3, we can assume that $C : yz = x^2$ and $P = [0 : 0 : 1]$. In the affine plane given by $z = 1$, let the equation of D be $ax^2 + by^2 + 2dxy + 2ex + 2fy = 0$. Substituting $y = x^2$ in this equation gives an equation in x, which has a zero at $x = 0$. Define $m_P(C, D)$ to be the multiplicity of the zero $x = 0$; it is clear that $m_P(C, D) \geq 1$. Of course, one can define $m_P(D, C)$ similarly and it can be proved that $m_P(C, D) = m_P(D, C)$. The strong form of Bézout's Theorem states that

$$\sum_{P \in C(F) \cap D(F)} m_P(C, D) = 4.$$

This sum is called the *intersection number* of C and D. Thus, if we consider the conics C and D to be plane projective curves of degree 2, their intersection number is equal to the product of their degrees.

(i) Prove that $m_P(C, D) = 1$ if and only if the tangent lines t_P^C and t_P^D are distinct.

(ii) Given the conics C and D over $\mathbb{C}$, calculate their intersection numbers explicitly:

- $C : x^2 + 4y^2 = z^2$, $D : 3x^2 - y^2 = z^2$.
- $C : yz = -x^2$, $D : x^2 + (y - z)^2 = z^2$.
- $C : yz = x^2$, $D : yz = 2x^2$.
- $C : x^2 + y^2 = z^2$, $D : 5x^2 + 5y^2 - 5z^2 + 6xy + 6yz = 0$.
- $C : x^2 + y^2 = z^2$, $D : 2x^2 + (y + z)^2 = 4z^2$.

3. Let F be an algebraically closed field. Let C and L be a conic and a line in $\mathbb{P}^2(F)$, respectively, and suppose that $P \in C(F) \cap L(F)$. We can define the intersection multiplicity of C and L at P, denoted $m_P(C, L)$, as in Exercise 2: we assume that $C : yz = x^2$ and $P = [0 : 0 : 1]$, and we let $m_P(C, L)$ be the multiplicity of the zero $x = 0$ in the equation obtained by substituting $y = x^2$ and $z = 1$ in the equation of L. Alternatively, we can assume that $D : y = 0$ and $P = [0 : 0 : 1]$, and we let $m_P(L, C)$ be the multiplicity of the zero $x = 0$ in the equation obtained by substituting $y = 0$ and $z = 1$ in the equation of C. It can be proved that $m_P(C, L) = m_P(L, C)$ and it is evident that $m_P(C, L) \geq 1$.

(i) Prove that $m_P(C, L) = 1$ if and only if the tangent line t_P^C is distinct from L; moreover, prove that if $t_P^C = L$, then $m_P(C, L) = 2$.

(ii) Prove that $\sum_{P \in C(F) \cap L(F)} m_P(C, L) = 2$. This sum is called the *intersection number* of C and L and it is equal to the product of the degrees of C and L, where the latter is considered to be a plane projective curve of degree 1.

(iii) Suppose that D is a degenerate conic in $\mathbb{P}^2(F)$. By Proposition 3.2.12, we consider D as the union of two lines L_1 and L_2, not necessarily distinct. We call these lines the *irreducible components* of D. Prove that

$$\sum_{P \in C(F) \cap L_1(F)} m_P(C, L_1) + \sum_{P \in C(F) \cap L_2(F)} m_P(C, L_2) = 4.$$

This sum is called the *intersection number* of C and D and it is equal to the product of the degrees of C and D.

(iv) Let $L, M \subset \mathbb{P}^2(F)$ be two distinct lines. Imitating the definitions of the intersection multiplicity given above, define $m_P(L, M)$ for $P \in L(F) \cap M(F)$ and prove that $m_P(L, M) = 1$. There is, of course, only one such P. Therefore, the sum

$$\sum_{P \in L(F) \cap M(F)} m_P(L, M)$$

is equal to 1. This sum is called the *intersection number* of L and M and it is equal to the product of the degrees of L and M.

(v) Let D be a degenerate conic in $\mathbb{P}^2(F)$, with irreducible components L_1 and L_2; let E be another degenerate conic in $\mathbb{P}^2(F)$, with irreducible components M_1 and M_2; and suppose that D and E have no irreducible components in common. Prove that

$$\sum_{j=1}^{2} \sum_{i=1}^{2} m_{L_i \cdot M_j}(L_i, M_j) = 4.$$

This sum is called the *intersection number* of D and E and it is equal to the product of the degrees of D and E.

3.4 Conics Through Five Given Points

As we have seen in Chaps. 1 and 2, through two distinct points in $\mathbb{P}^2(F)$—indeed, in any projective plane $\mathbb{P}^2$—there passes a unique line (**P1**). Through one point in $\mathbb{P}^2(F)$ there pass infinitely many lines. Given a collection of 3 or more points in $\mathbb{P}^2(F)$, whether a line passes through them is contingent on the configuration of the points, but the general answer is negative.

The set of all lines through a given point in $\mathbb{P}^2(F)$, which we call a *pencil*, corresponds to a line in the dual projective plane $(\mathbb{P}^2(F))^*$ to $\mathbb{P}^2(F)$. Given a second point, the pencil of lines through that point corresponds to another line in $(\mathbb{P}^2(F))^*$. Since $(\mathbb{P}^2(F))^*$ is itself a projective plane, two lines in it intersect in a unique point, which corresponds to a line in $\mathbb{P}^2(F)$. Thus, the fact that one needs *two* distinct points to define a line uniquely has to do with the fact that the intersection of *two* lines in $\mathbb{P}^2(F)$ is a unique point, a fact which, in its turn, has to do with the *dimension* of $\mathbb{P}^2(F)$, which is 2.

In this section, we shall generalize these considerations to conics by investigating conics in $\mathbb{P}^2(F)$ that pass through a collection of N points. Similarly to the case of lines, there is a value such that when N is below that value, infinitely many conics pass through a collection of N points; when N is equal to that value, a unique conic passes through the given N points (subject to some mild restrictions on the points); finally, when N is above that value, whether a conic passes through the given N points again depends on the configuration of the points, but the answer is generally negative. The crucial value is *five*, hence the title of this section.

In Sect. 1.1.3, we identified lines in $\mathbb{P}^2(F)$ with points in the dual projective plane $(\mathbb{P}^2(F))^*$ by using the coefficients of the equations of the lines. Specifically, a line $ax + by + cz = 0$ in $\mathbb{P}^2(F)$ corresponds to the point $[a : b : c] \in (\mathbb{P}^2(F))^*$. Since a line is defined by as many coefficients as there are projective coordinates

in $\mathbb{P}^2(F)$, the dual projective plane $(\mathbb{P}^2(F))^*$ also has dimension 2. For conics, the situation is different since a conic is defined by *six* coefficients. It is therefore necessary to define projective spaces of any dimension before we proceed. The following definition generalizes Definitions 1.2.1 and 2.3.5 to any dimension.

Definition 3.4.1 Given a vector space V over F with $\dim_F V = n + 1$, where $n \geq 1$, the *projective space* $\mathbb{P}(V)$ is the set of 1-dimensional vector subspaces of V. If $V = F^{n+1}$, $\mathbb{P}(V)$ shall also be denoted by $\mathbb{P}^n(F)$.

The *dimension* of $\mathbb{P}(V)$ is n. If $n = 1$, $\mathbb{P}(V)$ is called a *projective line*; if $n = 2$, it is called a *projective plane*.

Remark 3.4.2 Given a basis $\mathbf{e_0}, \ldots \mathbf{e_n} \in V$, any nonzero vector $\mathbf{v} \in V$ can be written uniquely as $\mathbf{v} = x_0\mathbf{e_0} + \cdots x_n\mathbf{e_n}$, and we use the coefficients $x_0, \ldots, x_n$ (not all zero) as projective coordinates of $[\mathbf{v}] = [x_0 : \cdots : x_n] \in \mathbb{P}(V)$.

Generalizing lines in $\mathbb{P}^2(F)$, hyperplanes in $\mathbb{P}^n(F)$ are defined by homogeneous polynomials of degree 1.

Definition 3.4.3 A *hyperplane* in $\mathbb{P}^n(F)$ is a subset defined by an equation of the form $a_0x_0 + \cdots + a_nx_n = 0$, where $a_0, \ldots, a_n$ are not all zero.

Remark 3.4.4 A hyperplane $\mathcal{P}$ with equation $a_0x_0 + \cdots + a_nx_n = 0$ will be simply denoted as $\mathcal{P} : a_0x_0 + \cdots + a_nx_n = 0$. Just as in Remark 1.2.8, the union of all the 1-dimensional subspaces of F^{n+1} corresponding to the points in $\mathcal{P}$ under Definition 3.4.3 forms an n-dimensional subspace in F^{n+1} and this subspace is also defined by the equation $a_0x_0 + \cdots + a_nx_n = 0$.

The definition of a hyperplane can be generalized to linear subspaces of $\mathbb{P}^n(F)$ of any dimension.

Definition 3.4.5 A *linear subspace* of $\mathbb{P}^n(F)$ is a subset consisting of the points $[\mathbf{v}] \in \mathbb{P}^n(F)$ for nonzero $\mathbf{v} \in W$, where W is a vector subspace of V. We denote this subspace by $\mathbb{P}(W)$. If $\dim_F W = k+1$, then k is the *dimension* of $\mathbb{P}(W)$. $\mathbb{P}(W)$ is called a *line* in $\mathbb{P}^n(F)$ if its dimension is 1 and is called a *plane* in $\mathbb{P}^n(F)$ if its dimension is 2.

Remark 3.4.6 A point is a linear subspace of $\mathbb{P}^n(F)$ under Definition 3.4.5, with dimension 0.

Remark 3.4.7 Note that a hyperplane in $\mathbb{P}^n(F)$ is simply a linear subspace of dimension $n - 1$.

Remark 3.4.8 Let hyperplanes $\mathcal{P}_i : a_{i,0}x_0 + \cdots + a_{i,n}x_n = 0$ for $i = 1, \ldots, m$ be given. Considering these equations as defining linear subspaces in F^{n+1} and taking

their intersections, we see that $\bigcap_{i=1}^{m} \mathcal{P}_i$ is a linear subspace of $\mathbb{P}^n(F)$ of dimension $n - r$, where r is the rank of the $m \times (n+1)$ matrix

$$\begin{bmatrix} a_{1,0} & a_{1,1} & \cdots & a_{1,n} \\ a_{2,0} & a_{2,1} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,0} & a_{m,1} & \cdots & a_{m,n} \end{bmatrix}.$$

Proposition 3.4.9 *Let F be a quadratically closed field. Then two conics C and D over F, with equations*

$$C : a_1x^2 + b_1y^2 + c_1z^2 + 2d_1xy + 2e_1xz + 2f_1yz = 0,$$
$$D : a_2x^2 + b_2y^2 + c_2z^2 + 2d_2xy + 2e_2xz + 2f_2yz = 0,$$

have identical sets of points $C(F) = D(F)$ if and only if

$$[a_1 : b_1 : c_1 : d_1 : e_1 : f_1] = [a_2 : b_2 : c_2 : d_2 : e_2 : f_2] \in \mathbb{P}^5(F).$$

Proof The "if" direction is clear.

For the "only if" direction, we start by making a simple observation. Two given points

$$[a_1 : b_1 : c_1 : d_1 : e_1 : f_1], [a_2 : b_2 : c_2 : d_2 : e_2 : f_2] \in \mathbb{P}^5(F)$$

are scalar multiples of each other if and only if the corresponding symmetric matrices

$$A_i = \begin{bmatrix} a_i & d_i & e_i \\ d_i & b_i & f_i \\ e_i & f_i & c_i \end{bmatrix},$$

for $i = 1, 2$, are scalar multiples of each other, and that happens if and only if $Q^t A_1 Q$ and $Q^t A_2 Q$ are scalar multiples of each other for any $Q \in \mathrm{GL}_3(F)$. It follows that we can transform the equations of the conics C and D by any $Q \in \mathrm{GL}_3(F)$ in order to prove the "only if" part.

By Proposition 3.2.11, both $C(F)$ and $D(F)$ are in one-to-one correspondence with the elements of F, plus one other point. Suppose now that $C(F) = D(F)$. By Exercise 3.2.10, C and D are either both nondegenerate or both degenerate. If they are both degenerate, Exercise 3.2.11 finishes the proof. Suppose now that C and D are both nondegenerate, and choose five distinct points P_i, $i = 1, \ldots, 5$. By Proposition 3.2.10, no three of the P_i are collinear, and we can assume, without loss of generality, that $P_1 = [1 : 0 : 0]$, $P_2 = [0 : 1 : 0]$, $P_3 = [0 : 0 : 1]$, $P_4 = [1 : 1 : 1]$, $P_5 = [x_0 : y_0 : z_0]$. This yields $a_i = b_i = c_i = 0$ for $i = 1, 2$, and

$d_1 + e_1 + f_1 = d_2 + e_2 + f_2 = 0$. Plugging in the coordinates of P_5, we get two further equations

$$x_0y_0d_1 + x_0z_0e_1 + y_0z_0f_1 = 0,$$
$$x_0y_0d_2 + x_0z_0e_2 + y_0z_0f_2 = 0.$$

It follows that the solutions (d_1, e_1, f_1) and (d_2, e_2, f_2) are scalar multiples of each other. Hence,

$$[a_1 : b_1 : c_1 : d_1 : e_1 : f_1] = [a_2 : b_2 : c_2 : d_2 : e_2 : f_2] \in \mathbb{P}^5(F).$$

□

Remark 3.4.10 We consider two conics C and D in $\mathbb{P}^2(F)$ to be different if their sets of points over a quadratic closure of F are different. Proposition 3.4.9 shows that conics in $\mathbb{P}^2(F)$ correspond to points of $\mathbb{P}^5(F)$ in a one-to-one manner. Nondegenerate conics in $\mathbb{P}^2(F)$ correspond to those points of $\mathbb{P}^5(F)$ that satisfy the determinantal equation

$$\begin{vmatrix} a & d & e \\ d & b & f \\ e & f & c \end{vmatrix} \neq 0.$$

We now come to the main topic of this section, namely the classification of conics that pass through a given set of points in $\mathbb{P}^2(F)$. The cases of one point, two points, and three points are simple enough and left to the reader. We start with the case of four points, under the assumption that no three of the points are collinear.

Example 3.4.11 Consider the four points $A = [-1 : -1 : 1]$, $B = [-1 : 1 : 1]$, $C = [1 : -1 : 1]$, $D = [1 : 1 : 1]$ in $\mathbb{P}^2(F)$. A conic with equation

$$ax^2 + by^2 + cz^2 + 2dxy + 2exz + 2fyz = 0$$

passes through A, B, C, D if and only if

$$a + b + c + 2d - 2e - 2f = 0,$$
$$a + b + c - 2d - 2e + 2f = 0,$$
$$a + b + c - 2d + 2e - 2f = 0,$$
$$a + b + c + 2d + 2e + 2f = 0,$$

which the reader can check to be equivalent to $d = e = f = 0$ and $a + b + c = 0$. The family of conics defined by these equations can be viewed as a line in $\mathbb{P}^5(F)$; the general member of this family is the conic $C_{a,b} : ax^2 + by^2 - (a + b)z^2 = 0$,

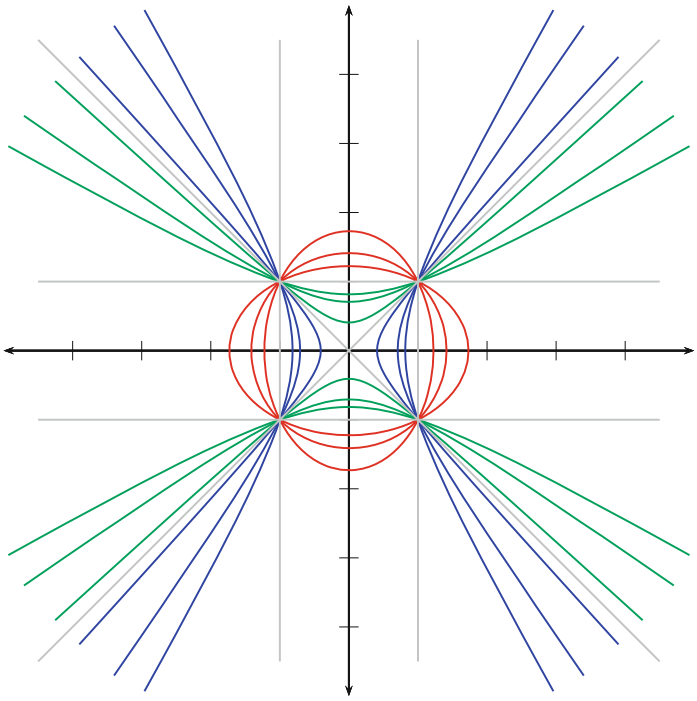


Fig. 3.8 Example 3.4.11 for $F = \mathbb{R}$. Only the affine plane with $z = 1$ is shown. The degenerate conics $C_{1,-1}$, $C_{1,0}$, $C_{0,1}$ are in gray

where $a, b \in F$ are not both zero. $C_{a,b}$ is degenerate if and only if $ab(a + b) = 0$; there are 3 such conics, with equations $C_{1,-1} : x^2 - y^2 = 0$, $C_{1,0} : x^2 - z^2 = 0$, $C_{0,1} : y^2 - z^2 = 0$.

Figure 3.8 shows some of the $C_{a,b}$ for $F = \mathbb{R}$, including $C_{1,-1}$, $C_{1,0}$, $C_{0,1}$.

Motivated by Example 3.4.11, we make the following definition.

Definition 3.4.12 A *pencil of conics* in $\mathbb{P}^2(F)$ is a family of conics given by a line in $\mathbb{P}^5(F)$.

In Proposition 3.2.5, we have seen that automorphisms of $\mathbb{P}^2(F)$ transform conics into conics. We recall that given a matrix $Q \in \mathrm{GL}_3(F)$, the conic

$$C : \mathbf{v}^t A \mathbf{v} = 0,$$

where

$$A = \begin{bmatrix} a & d & e \\ d & b & f \\ e & f & c \end{bmatrix}$$

and

$$\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$

is transformed into the conic

$$\phi_{Q^{-1}}(C) : \mathbf{v}^t (Q^t A Q)\mathbf{v} = 0.$$

Identifying the set of conics in $\mathbb{P}^2(F)$ with $\mathbb{P}^5(F)$ by Remark 3.4.10, we thus get a map $\Phi_Q : \mathbb{P}^5(F) \to \mathbb{P}^5(F)$, by sending C to $\phi_{Q^{-1}}(C)$.

Identifying points $(a, b, c, d, e, f) \in F^6$ with matrices of the form

$$A = \begin{bmatrix} a & d & e \\ d & b & f \\ e & f & c \end{bmatrix},$$

we see easily that Φ is induced by the vector space isomorphism $F^6 \to F^6$ that sends the matrix A as above to $Q^t A Q$. It is clear that Φ_Q sends linear subspaces of a given dimension to linear subspaces of the same dimension; in particular, it sends lines to lines. Hence, Q transforms a pencil of conics into another pencil of conics.

Theorem 3.4.13 *Let $A, B, C, D \in \mathbb{P}^2(F)$ be four points, no three of which are collinear. Then the conics in $\mathbb{P}^2(F)$ that pass through A, B, C, D form a pencil, and there are exactly three degenerate members of this pencil.*

Proof This follows from Proposition 1.4.6, Example 3.4.11, and the observation made above that an element of $\mathrm{GL}_3(F)$ transforms a pencil of conics into another pencil of conics. □

We now consider the case where five points in $\mathbb{P}^2(F)$ are given.

Theorem 3.4.14 *Let $A, B, C, D, E \in \mathbb{P}^2(F)$ be five points, no three of which are collinear. Then there exists a unique conic in $\mathbb{P}^2(F)$ that passes through all five, and it is nondegenerate.*

Proof Let four of the points be $A = [-1 : -1 : 1]$, $B = [-1 : 1 : 1]$, $C = [1 : -1 : 1]$, $D = [1 : 1 : 1]$, as in Example 3.4.11, and consider the pencil of conics

$$C_{a,b} : ax^2 + by^2 - (a+b)z^2 = 0,$$

where $a, b \in F$ are not both zero. Let now $E = [x_0 : y_0 : z_0]$. Then $E \in C_{a,b}(F)$ if and only if

$$a(x_0^2 - z_0^2) + b(y_0^2 - z_0^2) = 0,$$

which has one solution in a and b up to scalar multiple. In order to prove that $C_{a,b}$ is not one of the three degenerate conics listed in Example 3.4.11, one uses the hypothesis that E does not lie on any line passing through any two of A, B, C, D. We leave the details to the reader. □

Exercises

1. Prove that the intersection of a collection of linear subspaces of $\mathbb{P}^n(F)$ is a linear subspace of $\mathbb{P}^n(F)$ (possibly empty).
2. Let $\mathbb{P}(V)$ and $\mathbb{P}(W)$ be disjoint linear subspaces of $\mathbb{P}^n(F)$. Prove that $\dim \mathbb{P}(V) + \dim \mathbb{P}(W) \leq n - 1$.
3. Let $\mathbb{P}(V)$ and $\mathbb{P}(W)$ be disjoint linear subspaces of $\mathbb{P}^n(F)$ such that $\dim \mathbb{P}(V) + \dim \mathbb{P}(W) = n - 1$ and let $P \in \mathbb{P}^n(F) \setminus (\mathbb{P}(V) \cup \mathbb{P}(W))$. Prove that there exists a unique line through P that intersects $\mathbb{P}(V)$ and $\mathbb{P}(W)$.
4. Given three distinct lines in $\mathbb{P}^3(F)$, prove that there is a line that intersects all three given lines.
5. Given four distinct lines in $\mathbb{P}^3(F)$, when is there a line that intersects all four given lines?
6. Determine the sets of conics that pass through
   (i) one point,
   (ii) two points,
   (iii) three noncollinear points,
   (iv) three collinear points,
   (v) four points, three of which are collinear,
   (vi) four collinear points,
   as subsets of $\mathbb{P}^5(F)$.
7. Are there pencils of conics all of whose members are degenerate?
8. Prove that in any pencil of conics containing at least one nondegenerate conic there are at most 3 degenerate conics.
9. Finish the proof of Theorem 3.4.14.
10. Let $A, B \in \mathbb{P}^2(F)$ be two distinct points, and let $\ell, m \subset \mathbb{P}^2(F)$ be two distinct lines passing through A and B, respectively. Prove that the nondegenerate conics in $\mathbb{P}^2(F)$ that are tangent to ℓ at A and tangent to m at B are members of a uniquely defined pencil. How many degenerate members does this pencil have?
11. Pascal's Theorem (Theorem 3.3.1) gives necessary conditions for 6 points in $\mathbb{P}^2(F)$ to lie on a nondegenerate conic. Are these conditions sufficient?

3.5 Classification of Conics over $\mathbb{R}$

In this final section of this chapter, we establish the connections between the theory of conics in $\mathbb{P}^2(\mathbb{R})$ and the theory of conics in the Euclidean plane $\mathbb{A}^2(\mathbb{R})$ as is taught in courses on analytic geometry. Of course, every conic C in $\mathbb{P}^2(\mathbb{R})$ such that $C(\mathbb{R}) \neq \emptyset$ can be put, after a suitable change in coordinates, in the form $yz = x^2$ by Theorem 3.3.3. In that sense, every conic in $\mathbb{P}^2(\mathbb{R})$ with points over $\mathbb{R}$ "looks the same." But this is not the case in the theory of conics in the Euclidean plane, as there are three types of conics (hyperbola, parabola, ellipse) in the Euclidean plane. It turns out that the classification of the affine part of a nondegenerate conic in $\mathbb{P}^2(\mathbb{R})$ with points over $\mathbb{R}$ is determined by the intersection of the conic with the line at infinity (Theorems 3.5.2, 3.5.3, and 3.5.4).

Throughout this section, we consider $\mathbb{A}^2(\mathbb{R})$ as a subset of $\mathbb{P}^2(\mathbb{R})$ by setting $z = 1$. We work over the base field $F = \mathbb{R}$ unless otherwise mentioned, and we suppress the base field from the notation.

We first consider circles. A circle C with center (x_0, y_0) and radius $r > 0$ has the equation

$$C : (x - x_0)^2 + (y - y_0)^2 = r^2.$$

To define a conic in $\mathbb{P}^2$ that restricts to C on $\mathbb{A}^2$, we *homogenize* the equation of C with respect to z:

$$\overline{C} : (x - x_0 z)^2 + (y - y_0 z)^2 = r^2 z^2.$$

It is then clear that $C = \overline{C} \cap \mathbb{A}^2$ (Fig. 3.9).

What about the points of $\overline{C}$ that lie on the line at infinity $\ell_\infty : z = 0$? Setting $z = 0$ in the equation of $\overline{C}$, we get $x^2 + y^2 = 0$. There is only one real solution of this equation, namely $(x, y) = (0, 0)$, and it fails to yield a point in $\mathbb{P}^2$. However, over $\mathbb{C}$, $\overline{C}(\mathbb{C}) \cap \ell_\infty$ consists of the two points $[1 : i : 0]$ and $[1 : -i : 0]$.

Theorem 3.5.1 *A nondegenerate conic* $C : ax^2+by^2+cz^2+2dxy+2exz+2fyz = 0$ *with* $a \neq 0$ *describes a circle in* $\mathbb{A}^2$ *if and only if* $C(\mathbb{C}) \cap \ell_\infty$ *consists of the two points* $[1 : i : 0]$ *and* $[1 : -i : 0]$, *and* $e^2 + f^2 > ac$.

Proof One direction follows from the discussion above. For the other, the fact that $[1 : i : 0]$ and $[1 : -i : 0]$ lie on $C(\mathbb{C})$ implies that $a = b$ and $d = 0$. The equation of $C \cap \mathbb{A}^2$ is then

$$ax^2 + 2ex + ay^2 + 2fy + c = 0,$$

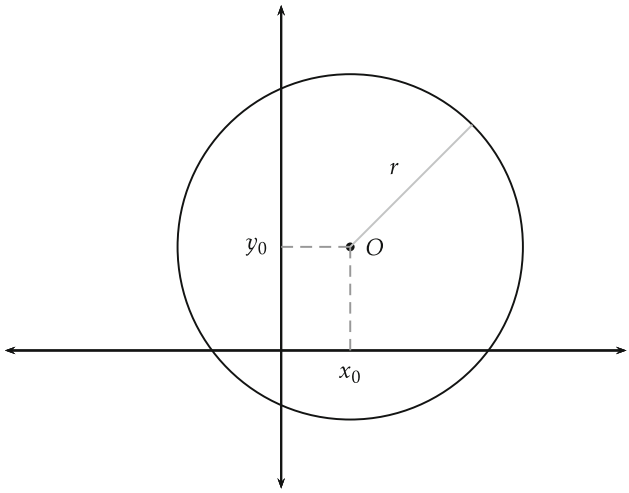


Fig. 3.9 The circle with center $O = (x_0, y_0)$ and radius $r > 0$

which, after completing the squares, can be rewritten as

$$\left(x + \frac{e}{a}\right)^2 + \left(y + \frac{f}{a}\right)^2 = \frac{e^2 + f^2 - ac}{a^2},$$

which is the equation of a circle in $\mathbb{A}^2$ since we have assumed $e^2 + f^2 > ac$. □

Before continuing, let us make a remark about the automorphisms of $\mathbb{A}^2$ (recall that the base field is $\mathbb{R}$). Recall, from Sect. 1.5, that the group of automorphisms of $\mathbb{A}^2$ is isomorphic to the semidirect product of a certain matrix group, denoted G, with the group of automorphisms of the field $\mathbb{R}$ (Theorem 1.5.13). Moreover, by Exercise 3 of Sect. 1.4, the group of automorphisms of $\mathbb{R}$ is trivial. Hence, by Proposition 1.5.7, every automorphism of $\mathbb{A}^2$ can be represented as a 3×3 nonsingular matrix of the form

$$\begin{bmatrix} & \mathrm{A} & x_0 \\ & & y_0 \\ 0 & 0 & 1 \end{bmatrix}$$

over $\mathbb{R}$, where its action on a point $(x, y) \in \mathbb{A}^2$ is defined by left multiplication with

$$\mathbf{v} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \in \mathbb{R}^3.$$

Since the equation of an ellipse, a parabola, or a hyperbola in $\mathbb{A}^2$ can be put into standard form using these automorphisms, we lose no generality if we work with standard forms. Note also that a 3×3 matrix of the form given above defines an element of $\mathrm{PGL}_3(\mathbb{R})$, hence an automorphism of $\mathbb{P}^2$, and that an automorphism of $\mathbb{P}^2$, necessarily given by an element of $\mathrm{PGL}_3(\mathbb{R})$, is of this form if and only if it preserves the line at infinity ℓ_∞ as a set. Thus, we consider automorphisms of $\mathbb{A}^2$ as automorphisms of $\mathbb{P}^2$ that preserve ℓ_∞.

We now move on to ellipses. The equation of an ellipse in $\mathbb{A}^2$ in standard form is

$$E : \frac{x^2}{\alpha^2} + \frac{y^2}{\beta^2} = 1,$$

where $\alpha, \beta \neq 0$. We homogenize the equation of E with respect to z:

$$\overline{E} : \beta^2 x^2 + \alpha^2 y^2 = \alpha^2 \beta^2 z^2.$$

It is clear that $E = \overline{E} \cap \mathbb{A}^2$. In order to obtain the points of $\overline{E}$ that lie on ℓ_∞, we set $z = 0$ in the equation of $\overline{E}$ and get $\beta^2 x^2 + \alpha^2 y^2 = 0$. As in the case of circles, the unique real solution of this equation does not yield a point in $\mathbb{A}^2$. We therefore consider the extension field $\mathbb{C}$ of $\mathbb{R}$, over which $\overline{E}(\mathbb{C}) \cap \ell_\infty$ consists of the two points $[\alpha : i\beta : 0]$ and $[\alpha : -i\beta : 0]$. Note that these two points are interchanged by complex conjugation (Fig. 3.10).

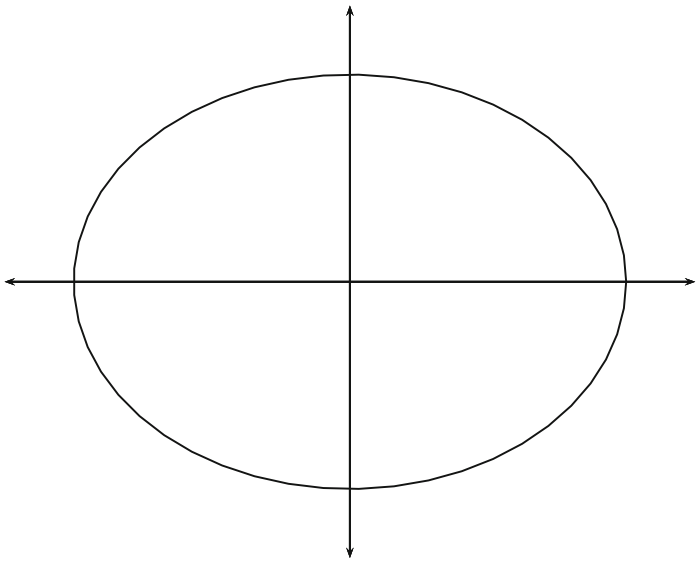


Fig. 3.10 An ellipse

Theorem 3.5.2 *A nondegenerate conic C over $\mathbb{R}$ with $C(\mathbb{R}) \neq \emptyset$ describes an ellipse in $\mathbb{A}^2$ if and only if $C(\mathbb{C}) \cap \ell_\infty$ consists of two $\mathbb{C}$-valued points that are interchanged by complex conjugation.*

The equation of a hyperbola in $\mathbb{A}^2$ in standard form is either

$$H : \frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1,$$

or

$$H : \frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = -1,$$

where $\alpha, \beta \neq 0$. The two asymptotes of the hyperbola have equations $\alpha y = \beta x$ and $\alpha y = -\beta x$. We homogenize the equation of E with respect to z, and we obtain either

$$\overline{H} : \beta^2 x^2 - \alpha^2 y^2 = \alpha^2 \beta^2 z^2,$$

or

$$\overline{H} : \beta^2 x^2 - \alpha^2 y^2 = -\alpha^2 \beta^2 z^2,$$

It is clear that $H = \overline{H} \cap \mathbb{A}^2$. In order to obtain the points of $\overline{H}$ that lie on ℓ_∞, we set $z = 0$ in the equation of $\overline{H}$ and get $\beta^2 x^2 - \alpha^2 y^2 = 0$. The two solutions to this equation give two $\mathbb{R}$-valued points on ℓ_∞, namely $[\alpha : \beta : 0]$ and $[\alpha : -\beta : 0]$. Note that these two points are, in fact, the ideal points of the two asymptotes of the hyperbola (Fig. 3.11).

Theorem 3.5.3 *A nondegenerate conic C over $\mathbb{R}$ with $C(\mathbb{R}) \neq \emptyset$ describes a hyperbola in $\mathbb{A}^2$ if and only if $C(\mathbb{R}) \cap \ell_\infty$ consists of two distinct points, which are then the ideal points of the asymptotes of the hyperbola.*

The equation of a parabola in $\mathbb{A}^2$ in standard form is

$$P : x^2 = 2py,$$

where $p \neq 0$. We homogenize the equation of P with respect to z:

$$\overline{P} : x^2 = 2pyz.$$

It is clear that $P = \overline{P} \cap \mathbb{A}^2$. In order to obtain the points of $\overline{P}$ that lie on ℓ_∞, we set $z = 0$ in the equation of $\overline{P}$ and get $x^2 = 0$. This equation has a unique real solution, namely $x = 0$, of multiplicity 2. This gives one $\mathbb{R}$-valued point on ℓ_∞, namely

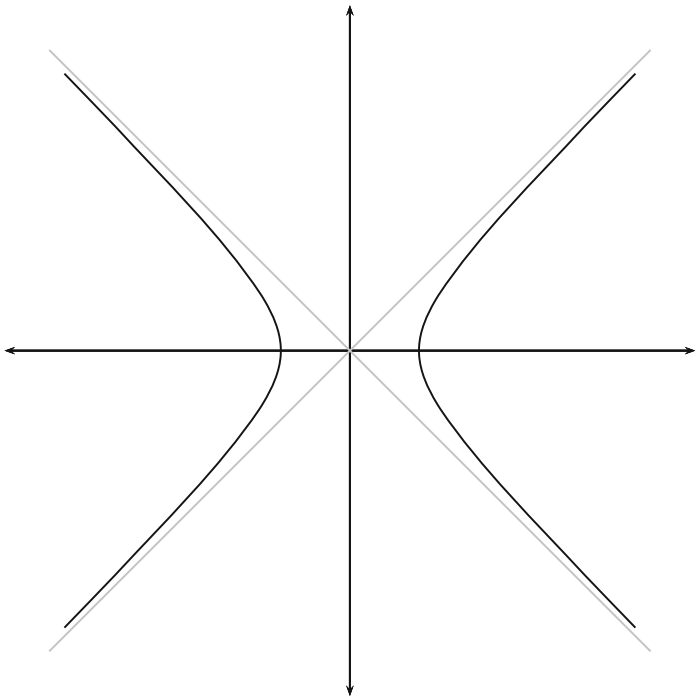
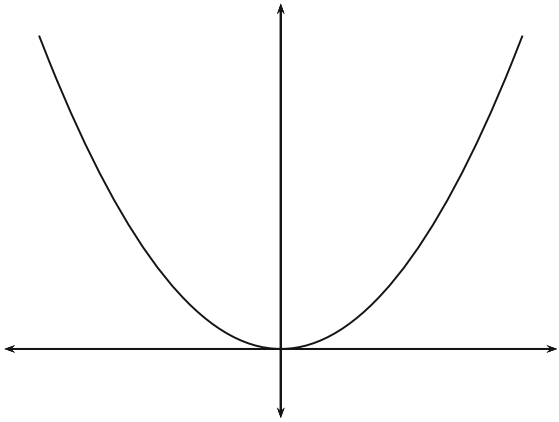


Fig. 3.11 A hyperbola and its asymptotes

Fig. 3.12 A parabola



$[0 : 1 : 0]$. It can be checked that the tangent line to P at this point is precisely ℓ_∞. Note also that $[0 : 1 : 0]$ is the ideal point of the axis of the parabola, which is the y-axis (Fig. 3.12).

Theorem 3.5.4 *A nondegenerate conic C over $\mathbb{R}$ with $C(\mathbb{R}) \neq \emptyset$ describes a parabola in $\mathbb{A}^2$ if and only if $C(\mathbb{R}) \cap \ell_\infty$ consists of a single point, necessarily over $\mathbb{R}$, which is the ideal point of the axis of the parabola.*

Suppose that C is a nondegenerate conic over $\mathbb{R}$ with $C(\mathbb{R}) \neq \emptyset$. By Proposition 3.2.10, there are three possibilities for the intersection of C with ℓ_∞:

(1) C intersects ℓ_∞ at two $\mathbb{C}$-valued points, which are then interchanged by complex conjugation. By Theorem 3.5.2, the affine part of C is an ellipse.
(2) C intersects ℓ_∞ at two distinct $\mathbb{R}$-valued points. By Theorem 3.5.3, the affine part of C is a hyperbola.
(3) C intersects ℓ_∞ at a unique $\mathbb{R}$-valued point, and this point is the point of tangency of C and ℓ_∞. By Theorem 3.5.4, the affine part of C is a parabola.

We therefore see that we have exhausted all the possibilities.

Exercise

1. We have only given the proofs of one direction of Theorems 3.5.2, 3.5.3, and 3.5.4. Complete the proofs of the other directions by showing that if C is a nondegenerate conic in $\mathbb{P}^2$ with points over $\mathbb{R}$, satisfying the conditions on the intersections with ℓ_∞, then its affine part is an ellipse, a hyperbola, or a parabola.

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E. Coşkun, *Projective Geometry*, Compact Textbooks in Mathematics,
https://doi.org/10.1007/978-3-032-02196-0

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